

TEST REPORT IEC 62368-1 Audio/video, information and communication technology equipment Part 1: Safety requirements	
Report Number	250623013SZN-001
Date of issue	Aug. 19, 2025 Amendment 2: May 20, 2026
Total number of pages	Refer to Page 3 for details
Name of Testing Laboratory preparing the Report	Intertek Testing Services Shenzhen Ltd. Longhua Branch
Applicant's name	Shen Zhen MTC Co., LTD
Address	MTC Industry Park, 1 st Lilang Road, Xialilang Community, Nanwan Street, Longgang District, Shenzhen, China
Test specification:	
Standard	IEC 62368-1:2018
Test procedure	General safety report
Non-standard test method	N/A
TRF template used	IECEE OD-2020-F1:2021, Ed.1.4
Test Report Form No.	IEC62368_1E
Test Report Form(s) Originator	UL(US)
Master TRF	Dated 2022-04-14
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Test item description	LED TV, LED LCD TV, TV	
Trade Mark(s)	AMTC, SHARP	
Manufacturer	Same as applicant	
Model/Type reference.....	MUDV50B0-T962K-S2, MUDV5080-T962K-S2, MUDV50****-T962K-##, 4T-C50HM5245EB, 4T-C50HM5245KB, 4T-C50GK4045KB, 4T-C50HK4245EB, 4T-C50HK4245KB, 4T-*50*****, 50HK4245K, 50HK4245E, 50*****, MUDV5080-T962K-S2 QLED, MUDV50****-T962K-## QLED, 4T-C50HM5246EB, 4T-C50HM5246KB, 4T-C50JM6245EB, 4T-C50JM6245KB ("*", "#" can be replaced by 0 to 9 or English letter from A to Z or blank)	
Ratings	220 - 240 V~, 50/60 Hz, 138 W, Class II	
Responsible Testing Laboratory (as applicable), testing procedure and testing location(s):		
☐ CB Testing Laboratory:	Intertek Testing Services Shenzhen Ltd. Longhua Branch	
Testing location/ address	No.101&201, Building B, No.308, Wuhe Avenue, Zhangkengjing, Guanhu Subdistrict, Longhua District, Shenzhen, Guangdong, China	
Tested by (name, function, signature)	Aline Li / Engineer	
Approved by (name, function, signature) ..	Jacky Chen / Sr. Project Engineer	
Testing procedure: CTF Stage 1:		
Testing location/ address		
Tested by (name, function, signature)		
Approved by (name, function, signature) ..		
Testing procedure: CTF Stage 2:		
Testing location/ address		
Tested by (name + signature)		
Witnessed by (name, function, signature) .		
Approved by (name, function, signature) ..		
Testing procedure: CTF Stage 3:		
Testing procedure: CTF Stage 4:		
Testing location/ address		
Tested by (name, function, signature)		
Witnessed by (name, function, signature) .		
Approved by (name, function, signature) ..		
Supervised by (name, function, signature) :		

List of Attachments (including a total number of pages in each attachment):

- Pages 1 to 86 for IEC 62368-1 TRF (main report)
- Appendix 1 (20 pages): Group and national differences of all CENELEC countries
- Appendix 2 (11 pages): Product Photos
- Appendix 3 (2 pages): Transformer Specifications

Summary of testing:

The sample(s) tested complies with the requirements of IEC 62368-1:2018 / EN IEC 62368-1:2020+A11:2020.

The tropical condition was considered and complied according to clause 5.4.8 humidity test.

The model of MUDV50B0-T962K-S2 was selected as representative for test unless other specified.

Tests performed (name of test and test clause):

Refer to content of this test report

Testing location:

Intertek Testing Services Shenzhen Ltd. Longhua Branch

No.101&201, Building B, No.308, Wuhe Avenue, Zhangkengjing, Guanhu Subdistrict, Longhua District, Shenzhen, Guangdong, China

Summary of compliance with National Differences (List of countries addressed):

Group and national differences of all CENELEC countries have been considered.

☒ **The product fulfils the requirements of IEC 62368-1:2018 / EN IEC 62368-1:2020+A11:2020**

Use of uncertainty of measurement for decisions on conformity (decision rule):

☒ No decision rule is specified by the IEC standard, when comparing the measurement result with the applicable limit according to the specification in that standard. The decisions on conformity are made without applying the measurement uncertainty ("simple acceptance" decision rule, previously known as "accuracy method").

☐ Other: (to be specified, for example when required by the standard or client, or if national accreditation requirements apply)

Information on uncertainty of measurement:

The uncertainties of measurement are calculated by the laboratory based on application of criteria given by OD-5014 for test equipment and application of test methods, decision sheets and operational procedures of IECEE.

IEC Guide 115 provides guidance on the application of measurement uncertainty principles and applying the decision rule when reporting test results within IECEE scheme, noting that the reporting of the measurement uncertainty for measurements is not necessary unless required by the test standard or customer.

Calculations leading to the reported values are on file with the NCB and testing laboratory that conducted the testing.

Copy of marking plate (representative):

The artwork below may be only a draft. The use of certification marks on a product must be authorized by the respective NCBs that own these marks.

**Marking label****Remark:**

- The above markings are the minimum requirements required by the safety standard. For the final production samples, the additional markings which do not give rise to misunderstanding may be added.
- Above information was labelled or silk-screened on rear enclosure.
- The CE and UKCA marking should be at least 5.0 mm in height.
- WEEE symbol should be at least 7.0 mm in height.
- The product name, model No. and trade mark can be replaced by others listed in this report.
- When placing electrical equipment on the EU market, according to EU NLF (new legislative framework), the importer (and manufacturer, if it is different from importer)'s name registered trade name or registered trade marks and the postal address at which he can be contacted shall be marked on the product or, where that is not possible, on its packaging or in a document accompanying the product. The contact details shall be in a language easily understood by end-users and market surveillance authorities.

Test item particulars:			
Product group	☒ end product	☐ built-in component	
Classification of use by	☒ Ordinary person ☒ Children likely present ☐ Instructed person ☐ Skilled person		
Supply connection	☒ AC mains ☐ DC mains ☐ not mains connected: ☐ ES1 ☐ ES2 ☒ ES3		
Supply tolerance	☒ +10%/-10% ☐ +20%/-15% ☐ + %/ - % ☐ None		
Supply connection – type	☒ pluggable equipment type A - ☒ non-detachable supply cord ☐ appliance coupler ☐ direct plug-in ☐ pluggable equipment type B - ☐ non-detachable supply cord ☐ appliance coupler ☐ permanent connection ☐ mating connector ☐ other:		
Considered current rating of protective device	☒ 16 A; Location: ☒ building ☐ equipment ☐ N/A		
Equipment mobility	☒ movable ☐ hand-held ☐ transportable ☐ direct plug-in ☐ stationary ☐ for building-in ☒ wall/ceiling-mounted ☐ SRME/rack-mounted ☐ other:		
Overvoltage category (OVC)	☐ OVC I ☒ OVC II ☐ OVC III ☐ OVC IV ☐ other:		
Class of equipment	☐ Class I ☒ Class II ☐ Class III ☐ Not classified ☐		
Special installation location	☒ N/A ☐ restricted access area ☐ outdoor location ☐		
Pollution degree (PD)	☐ PD 1 ☒ PD 2 ☐ PD 3		
Manufacturer's specified T_{ma}	40 °C ☐ Outdoor: minimum °C		
IP protection class	☒ IPX0 ☐ IP___		
Power systems	☒ TN ☐ TT ☐ IT - V _{L-L} ☐ not AC mains		
Altitude during operation (m)	☒ 2000 m or less ☐ m		
Altitude of test laboratory (m)	☒ 2000 m or less ☐ m		
Mass of equipment (kg)	☒ with stand base A: 8.7kg, stand base A: 0.16kg; with stand base B: 9.18kg, stand base B: 0.64kg		

Possible test case verdicts:	
- test case does not apply to the test object.....: N/A	
- test object does meet the requirement.....: P (Pass)	
- test object does not meet the requirement.....: F (Fail)	
Testing:	
Date of receipt of test item	Jun. 23, 2025 Amendment 1: N/A; Amendment 2: May 12, 2026
Date (s) of performance of tests	Jun. 23, 2025 – Jul. 15, 2025 Amendment 1: N/A; Amendment 2: May 12, 2026 to May 19, 2026
General remarks:	
"(See Enclosure #)" refers to additional information appended to the report. "(See appended table)" refers to a table appended to the report. Throughout this report a ☐ comma / ☒ point is used as the decimal separator.	
Manufacturer's Declaration per sub-clause 4.2.5 of IEC62368-1:	
The application for obtaining a CB Test Certificate includes more than one factory location and a declaration from the Manufacturer stating that the sample(s) submitted for evaluation is (are) representative of the products from each factory has been provided:	☒ Yes ☐ Not applicable
When differences exist; they shall be identified in the General product information section.	
Name and address of factory (ies):	1. Name: Shen Zhen MTC Co., LTD 1. Address: MTC Industry Park, 1 st Lilang Road, Xialilang Community, Nanwan Street, Longgang District, Shenzhen, China 2. Name: MTC TECHNOLOGY (VIETNAM) COMPANY LIMITED 2. Address: Road No. D3, Nhon Trach II–Loc Khang Industrial Park, Nhon Trach Commune, Dong Nai Province, Vietnam
General product information and other remarks:	
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Product Description:	
The product covered by this report is LCD TV with LED backlight which supplied by AC mains via non-detachable power cord set and classified as Class II equipment for pollution degree 2 and intended for indoor use only.	

The maximum output voltage of the LNB output can reach 18V, but when a signal is detected, it can only output 13V.

The LCD panels with all the models covered in this report are identical except for model No. and appearance (only for silk-screen and color, decorating parts).

The product has two configurations for rear enclosure, either configuration 1 or configuration 2 can be used for each model in end mass production. See below for detail:

Configurations	Difference
Configuration 1	Top rear enclosure used V-0 enclosure, bottom rear enclosure used HB enclosure, and internal use V-0 insulation sheet as fire barrier near power board.
Configuration 2	Both top and bottom rear enclosure used V-0 enclosure, and internal no need use V-0 insulation sheet as fire barrier near power board.

The product has two configurations for internal design, the difference between them are only conductive cloth on the flexible flat cable and conductive foam on the main board, see photo 4 and photo 12 on appendix 2 for details.

Either configuration A or configuration B can be used for each model in end mass production.

Model Differences:

The product with all the models covered in this report are the same as each other, except for different product name, model no., brand name and the appearance (for color, silk-screen only) for trading purpose.

Revision History:

Amendment 1:

This report was revision report basing on previous report with report No. 250623013SZN-001 dated on Aug. 19, 2025 and supersedes it due to the amendments as following:

- Added some new models: MUDV5080-T962K-S2, 4T-C50GK4045KB, 50HK4245K, 50HK4245E, 4T-C50HK4245EB, 4T-C50HK4245KB, 50***** ("*" can be replaced by 0 to 9 or English letter from A to Z or blank).
- Added factory 2 and its related information, refer to page 6 for details.

Basing on above amendments and based on previously conducted testing and the review of product construction, no additional clause/testing need to be re-considered and conducted and still complied.

Amendment 2:

This report is revision report and basing on previous report 250623013SZN-001 Amendment 1 dated on Dec. 02, 2025 and superseded it due to the following amendments:

- Added alternative product name: TV.
- Added alternative model No. MUDV5080-T962K-S2 QLED, MUDV50****-T962K-## QLED, 4T-C50HM5246EB, 4T-C50HM5246KB, 4T-C50JM6245EB, 4T-C50JM6245KB ("*", "#" can be replaced by 0 to 9 or English letter from A to Z or blank).
- Added stand base B, added some photos, see Fig. 20 to Fig. 23 of appendix 2 for details.

Basing on above amendments and based on previously conducted testing and the review of product construction, only the following clause/testing need to be re-considered and conducted and still complied: Clause 8.6.

OVERVIEW OF ENERGY SOURCES AND SAFEGUARDS				
Clause	Possible Hazard			
5	Electrically-caused injury			
Class and Energy Source (e.g. ES3: Primary circuit)	Body Part (e.g. Ordinary)	Safeguards		
		B	S	R
ES3: All primary circuits and components	Ordinary person	N/A	N/A	1. Plastic enclosure; 2. Approved Y-cap. and optocoupler used; 3. clearances and creepage distances; 4. Verified the test of prospective touch voltage and touch Current; 5. Isolated transformer.
ES1: All accessible connectors and parts		N/A	N/A	N/A

6		Electrically-caused fire		
Class and Energy Source (e.g. PS2: 100 Watt circuit)	Material part (e.g. Printed board)	Safeguards		
		B	1 st S	2 nd S
PS3: All circuits and components on power board PS3: LED circuit output of transformer T101	Combustible material	1. Ignition Not occurred; 2. No part of The equipment attained a temperature value greater than 90 % of the spontaneous ignition temperature limit; 3. Plastic enclosure not inside fire enclosure complied with HB or better	1. Refer to configurations in general product information field; 2. V-0 or better class PCB used; 3. All components mounted on V-0 class material; 4. Transformer comply with clause G.5.3; 5. AC connector made of V-0 class material.	N/A
PS2: +12V output of transformer T101 PS2: USB output		N/A	1. V-0 class PCB used; 2. All components mounted on V-0 class material; 3. Transformer comply with clause G.5.3.	N/A
PS1: Speaker output PS1: LNB output		N/A	N/A	N/A
7		Injury caused by hazardous substances		
Class and Energy Source (e.g. Ozone)	Body Part (e.g., Skilled)	Safeguards		
		B	S	R
N/A	N/A	N/A	N/A	N/A
8		Mechanically-caused injury		
Class and Energy Source (e.g. MS3: Plastic fan blades)	Body Part (e.g. Ordinary)	Safeguards		
		B	S	R
MS3: Mounting to wall (The Equipment >1kg, mounting height not declared, defined as mounted > 2m)	Ordinary person	Fixing by screws and brackets specified by manufacture	1.Screw sizes and numbers were advised in user manual 2. Threaded hole complied with 8.7.2, Test 2, Test 3	N/A
MS2: Mass of product >7kg & <25kg		see subclause 8.6 (skid proof pad provided	N/A	N/A

		for stand base)		
MS1: Rounded edges and corners; Equipment mass		N/A	N/A	N/A
9	Thermal burn			
Class and Energy Source (e.g. TS1: Keyboard caps)	Body Part (e.g., Ordinary)	Safeguards		
		B	S	R
TS1: All accessible external surface (> 1 s and < 10 s)	Ordinary person	N/A	N/A	N/A
10	Radiation			
Class and Energy Source (e.g. RS1: PMP sound output)	Body Part (e.g., Ordinary)	Safeguards		
		B	S	R
RS1: LED indicator	Ordinary person	N/A	N/A	N/A
Supplementary Information:				
"B" – Basic Safeguard; "S" – Supplementary Safeguard; "R" – Reinforced Safeguard				

ENERGY SOURCE DIAGRAM
Optional. Manufacturers are to provide the energy sources diagram identify declared energy sources and identifying the demarcations are between power sources. Recommend diagram be provided included in power supply and multipart systems. Insert diagram below. Example diagram designs are; Block diagrams; image(s) with layered data; mechanical drawings
☐ ES ☐ PS ☐ MS ☐ TS ☐ RS

IEC 62368-1			
Clause	Requirement + Test	Result - Remark	Verdict
4	GENERAL REQUIREMENTS		P
4.1.1	Acceptance of materials, components and subassemblies		P
4.1.2	Use of components		P
4.1.3	Equipment design and construction		P
4.1.4	Specified ambient temperature for outdoor use (°C) :	Indoor use only	N/A
4.1.5	Constructions and components not specifically covered		P
4.1.8	Liquids and liquid filled components (LFC)		N/A
4.1.15	Markings and instructions	(See Annex F)	P
4.4.3	Safeguard robustness		P
4.4.3.1	General		P
4.4.3.2	Steady force tests	(See Annex T.5)	P
4.4.3.3	Drop tests		N/A
4.4.3.4	Impact tests	(See Annex T.6)	P
4.4.3.5	Internal accessible safeguard tests		N/A
4.4.3.6	Glass impact tests		N/A
4.4.3.7	Glass fixation tests		N/A
	Glass impact test (1J)		N/A
	Push/pull test (10 N)		N/A
4.4.3.8	Thermoplastic material tests	(See Clause T.8)	P
4.4.3.9	Air comprising a safeguard		N/A
4.4.3.10	Accessibility, glass, safeguard effectiveness		P
4.4.4	Displacement of a safeguard by an insulating liquid		N/A
4.4.5	Safety interlocks		N/A
4.5	Explosion		P
4.5.1	General		P
4.5.2	No explosion during normal/abnormal operating condition	(See Clause B.2, B.3)	P
	No harm by explosion during single fault conditions	(See Clause B.4)	P
4.6	Fixing of conductors		P
	Fix conductors not to defeat a safeguard		P
	Compliance is checked by test..... :	(See Clause T.2)	P
4.7	Equipment for direct insertion into mains socket-outlets		N/A
4.7.2	Mains plug part complies with relevant standard .. :		N/A
4.7.3	Torque (Nm) :		N/A
4.8	Equipment containing coin/button cell batteries		N/A

IEC 62368-1			
Clause	Requirement + Test	Result - Remark	Verdict
4.8.1	General		N/A
4.8.2	Instructional safeguard		N/A
4.8.3	Battery compartment door/cover construction		N/A
	Open torque test		N/A
4.8.4.2	Stress relief test		N/A
4.8.4.3	Battery replacement test		N/A
4.8.4.4	Drop test		N/A
4.8.4.5	Impact test		N/A
4.8.4.6	Crush test		N/A
4.8.5	Compliance		N/A
	30N force test with test probe		N/A
	20N force test with test hook		N/A
4.9	Likelihood of fire or shock due to entry of conductive object		P
4.10	Component requirements		P
4.10.1	Disconnect Device	(See Annex L)	P
4.10.2	Switches and relays	No such component used	N/A
5	ELECTRICALLY-CAUSED INJURY		P
5.2	Classification and limits of electrical energy sources		P
5.2.2	ES1, ES2 and ES3 limits		P
5.2.2.2	Steady-state voltage and current limits	(See appended table 5.2)	P
5.2.2.3	Capacitance limits	(See appended table 5.2)	P
5.2.2.4	Single pulse limits		N/A
5.2.2.5	Limits for repetitive pulses		N/A
5.2.2.6	Ring signals		N/A
5.2.2.7	Audio signals	(See Clause E.1)	P
5.3	Protection against electrical energy sources		P
5.3.1	General Requirements for accessible parts to ordinary, instructed and skilled persons		P
5.3.1 a)	Accessible ES1/ES2 derived from ES2/ES3 circuits		P
5.3.1 b)	Skilled persons not unintentional contact ES3 bare conductors		P
5.3.2.1	Accessibility to electrical energy sources and safeguards		P
	Accessibility to outdoor equipment bare parts		N/A
5.3.2.2	Contact requirements		P

IEC 62368-1			
Clause	Requirement + Test	Result - Remark	Verdict
	Test with test probe from Annex V	For ES3 voltages up to 420 V _{peak} and above 420 V _{peak} , the test probe from Annex V cannot contact a bare internal conductive part	-
5.3.2.2 a)	Air gap – electric strength test potential (V)	See below	N/A
5.3.2.2 b)	Air gap – distance (mm)	> 0.2 mm (measured working voltage: 500V _{peak})	P
5.3.2.3	Compliance		P
5.3.2.4	Terminals for connecting stripped wire		P
5.4	Insulation materials and requirements		P
5.4.1.2	Properties of insulating material		P
5.4.1.3	Material is non-hygroscopic	No hygroscopic material used (See sub-clause 5.4.8)	P
5.4.1.4	Maximum operating temperature for insulating materials	(See appended table)	P
5.4.1.5	Pollution degrees	PD2	P
5.4.1.5.2	Test for pollution degree 1 environment and for an insulating compound		N/A
5.4.1.5.3	Thermal cycling test		N/A
5.4.1.6	Insulation in transformers with varying dimensions		N/A
5.4.1.7	Insulation in circuits generating starting pulses		N/A
5.4.1.8	Determination of working voltage	(See appended table 5.4.1.8)	P
5.4.1.9	Insulating surfaces		P
5.4.1.10	Thermoplastic parts on which conductive metallic parts are directly mounted	No conductive metallic parts are directly mounted on thermoplastic parts	N/A
5.4.1.10.2	Vicat test		N/A
5.4.1.10.3	Ball pressure test		N/A
5.4.2	Clearances		P
5.4.2.1	General requirements		P
	Clearances in circuits connected to AC Mains, Alternative method		N/A
5.4.2.2	Procedure 1 for determining clearance		P
	Temporary overvoltage	2000 V _{peak}	—
5.4.2.3	Procedure 2 for determining clearance		P
5.4.2.3.2.2	a.c. mains transient voltage	2500 V _{peak}	—
5.4.2.3.2.3	d.c. mains transient voltage		—
5.4.2.3.2.4	External circuit transient voltage		—
5.4.2.3.2.5	Transient voltage determined by measurement		—

IEC 62368-1			
Clause	Requirement + Test	Result - Remark	Verdict
5.4.2.4	Determining the adequacy of a clearance using an electric strength test	Clause 5.4.2.2 and 5.4.2.3 considered	N/A
5.4.2.5	Multiplication factors for clearances and test voltages	Up to 2000m sea level, the multiplication factor for clearances and electric strength test voltages is 1.00.	P
5.4.2.6	Clearance measurement		N/A
5.4.3	Creepage distances		P
5.4.3.1	General		P
5.4.3.3	Material group	IIIb	—
5.4.3.4	Creepage distances measurement	(See appended table 5.4.3)	P
5.4.4	Solid insulation		P
5.4.4.1	General requirements		P
5.4.4.2	Minimum distance through insulation	(See appended table 5.4.4.2)	P
5.4.4.3	Insulating compound forming solid insulation		N/A
5.4.4.4	Solid insulation in semiconductor devices	Approved optocoupler used, see appended table 4.1.2	P
5.4.4.5	Insulating compound forming cemented joints		N/A
5.4.4.6	Thin sheet material		P
5.4.4.6.1	General requirements		P
5.4.4.6.2	Separable thin sheet material		P
	Number of layers (pcs)	Two or more layers used.	P
5.4.4.6.3	Non-separable thin sheet material		N/A
	Number of layers (pcs)		N/A
5.4.4.6.4	Standard test procedure for non-separable thin sheet material		N/A
5.4.4.6.5	Mandrel test		N/A
5.4.4.7	Solid insulation in wound components	Approved TIW used inside transformer, see appended table 4.1.2	P

IEC 62368-1			
Clause	Requirement + Test	Result - Remark	Verdict
5.4.4.9	Solid insulation at frequencies >30 kHz, E_P , K_R , d , V_{PW} (V)	Transformer T101: operation frequency: 147.9 kHz, For bobbin transformer (Phenolic) E_P : 17 kV/mm K_R : 0.71 $E_F = 17 \text{ k} \times 0.71 = 12.07 \text{ kV/mm}$ d : 0.70 mm $V_w = 12.07 \text{ k} \times 0.70 = 8.449 \text{ kV}$ $V_w (9.052 \text{ kV}) > 1.2 \times 500 \text{ V} / 1.41$	P
	Alternative by electric strength test, tested voltage (V), K_R		N/A
5.4.5	Antenna terminal insulation		P
5.4.5.1	General		P
5.4.5.2	Voltage surge test		P
5.4.5.3	Insulation resistance ($M\Omega$)	> 4 $M\Omega$	P
	Electric strength test		N/A
5.4.6	Insulation of internal wire as part of supplementary safeguard		N/A
5.4.7	Tests for semiconductor components and for cemented joints		N/A
5.4.8	Humidity conditioning		P
	Relative humidity (%), temperature ($^{\circ}\text{C}$), duration (h)	Relative humidity: 93%; temperature: 40 $^{\circ}\text{C}$; duration: 120 h	—
5.4.9	Electric strength test		P
5.4.9.1	Test procedure for type test of solid insulation	(See appended table 5.4.9)	P
5.4.9.2	Test procedure for routine test	Conducted by manufacturer	P
5.4.10	Safeguards against transient voltages from external circuits		N/A
5.4.10.1	Parts and circuits separated from external circuits		N/A
5.4.10.2	Test methods		N/A
5.4.10.2.1	General		N/A
5.4.10.2.2	Impulse test		N/A
5.4.10.2.3	Steady-state test		N/A
5.4.10.3	Verification for insulation breakdown for impulse test		N/A
5.4.11	Separation between external circuits and earth		N/A

IEC 62368-1			
Clause	Requirement + Test	Result - Remark	Verdict
5.4.11.1	Exceptions to separation between external circuits and earth		N/A
5.4.11.2	Requirements		N/A
	SPDs bridge separation between external circuit and earth		N/A
	Rated operating voltage U_{op} (V)..... :		—
	Nominal voltage U_{peak} (V)..... :		—
	Max increase due to variation ΔU_{sp} :		—
	Max increase due to ageing ΔU_{sa} :		—
5.4.11.3	Test method and compliance :		N/A
5.4.12	Insulating liquid		N/A
5.4.12.1	General requirements		N/A
5.4.12.2	Electric strength of an insulating liquid :		N/A
5.4.12.3	Compatibility of an insulating liquid :		N/A
5.4.12.4	Container for insulating liquid :		N/A
5.5	Components as safeguards		P
5.5.1	General		P
5.5.2	Capacitors and RC units	Y-capacitor used as safeguard.	P
5.5.2.1	General requirement	Approved Y1-capacitor used to bridge between primary and secondary circuits as reinforced insulation.	P
5.5.2.2	Safeguards against capacitor discharge after disconnection of a connector :	(See appended table 5.5.2.2)	P
5.5.3	Transformers	(See Annex G.5.3)	P
5.5.4	Optocouplers	(See subclause 5.4 or Annex G.12)	P
5.5.5	Relays		N/A
5.5.6	Resistors		N/A
5.5.7	SPDs		N/A
5.5.8	Insulation between the mains and an external circuit consisting of a coaxial cable :		N/A
5.5.9	Safeguards for socket-outlets in outdoor equipment		N/A
	RCD rated residual operating current (mA) :		—
5.6	Protective conductor		N/A
5.6.2	Requirement for protective conductors		N/A
5.6	Protective conductor		N/A
5.6.2	Requirement for protective conductors		N/A

IEC 62368-1			
Clause	Requirement + Test	Result - Remark	Verdict
5.6.2.1	General requirements		N/A
5.6.2.2	Colour of insulation		N/A
5.6.3	Requirement for protective earthing conductors		N/A
	Protective earthing conductor size (mm ²)		—
	Protective earthing conductor serving as a reinforced safeguard		N/A
	Protective earthing conductor serving as a double safeguard		N/A
5.6.4	Requirements for protective bonding conductors		N/A
5.6.4.1	Protective bonding conductors		N/A
	Protective bonding conductor size (mm ²)		N/A
5.6.4.2	Protective current rating (A)		N/A
5.6.5	Terminals for protective conductors		N/A
5.6.5.1	Terminal size for connecting protective earthing conductors (mm)		N/A
	Terminal size for connecting protective bonding conductors (mm)		N/A
5.6.5.2	Corrosion		N/A
5.6.6	Resistance of the protective bonding system		N/A
5.6.6.1	Requirements		N/A
5.6.6.2	Test Method		N/A
5.6.6.3	Resistance (Ω) or voltage drop		N/A
5.6.7	Reliable connection of a protective earthing conductor		N/A
5.6.8	Functional earthing		N/A
	Conductor size (mm ²)		N/A
	Class II with functional earthing marking		N/A
	Appliance inlet cl & cr (mm)		N/A
5.7	Prospective touch voltage, touch current and protective conductor current		P
5.7.2	Measuring devices and networks		P
5.7.2.1	Measurement of touch current	See appended table 5.2	P
5.7.2.2	Measurement of voltage		P
5.7.3	Equipment set-up, supply connections and earth connections		P
5.7.4	Unearthed accessible parts	See appended table 5.2	P
5.7.5	Earthed accessible conductive parts		N/A
5.7.6	Requirements when touch current exceeds ES2 limits		N/A
	Protective conductor current (mA)		N/A

IEC 62368-1			
Clause	Requirement + Test	Result - Remark	Verdict
	Instructional Safeguard..... :	Refer to the user manual	N/A
5.7.7	Prospective touch voltage and touch current associated with external circuits		N/A
5.7.7.1	Touch current from coaxial cables		N/A
5.7.7.2	Prospective touch voltage and touch current associated with paired conductor cables		N/A
5.7.8	Summation of touch currents from external circuits		N/A
	a) Equipment connected to earthed external circuits, current (mA)		N/A
	b) Equipment connected to unearthed external circuits, current (mA)		N/A
5.8	Backfeed safeguard in battery backed up supplies		N/A
	Mains terminal ES		N/A
	Air gap (mm).....		N/A

6	ELECTRICALLY- CAUSED FIRE		P
6.2	Classification of PS and PIS		P
6.2.2	Power source circuit classifications	(See appended table 6.2.2)	P
6.2.3	Classification of potential ignition sources	See below	P
6.2.3.1	Arcing PIS	(See appended table 6.2.3.1)	P
6.2.3.2	Resistive PIS	(See appended table 6.2.3.2)	P
6.3	Safeguards against fire under normal operating and abnormal operating conditions		P
6.3.1	No ignition and attainable temperature value less than 90 % defined by ISO 871 or less than 300 °C for unknown materials	(See appended table B.1.5 and B.3)	P
	Combustible materials outside fire enclosure	HB or other non-combustible material	P
6.4	Safeguards against fire under single fault conditions		P
6.4.1	Safeguard method	Control fire spread used	P
6.4.2	Reduction of the likelihood of ignition under single fault conditions in PS1 circuits		N/A
6.4.3	Reduction of the likelihood of ignition under single fault conditions in PS2 and PS3 circuits		N/A
6.4.3.1	Supplementary safeguards		N/A
6.4.3.2	Single Fault Conditions		N/A
	Special conditions for temperature limited by fuse		N/A
6.4.4	Control of fire spread in PS1 circuits		N/A
6.4.5	Control of fire spread in PS2 circuits		P

IEC 62368-1			
Clause	Requirement + Test	Result - Remark	Verdict
6.4.5.2	Supplementary safeguards	- All components mounted on PCB with V-0 classified material; - Internal wiring rated VW-1.	P
6.4.6	Control of fire spread in PS3 circuits	Refer to configurations in general product information field.	P
6.4.7	Separation of combustible materials from a PIS		N/A
6.4.7.2	Separation by distance		N/A
6.4.7.3	Separation by a fire barrier		N/A
6.4.8	Fire enclosures and fire barriers		P
6.4.8.2	Fire enclosure and fire barrier material properties		P
6.4.8.2.1	Requirements for a fire barrier	Refer to configurations in general product information field.	P
6.4.8.2.2	Requirements for a fire enclosure	Refer to configurations in general product information field.	P
6.4.8.3	Constructional requirements for a fire enclosure and a fire barrier	Refer to configurations in general product information field.	P
6.4.8.3.1	Fire enclosure and fire barrier openings		P
6.4.8.3.2	Fire barrier dimensions		N/A
6.4.8.3.3	Top openings and properties		P
	Openings dimensions (mm)..... :	Openings located on top surface and rear surface of rear enclosure: oval-shaped, max. 5 mm in any dimension	P
6.4.8.3.4	Bottom openings and properties		P
	Openings dimensions (mm)..... :	Openings located on bottom surface of rear enclosure: max. 3 mm in any dimension	P
	Flammability tests for the bottom of a fire enclosure		N/A
	Instructional Safeguard..... :		N/A
6.4.8.3.5	Side openings and properties		P
	Openings dimensions (mm)..... :	Openings located on side surface of rear enclosure: round, not exceed 3 mm.	P
6.4.8.3.6	Integrity of a fire enclosure, condition met: a), b) or c)..... :		N/A

IEC 62368-1			
Clause	Requirement + Test	Result - Remark	Verdict
6.4.8.4	Separation of a PIS from a fire enclosure and a fire barrier distance (mm) or flammability rating :	Refer to configurations in general product information field.	P
6.4.9	Flammability of insulating liquid..... :		N/A
6.5	Internal and external wiring		P
6.5.1	General requirements	Internal wiring: rated VW-1	P
6.5.2	Requirements for interconnection to building wiring :		N/A
6.5.3	Internal wiring size (mm ²) for socket-outlets..... :		N/A
6.6	Safeguards against fire due to the connection to additional equipment		P
7	INJURY CAUSED BY HAZARDOUS SUBSTANCES		N/A
7.2	Reduction of exposure to hazardous substances		N/A
7.3	Ozone exposure		N/A
7.4	Use of personal safeguards or personal protective equipment (PPE)		N/A
	Personal safeguards and instructions :		—
7.5	Use of instructional safeguards and instructions		N/A
	Instructional safeguard (ISO 7010)..... :		—
7.6	Batteries and their protection circuits		N/A
8	MECHANICALLY-CAUSED INJURY		P
8.2	Mechanical energy source classifications		P
8.3	Safeguards against mechanical energy sources		P
8.4	Safeguards against parts with sharp edges and corners		P
8.4.1	Safeguards	Edges and corners are rounded, MS1	P
	Instructional Safeguard..... :		N/A
8.4.2	Sharp edges or corners		N/A
8.5	Safeguards against moving parts		N/A
8.5.1	Fingers, jewellery, clothing, hair, etc., contact with MS2 or MS3 parts	No such moving part	N/A
	MS2 or MS3 part required to be accessible for the function of the equipment		N/A
	Moving MS3 parts only accessible to skilled person		N/A
8.5.2	Instructional safeguard..... :		N/A
8.5.4	Special categories of equipment containing moving parts		N/A
8.5.4.1	General		N/A
8.5.4.2	Equipment containing work cells with MS3 parts		N/A

IEC 62368-1			
Clause	Requirement + Test	Result - Remark	Verdict
8.5.4.2.1	Protection of persons in the work cell		N/A
8.5.4.2.2	Access protection override		N/A
8.5.4.2.2.1	Override system		N/A
8.5.4.2.2.2	Visual indicator		N/A
8.5.4.2.3	Emergency stop system		N/A
	Maximum stopping distance from the point of activation (m).....:		N/A
	Space between end point and nearest fixed mechanical part (mm)		N/A
8.5.4.2.4	Endurance requirements		N/A
	Mechanical system subjected to 100 000 cycles of operation		N/A
	- Mechanical function check and visual inspection		N/A
	- Cable assembly		N/A
8.5.4.3	Equipment having electromechanical device for destruction of media		N/A
8.5.4.3.1	Equipment safeguards		N/A
8.5.4.3.2	Instructional safeguards against moving parts		N/A
8.5.4.3.3	Disconnection from the supply		N/A
8.5.4.3.4	Cut type and test force (N).....:		N/A
8.5.4.3.5	Compliance		N/A
8.5.5	High pressure lamps		N/A
	Explosion test.....:		N/A
8.5.5.3	Glass particles dimensions (mm)		N/A
8.6	Stability of equipment		P
8.6.1	General	MS2, Max. 8.7 kg with stand base A, with stand base B: 9.18kg, stand base B: 0.64kg	P
	Instructional safeguard.....:	Mentioned in user manual	P
8.6.2	Static stability		P
8.6.2.2	Static stability test	Rotated 360° at an angle of 10°. During the test, the EUT did not tip over.	P
8.6.2.3	Downward force test		N/A
8.6.3	Relocation stability		N/A
	Wheels diameter (mm)		—
	Tilt test		N/A

IEC 62368-1			
Clause	Requirement + Test	Result - Remark	Verdict
8.6.4	Glass slide test	During the test, the EUT did not tip over or slide.	P
8.6.5	Horizontal force test		N/A
8.7	Equipment mounted to wall, ceiling or other structure		P
8.7.1	Mount means type	Wall mounted	P
8.7.2	Test methods	Test 2 and Test 3	P
	Test 1, additional downwards force (N).....		N/A
	Test 2, number of attachment points and test force (N).....	4 threaded holes provided, 83.69N for each point.	P
	Test 3 Nominal diameter (mm) and applied torque (Nm).....	2.5 Nm	P
8.8	Handles strength		N/A
8.8.1	General		N/A
8.8.2	Handle strength test		N/A
	Number of handles.....	No handles used	—
	Force applied (N)		—
8.9	Wheels or casters attachment requirements		N/A
8.9.2	Pull test	No such devices used	N/A
8.10	Carts, stands and similar carriers		N/A
8.10.1	General	No such devices used	N/A
8.10.2	Marking and instructions.....		N/A
8.10.3	Cart, stand or carrier loading test		N/A
	Loading force applied (N)		N/A
8.10.4	Cart, stand or carrier impact test		N/A
8.10.5	Mechanical stability		N/A
	Force applied (N)		—
8.10.6	Thermoplastic temperature stability		N/A
8.11	Mounting means for slide-rail mounted equipment (SRME)		N/A
8.11.1	General	No such devices used	N/A
8.11.2	Requirements for slide rails		N/A
	Instructional Safeguard.....		N/A
8.11.3	Mechanical strength test		N/A
8.11.3.1	Downward force test, force (N) applied.....		N/A
8.11.3.2	Lateral push force test		N/A
8.11.3.3	Integrity of slide rail end stops		N/A
8.11.4	Compliance		N/A
8.12	Telescoping or rod antennas		N/A

IEC 62368-1			
Clause	Requirement + Test	Result - Remark	Verdict

	Button/ball diameter (mm)		—
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9	THERMAL BURN INJURY		P
9.2	Thermal energy source classifications		P
9.3	Touch temperature limits		P
9.3.1	Touch temperatures of accessible parts	(See appended table)	P
9.3.2	Test method and compliance		P
9.4	Safeguards against thermal energy sources		N/A
9.5	Requirements for safeguards		N/A
9.5.1	Equipment safeguard	TS1 for accessible enclosure	N/A
9.5.2	Instructional safeguard.....		N/A
9.6	Requirements for wireless power transmitters		N/A
9.6.1	General	No such function	N/A
9.6.2	Specification of the foreign objects		N/A
9.6.3	Test method and compliance		N/A

10	RADIATION		P
10.2	Radiation energy source classification		P
10.2.1	General classification		P
	Lasers	No laser diode used	—
	Lamps and lamp systems	Low power LED indicator and optical output terminal	—
	Image projectors	No such LED used	—
	X-Ray	No X-ray	—
	Personal music player	Not such appliance	—
10.3	Safeguards against laser radiation		N/A
	The standard(s) equipment containing laser(s) comply	No laser diode used	N/A
10.4	Safeguards against optical radiation from lamps and lamp systems (including LED types)		N/A
10.4.1	General requirements	Low power LED used	N/A
	Instructional safeguard provided for accessible radiation level needs to exceed		N/A
	Risk group marking and location		N/A
	Information for safe operation and installation		N/A
10.4.2	Requirements for enclosures		N/A
	UV radiation exposure		N/A
10.4.3	Instructional safeguard		N/A

IEC 62368-1			
Clause	Requirement + Test	Result - Remark	Verdict

10.5	Safeguards against X-radiation		N/A
10.5.1	Requirements	No X-ray	N/A
	Instructional safeguard for skilled persons		—
10.5.3	Maximum radiation (pA/kg)		—
10.6	Safeguards against acoustic energy sources		N/A
10.6.1	General	Not such appliance	N/A
10.6.2	Classification		N/A
	Acoustic output $L_{Aeq,T}$, dB(A)		N/A
	Unweighted RMS output voltage (mV)		N/A
	Digital output signal (dBFS)		N/A
10.6.3	Requirements for dose-based systems		N/A
10.6.3.1	General requirements		N/A
10.6.3.2	Dose-based warning and automatic decrease		N/A
10.6.3.3	Exposure-based warning and requirements		N/A
	30 s integrated exposure level (MEL30)		N/A
	Warning for MEL ≥ 100 dB(A)		N/A
10.6.4	Measurement methods		N/A
10.6.5	Protection of persons		N/A
	Instructional safeguards		N/A
10.6.6	Requirements for listening devices (headphones, earphones, etc.)		N/A
10.6.6.1	Corded listening devices with analogue input		N/A
	Listening device input voltage (mV)		N/A
10.6.6.2	Corded listening devices with digital input		N/A
	Max. acoustic output $L_{Aeq,T}$, dB(A)		N/A
10.6.6.3	Cordless listening devices		N/A
	Max. acoustic output $L_{Aeq,T}$, dB(A)		N/A

B	NORMAL OPERATING CONDITION TESTS, ABNORMAL OPERATING CONDITION TESTS AND SINGLE FAULT CONDITION TESTS		P
B.1	General		P
B.1.5	Temperature measurement conditions	(See appended table B.1.5)	P
B.2	Normal operating conditions		P
B.2.1	General requirements	(See Test Item Particulars and appended test tables)	P
	Audio Amplifiers and equipment with audio amplifiers	(See Annex E)	P
B.2.3	Supply voltage and tolerances		P

IEC 62368-1			
Clause	Requirement + Test	Result - Remark	Verdict
B.2.5	Input test	(See appended table B.2.5)	P
B.3	Simulated abnormal operating conditions		P
B.3.1	General		P
B.3.2	Covering of ventilation openings		P
	Instructional safeguard	Not used on soft support	N/A
B.3.3	DC mains polarity test		N/A
B.3.4	Setting of voltage selector		N/A
B.3.5	Maximum load at output terminals		P
B.3.6	Reverse battery polarity		P
B.3.7	Audio amplifier abnormal operating conditions		P
B.3.8	Safeguards functional during and after abnormal operating conditions	(See appended table B.3)	P
B.4	Simulated single fault conditions		P
B.4.1	General		P
B.4.2	Temperature controlling device		N/A
B.4.3	Blocked motor test		N/A
B.4.4	Functional insulation		P
B.4.4.1	Short circuit of clearances for functional insulation	Line and Neutral comply with the clearance for basic insulation. The other functional insulation complies with short circuit test	P
B.4.4.2	Short circuit of creepage distances for functional insulation	Line and Neutral comply with the creepage for basic insulation. The other functional insulation complies with short circuit test	P
B.4.4.3	Short circuit of functional insulation on coated printed boards		N/A
B.4.5	Short-circuit and interruption of electrodes in tubes and semiconductors	(See appended table B.4)	P
B.4.6	Short circuit or disconnection of passive components	(See appended table B.4)	P
B.4.7	Continuous operation of components		N/A
B.4.8	Compliance during and after single fault conditions	(See appended table B.4)	P
B.4.9	Battery charging and discharging under single fault conditions		N/A
C	UV RADIATION		N/A
C.1	Protection of materials in equipment from UV radiation		N/A

IEC 62368-1			
Clause	Requirement + Test	Result - Remark	Verdict
C.1.2	Requirements		N/A
C.1.3	Test method		N/A
C.2	UV light conditioning test		N/A
C.2.1	Test apparatus..... :		N/A
C.2.2	Mounting of test samples		N/A
C.2.3	Carbon-arc light-exposure test		N/A
C.2.4	Xenon-arc light-exposure test		N/A
D	TEST GENERATORS		N/A
D.1	Impulse test generators		N/A
D.2	Antenna interface test generator		N/A
D.3	Electronic pulse generator		N/A
E	TEST CONDITIONS FOR EQUIPMENT CONTAINING AUDIO AMPLIFIERS		P
E.1	Electrical energy source classification for audio signals		P
	Maximum non-clipped output power (W)..... :	9.25W x 2	—
	Rated load impedance (Ω)	6 Ω x 2	—
	Open-circuit output voltage (V)..... :	7.45Va.c	—
	Instructional safeguard	ES1, not applicable	—
E.2	Audio amplifier normal operating conditions		P
	Audio signal source type	1kHz	—
	Audio output power (W)..... :	1.15 W x 2	—
	Audio output voltage (V)	2.63 V x 2	—
	Rated load impedance (Ω)	6 Ω x 2	—
	Requirements for temperature measurement	(See Table B.1.5)	P
E.3	Audio amplifier abnormal operating conditions	(See Table B.3, B.4)	P
F	EQUIPMENT MARKINGS, INSTRUCTIONS, AND INSTRUCTIONAL SAFEGUARDS		P
F.1	General		P
	Language	English checked	—
F.2	Letter symbols and graphical symbols		P
F.2.1	Letter symbols according to IEC60027-1		P
F.2.2	Graphic symbols according to IEC, ISO or manufacturer specific		P
F.3	Equipment markings		P
F.3.1	Equipment marking locations		P
F.3.2	Equipment identification markings		P
F.3.2.1	Manufacturer identification	See copy of marking plate	P
F.3.2.2	Model identification	See copy of marking plate	P
F.3.3	Equipment rating markings		P

IEC 62368-1			
Clause	Requirement + Test	Result - Remark	Verdict
F.3.3.1	Equipment with direct connection to mains		P
F.3.3.2	Equipment without direct connection to mains		N/A
F.3.3.3	Nature of the supply voltage	See copy of marking plate	P
F.3.3.4	Rated voltage	See copy of marking plate	P
F.3.3.5	Rated frequency	See copy of marking plate	P
F.3.3.6	Rated current or rated power	See copy of marking plate	P
F.3.3.7	Equipment with multiple supply connections		N/A
F.3.4	Voltage setting device		N/A
F.3.5	Terminals and operating devices		P
F.3.5.1	Mains appliance outlet and socket-outlet markings		N/A
F.3.5.2	Switch position identification marking		N/A
F.3.5.3	Replacement fuse identification and rating markings	The fuse is not replaceable by an ordinary person or an instructed person. The identification of a suitable replacement fuse marked adjacent to the fuse Fuse F101: T5AL 250V	P
	Instructional safeguards for neutral fuse		N/A
F.3.5.4	Replacement battery identification marking		N/A
F.3.5.5	Neutral conductor terminal		N/A
F.3.5.6	Terminal marking location		P
F.3.6	Equipment markings related to equipment classification		P
F.3.6.1	Class I equipment		N/A
F.3.6.1.1	Protective earthing conductor terminal		N/A
F.3.6.1.2	Protective bonding conductor terminals		N/A
F.3.6.2	Equipment class marking	Class II equipment, use symbol "□" in label.	P
F.3.6.3	Functional earthing terminal marking		N/A
F.3.7	Equipment IP rating marking	IPX0	N/A
F.3.8	External power supply output marking	See copy of marking plate for detailed	P
F.3.9	Durability, legibility and permanence of marking		P
F.3.10	Test for permanence of markings		P
F.4	Instructions		P
	a) Information prior to installation and initial use		P

IEC 62368-1			
Clause	Requirement + Test	Result - Remark	Verdict
	b) Equipment for use in locations where children not likely to be present		N/A
	c) Instructions for installation and interconnection		P
	d) Equipment intended for use only in restricted access area		N/A
	e) Equipment intended to be fastened in place		N/A
	f) Instructions for audio equipment terminals		N/A
	g) Protective earthing used as a safeguard		N/A
	h) Protective conductor current exceeding ES2 limits		N/A
	i) Graphic symbols used on equipment		P
	j) Permanently connected equipment not provided with all-pole mains switch		N/A
	k) Replaceable components or modules providing safeguard function		N/A
	l) Equipment containing insulating liquid		N/A
	m) Installation instructions for outdoor equipment		N/A
F.5	Instructional safeguards		P
G	COMPONENTS		P
G.1	Switches		N/A
G.1.1	General	No such component used	N/A
G.1.2	Ratings, endurance, spacing, maximum load		N/A
G.1.3	Test method and compliance		N/A
G.2	Relays		N/A
G.2.1	Requirements	No such component used	N/A
G.2.2	Overload test		N/A
G.2.3	Relay controlling connectors supplying power to other equipment		N/A
G.2.4	Test method and compliance		N/A
G.3	Protective devices		P
G.3.1	Thermal cut-offs		N/A
	Thermal cut-outs separately approved according to IEC 60730 with conditions indicated in a) & b)		N/A
	Thermal cut-outs tested as part of the equipment as indicated in c)		N/A
G.3.1.2	Test method and compliance		N/A
G.3.2	Thermal links		N/A
G.3.2.1	a) Thermal links tested separately according to IEC 60691 with specifics		N/A
	b) Thermal links tested as part of the equipment		N/A

IEC 62368-1			
Clause	Requirement + Test	Result - Remark	Verdict
G.3.2.2	Test method and compliance		N/A
G.3.3	PTC thermistors		N/A
G.3.4	Overcurrent protection devices	Approved fuse used, see table 4.1.2 for details	P
G.3.5	Safeguards components not mentioned in G.3.1 to G.3.4		N/A
G.3.5.1	Non-resettable devices suitably rated and marking provided		N/A
G.3.5.2	Single faults conditions		N/A
G.4	Connectors		P
G.4.1	Spacings		P
G.4.2	Mains connector configuration.....	Approved mains connector used, see table 4.1.2 for details	P
G.4.3	Plug is shaped that insertion into mains socket-outlets or appliance coupler is unlikely		P
G.5	Wound components		P
G.5.1	Wire insulation in wound components	Approved TIW used as inside transformer	P
G.5.1.2	Protection against mechanical stress	Protection against mechanical stress by providing physical separation in the form of insulating tube or insulating tape	P
G.5.2	Endurance test		N/A
G.5.2.1	General test requirements		N/A
G.5.2.2	Heat run test		N/A
	Test time (days per cycle)		—
	Test temperature (°C).....		—
G.5.2.3	Wound components supplied from the mains		N/A
G.5.2.4	No insulation breakdown		N/A
G.5.3	Transformers		P
G.5.3.1	Compliance method.....	Meet the requirements given in G.5.3.2 and G.5.3.3	P
	Position	Part No. T101 separate primary from secondary circuits	P
	Method of protection	Protection by inherent or external impedance.	P
G.5.3.2	Insulation		P

IEC 62368-1			
Clause	Requirement + Test	Result - Remark	Verdict
	Protection from displacement of windings	Concentric windings on insulation applied in thin sheet form to the transformer core	—
G.5.3.3	Transformer overload tests		P
G.5.3.3.1	Test conditions		P
G.5.3.3.2	Winding temperatures	(See appended table B.3)	P
G.5.3.3.3	Winding temperatures - alternative test method		N/A
G.5.3.4	Transformers using FIW	No FIW used	N/A
G.5.3.4.1	General		N/A
	FIW wire nominal diameter		—
G.5.3.4.2	Transformers with basic insulation only		N/A
G.5.3.4.3	Transformers with double insulation or reinforced insulation		N/A
G.5.3.4.4	Transformers with FIW wound on metal or ferrite core		N/A
G.5.3.4.5	Thermal cycling test and compliance		N/A
G.5.3.4.6	Partial discharge test		N/A
G.5.3.4.7	Routine test		N/A
G.5.4	Motors	No such component used	N/A
G.5.4.1	General requirements		N/A
G.5.4.2	Motor overload test conditions		N/A
G.5.4.3	Running overload test		N/A
G.5.4.4.2	Locked-rotor overload test		N/A
	Test duration (days)		—
G.5.4.5	Running overload test for DC motors		N/A
G.5.4.5.2	Tested in the unit		N/A
G.5.4.5.3	Alternative method		N/A
G.5.4.6	Locked-rotor overload test for DC motors		N/A
G.5.4.6.2	Tested in the unit		N/A
	Maximum Temperature		N/A
G.5.4.6.3	Alternative method		N/A
G.5.4.7	Motors with capacitors		N/A
G.5.4.8	Three-phase motors		N/A
G.5.4.9	Series motors		N/A
	Operating voltage		—
G.6	Wire Insulation		P
G.6.1	General		P
G.6.2	Enamelled winding wire insulation		N/A

IEC 62368-1			
Clause	Requirement + Test	Result - Remark	Verdict
G.7	Mains supply cords		P
G.7.1	General requirements		P
	Type..... :	See appended table 4.1.2	—
G.7.2	Cross sectional area (mm ² or AWG) :	See appended table 4.1.2	P
G.7.3	Cord anchorages and strain relief for non-detachable power supply cords		P
G.7.3.21	Cord strain relief		P
G.7.3.2.1	Requirements		P
	Strain relief test force (N)..... :	100N applied, displacement: 1.3mm	P
G.7.3.2.2	Strain relief mechanism failure		N/A
G.7.3.2.3	Cord sheath or jacket position, distance (mm) :		N/A
G.7.3.2.4	Strain relief and cord anchorage material	Approved double insulated power cord used	P
G.7.4	Cord Entry	Plastic bushing used	P
G.7.5	Non-detachable cord bend protection		N/A
G.7.5.1	Requirements		N/A
G.7.5.2	Test method and compliance		N/A
	Overall diameter or minor overall dimension, <i>D</i> (mm) :		—
	Radius of curvature after test (mm) :		—
G.7.6	Supply wiring space		N/A
G.7.6.1	General requirements		N/A
G.7.6.2	Stranded wire		N/A
G.7.6.2.1	Requirements		N/A
G.7.6.2.2	Test with 8 mm strand		N/A
G.8	Varistors		P
G.8.1	General requirements	Approved varistor used, see table 4.1.2 for details	P
G.8.2	Safeguards against fire		P
G.8.2.1	General		P
G.8.2.2	Varistor overload test		P
G.8.2.3	Temporary overvoltage test		N/A
G.9	Integrated circuit (IC) current limiters		N/A
G.9.1	Requirements	No such component used	N/A
	IC limiter output current (max. 5A)..... :		—
	Manufacturers' defined drift :		—
G.9.2	Test Program		N/A
G.9.3	Compliance		N/A

IEC 62368-1			
Clause	Requirement + Test	Result - Remark	Verdict
G.10	Resistors		N/A
G.10.1	General	No such component used	N/A
G.10.2	Conditioning		N/A
G.10.3	Resistor test		N/A
G.10.4	Voltage surge test		N/A
G.10.5	Impulse test		N/A
G.10.6	Overload test		N/A
G.11	Capacitors and RC units		P
G.11.1	General requirements	X-capacitor and Y-capacitor is complied with IEC 60384-14.	P
G.11.2	Conditioning of capacitors and RC units		N/A
G.11.3	Rules for selecting capacitors		P
G.12	Optocouplers		P
	Optocouplers comply with IEC 60747-5-5 with specifics	Approved optocoupler, see appended table 4.1.2 for details	P
	Type test voltage $V_{ini,a}$		—
	Routine test voltage, $V_{ini,b}$		—
G.13	Printed boards		P
G.13.1	General requirements		P
G.13.2	Uncoated printed boards		P
G.13.3	Coated printed boards	Not such printed board	N/A
G.13.4	Insulation between conductors on the same inner surface		N/A
G.13.5	Insulation between conductors on different surfaces		N/A
	Distance through insulation		N/A
	Number of insulation layers (pcs)		—
G.13.6	Tests on coated printed boards		N/A
G.13.6.1	Sample preparation and preliminary inspection		N/A
G.13.6.2	Test method and compliance		N/A
G.14	Coating on components terminals		N/A
G.14.1	Requirements		N/A
G.15	Pressurized liquid filled components		N/A
G.15.1	Requirements		N/A
G.15.2	Test methods and compliance		N/A
G.15.2.1	Hydrostatic pressure test		N/A
G.15.2.2	Creep resistance test		N/A
G.15.2.3	Tubing and fittings compatibility test		N/A

IEC 62368-1			
Clause	Requirement + Test	Result - Remark	Verdict
G.15.2.4	Vibration test		N/A
G.15.2.5	Thermal cycling test		N/A
G.15.2.6	Force test		N/A
G.15.3	Compliance		N/A
G.16	IC including capacitor discharge function (ICX)		N/A
G.16.1	Condition for fault tested is not required	No such component used	N/A
	ICX with associated circuitry tested in equipment		N/A
	ICX tested separately		N/A
G.16.2	Tests		N/A
	Smallest capacitance and smallest resistance specified by ICX manufacturer for impulse test..... :		—
	Mains voltage that impulses to be superimposed on :		—
	Largest capacitance and smallest resistance for ICX tested by itself for 10000 cycles test :		—
G.16.3	Capacitor discharge test..... :		N/A
H	CRITERIA FOR TELEPHONE RINGING SIGNALS		N/A
H.1	General		N/A
H.2	Method A		N/A
H.3	Method B		N/A
H.3.1	Ringling signal		N/A
H.3.1.1	Frequency (Hz) :		—
H.3.1.2	Voltage (V) :		—
H.3.1.3	Cadence; time (s) and voltage (V) :		—
H.3.1.4	Single fault current (mA): :		—
H.3.2	Tripping device and monitoring voltage		N/A
H.3.2.1	Conditions for use of a tripping device or a monitoring voltage		N/A
H.3.2.2	Tripping device		N/A
H.3.2.3	Monitoring voltage (V) :		N/A
J	INSULATED WINDING WIRES FOR USE WITHOUT INTERLEAVED INSULATION		N/A
J.1	General		N/A
	Winding wire insulation :	Approved TIW used, see appended table 4.1.2 for details	—
	Solid round winding wire, diameter (mm) :		N/A
	Solid square and rectangular (flatwise bending) winding wire, cross-sectional area (mm ²)..... :		N/A

IEC 62368-1			
Clause	Requirement + Test	Result - Remark	Verdict
J.2/J.3	Tests and Manufacturing		N/A
K	SAFETY INTERLOCKS		N/A
K.1	General requirements		N/A
	Instructional safeguard		N/A
K.2	Components of safety interlock safeguard mechanism		N/A
K.3	Inadvertent change of operating mode		N/A
K.4	Interlock safeguard override		N/A
K.5	Fail-safe		N/A
K.5.1	Under single fault condition		N/A
K.6	Mechanically operated safety interlocks		N/A
K.6.1	Endurance requirement		N/A
K.6.2	Test method and compliance		N/A
K.7	Interlock circuit isolation		N/A
K.7.1	Separation distance for contact gaps & interlock circuit elements		N/A
	In circuit connected to mains, separation distance for contact gaps (mm).....		N/A
	In circuit isolated from mains, separation distance for contact gaps (mm).....		N/A
	Electric strength test before and after the test of K.7.2		N/A
K.7.2	Overload test, Current (A)		N/A
K.7.3	Endurance test		N/A
K.7.4	Electric strength test		N/A
L	DISCONNECT DEVICES		P
L.1	General requirements	Plug used as disconnect device	P
L.2	Permanently connected equipment		N/A
L.3	Parts that remain energized		N/A
L.4	Single-phase equipment	Disconnect both poles simultaneously	P
L.5	Three-phase equipment		N/A
L.6	Switches as disconnect devices		N/A
L.7	Plugs as disconnect devices	Mentioned in user manual	P
L.8	Multiple power sources		N/A
	Instructional safeguard	No such power sources	N/A
M	EQUIPMENT CONTAINING BATTERIES AND THEIR PROTECTION CIRCUITS		P
M.1	General requirements		P
M.2	Safety of batteries and their cells		N/A

IEC 62368-1			
Clause	Requirement + Test	Result - Remark	Verdict
M.2.1	Batteries and their cells comply with relevant IEC standards		N/A
M.3	Protection circuits for batteries provided within the equipment		P
M.3.1	Requirements	Non-rechargeable carbon-zinc or alkaline batteries are considered safe under short-circuiting conditions and therefore are not tested for discharge	P
M.3.2	Test method		N/A
	Overcharging of a rechargeable battery		N/A
	Excessive discharging		N/A
	Unintentional charging of a non-rechargeable battery		N/A
	Reverse charging of a rechargeable battery		N/A
M.3.3	Compliance		N/A
M.4	Additional safeguards for equipment containing a portable secondary lithium battery		N/A
M.4.1	General		N/A
M.4.2	Charging safeguards		N/A
M.4.2.1	Requirements		N/A
M.4.2.2	Compliance		N/A
M.4.3	Fire enclosure		N/A
M.4.4	Drop test of equipment containing a secondary lithium battery		N/A
M.4.4.2	Preparation and procedure for the drop test		N/A
M.4.4.3	Drop, Voltage on reference and dropped batteries (V); voltage difference during 24 h period (%):		N/A
M.4.4.4	Check of the charge/discharge function		N/A
M.4.4.5	Charge / discharge cycle test		N/A
M.4.4.6	Compliance		N/A
M.5	Risk of burn due to short-circuit during carrying		N/A
M.5.1	Requirement		N/A
M.5.2	Test method and compliance		N/A
M.6	Safeguards against short-circuits		N/A
M.6.1	External and internal faults		N/A
M.6.2	Compliance		N/A
M.7	Risk of explosion from lead acid and NiCd batteries		N/A
M.7.1	Ventilation preventing explosive gas concentration		N/A

IEC 62368-1			
Clause	Requirement + Test	Result - Remark	Verdict
	Calculated hydrogen generation rate		N/A
M.7.2	Test method and compliance		N/A
	Minimum air flow rate, Q (m ³ /h)		N/A
M.7.3	Ventilation tests		N/A
M.7.3.1	General		N/A
M.7.3.2	Ventilation test – alternative 1		N/A
	Hydrogen gas concentration (%)		N/A
M.7.3.3	Ventilation test – alternative 2		N/A
	Obtained hydrogen generation rate		N/A
M.7.3.4	Ventilation test – alternative 3		N/A
	Hydrogen gas concentration (%)		N/A
M.7.4	Marking		N/A
M.8	Protection against internal ignition from external spark sources of batteries with aqueous electrolyte		N/A
M.8.1	General		N/A
M.8.2	Test method		N/A
M.8.2.1	General		N/A
M.8.2.2	Estimation of hypothetical volume V_z (m ³ /s)		—
M.8.2.3	Correction factors		—
M.8.2.4	Calculation of distance d (mm)		—
M.9	Preventing electrolyte spillage		N/A
M.9.1	Protection from electrolyte spillage		N/A
M.9.2	Tray for preventing electrolyte spillage		N/A
M.10	Instructions to prevent reasonably foreseeable misuse		P
	Instructional safeguard	Mentioned in user manual	P
N	ELECTROCHEMICAL POTENTIALS		N/A
	Material(s) used		—
O	MEASUREMENT OF CREEPAGE DISTANCES AND CLEARANCES		P
	Value of X (mm)	1.0 mm, PD2	—
P	SAFEGUARDS AGAINST CONDUCTIVE OBJECTS		P
P.1	General		P
P.2	Safeguards against entry or consequences of entry of a foreign object		P
P.2.1	General		P
P.2.2	Safeguards against entry of a foreign object		P

IEC 62368-1			
Clause	Requirement + Test	Result - Remark	Verdict
	Location and Dimensions (mm)	Openings located on top surface and rear surface of rear enclosure: oval-shaped, max. 5mm in any dimension	—
P.2.3	Safeguards against the consequences of entry of a foreign object		N/A
P.2.3.1	Safeguard requirements		N/A
	The ES3 and PS3 keep-out volume in Figure P.3 not applicable to transportable equipment		N/A
	Transportable equipment with metalized plastic parts		N/A
P.2.3.2	Consequence of entry test.....		N/A
P.3	Safeguards against spillage of internal liquids		N/A
P.3.1	General		N/A
P.3.2	Determination of spillage consequences		N/A
P.3.3	Spillage safeguards		N/A
P.3.4	Compliance		N/A
P.4	Metallized coatings and adhesives securing parts		N/A
P.4.1	General	No such adhesives used as safeguard	N/A
P.4.2	Tests		N/A
	Conditioning, T _c (°C)		—
	Duration (weeks).....		—
Q	CIRCUITS INTENDED FOR INTERCONNECTION WITH BUILDING WIRING		P
Q.1	Limited power sources	(See appended table Q.1)	P
Q.1.1	Requirements		P
	a) Inherently limited output		N/A
	b) Impedance limited output		N/A
	c) Regulating network limited output		P
	d) Overcurrent protective device limited output		N/A
	e) IC current limiter complying with G.9		N/A
Q.1.2	Test method and compliance	(See appended table Annex Q.1)	P
	Current rating of overcurrent protective device (A)		N/A
Q.2	Test for external circuits – paired conductor cable		N/A
	Maximum output current (A)		N/A
	Current limiting method		—

IEC 62368-1			
Clause	Requirement + Test	Result - Remark	Verdict
R	LIMITED SHORT CIRCUIT TEST		N/A
R.1	General		N/A
R.2	Test setup		N/A
	Overcurrent protective device for test..... :		—
R.3	Test method		N/A
	Cord/cable used for test :		—
R.4	Compliance		N/A
S	TESTS FOR RESISTANCE TO HEAT AND FIRE		N/A
S.1	Flammability test for fire enclosures and fire barrier materials of equipment where the steady state power does not exceed 4 000 W		N/A
	Samples, material :		—
	Wall thickness (mm) :		—
	Conditioning (°C) :		—
	Test flame according to IEC 60695-11-5 with conditions as set out		N/A
	- Material not consumed completely		N/A
	- Material extinguishes within 30s		N/A
	- No burning of layer or wrapping tissue		N/A
S.2	Flammability test for fire enclosure and fire barrier integrity		N/A
	Samples, material :		—
	Wall thickness (mm) :		—
	Conditioning (°C) :		—
S.3	Flammability test for the bottom of a fire enclosure		N/A
S.3.1	Mounting of samples		N/A
S.3.2	Test method and compliance		N/A
	Mounting of samples :		—
	Wall thickness (mm) :		—
S.4	Flammability classification of materials		N/A
S.5	Flammability test for fire enclosure materials of equipment with a steady state power exceeding 4 000 W		N/A
	Samples, material :		—
	Wall thickness (mm) :		—
	Conditioning (°C) :		—
T	MECHANICAL STRENGTH TESTS		P
T.1	General		P
T.2	Steady force test, 10 N :	(See appended table T.2)	P
T.3	Steady force test, 30 N :		N/A
T.4	Steady force test, 100 N :		N/A

IEC 62368-1			
Clause	Requirement + Test	Result - Remark	Verdict
T.5	Steady force test, 250 N :	(See appended table T.5)	P
T.6	Enclosure impact test	(See appended table T.6)	P
	Fall test		P
	Swing test		N/A
T.7	Drop test :		N/A
T.8	Stress relief test..... :	(See appended table T.8)	P
T.9	Glass Impact Test :		N/A
T.10	Glass fragmentation test		N/A
	Number of particles counted.....:		N/A
T.11	Test for telescoping or rod antennas		N/A
	Torque value (Nm)		N/A
U	MECHANICAL STRENGTH OF CATHODE RAY TUBES (CRT) AND PROTECTION AGAINST THE EFFECTS OF IMPLOSION		N/A
U.1	General		N/A
	Instructional safeguard :		N/A
U.2	Test method and compliance for non-intrinsically protected CRTs		N/A
U.3	Protective screen		N/A
V	DETERMINATION OF ACCESSIBLE PARTS		P
V.1	Accessible parts of equipment		P
V.1.1	General		P
V.1.2	Surfaces and openings tested with jointed test probes		P
V.1.3	Openings tested with straight unjointed test probes		P
V.1.4	Plugs, jacks, connectors tested with blunt probe		P
V.1.5	Slot openings tested with wedge probe		N/A
V.1.6	Terminals tested with rigid test wire		P
V.2	Accessible part criterion		P
X	ALTERNATIVE METHOD FOR DETERMINING CLEARANCES FOR INSULATION IN CIRCUITS CONNECTED TO AN AC MAINS NOT EXCEEDING 420 V PEAK (300 V RMS)		N/A
	Clearance		N/A
Y	CONSTRUCTION REQUIREMENTS FOR OUTDOOR ENCLOSURES		N/A
Y.1	General	Indoor used only	N/A
Y.2	Resistance to UV radiation		N/A
Y.3	Resistance to corrosion		N/A
Y.3	Resistance to corrosion		N/A

IEC 62368-1			
Clause	Requirement + Test	Result - Remark	Verdict
Y.3.1	Metallic parts of outdoor enclosures are resistant to effects of water-borne contaminants by :		N/A
Y.3.2	Test apparatus		N/A
Y.3.3	Water – saturated sulphur dioxide atmosphere		N/A
Y.3.4	Test procedure :		N/A
Y.3.5	Compliance		N/A
Y.4	Gaskets		N/A
Y.4.1	General		N/A
Y.4.2	Gasket tests		N/A
Y.4.3	Tensile strength and elongation tests		N/A
	Alternative test methods :		N/A
Y.4.4	Compression test		N/A
Y.4.5	Oil resistance		N/A
Y.4.6	Securing means		N/A
Y.5	Protection of equipment within an outdoor enclosure		N/A
Y.5.1	General		N/A
Y.5.2	Protection from moisture		N/A
	Relevant tests of IEC 60529 or Y.5.3 :		N/A
Y.5.3	Water spray test		N/A
Y.5.4	Protection from plants and vermin		N/A
Y.5.5	Protection from excessive dust		N/A
Y.5.5.1	General		N/A
Y.5.5.2	IP5X equipment		N/A
Y.5.5.3	IP6X equipment		N/A
Y.6	Mechanical strength of enclosures		N/A
Y.6.1	General		N/A
Y.6.2	Impact test :		N/A

IEC 62368-1			
Clause	Requirement + Test	Result - Remark	Verdict

5.2	TABLE: Classification of electrical energy sources						P
Supply Voltage	Location (e.g. circuit designation)	Test conditions	Parameters				ES Class
			U (V)	I (mA)	Type ¹⁾	Additional Info ²⁾	
264Vac	All primary circuits	--	--	--	--	--	ES3 (Declared)
264Vac	USB 1/2 output+ to –	Normal	4.94Vdc	--	SS	DC	ES1
		Abnormal: OL	3.88Vdc	--	SS	DC	
		Single fault: UU2 PIN 1-5 SC	4.98Vdc	--	SS	DC	
		Single fault: UD21 PIN 2-3 SC	0	--	SS	DC	
264Vac	LNB output+ to –	Normal	17.70Vdc	--	SS	DC	ES1
		Abnormal: OL	17.42Vdc	--	SS	DC	
		Single fault: UT101 1-2 SC	11.86Vdc	--	SS	DC	
264Vac	Speaker + to -	Normal	2.36Vrms	--	SS	1K	ES1
		Abnormal-MAX	7.45Vrms	--	SS	1K	
		Single fault-SC	0	--	SS	--	
264Vac	All accessible port and metal enclosure to earth	Normal	--	0.198mA _{pk}	SS	60	ES1
		Abnormal Output overload	--	0.198mA _{pk}	SS	60	
		Single fault: EC101 SC	--	0.230mA _{pk}	SS	60	
264Vac	Plastic enclosure with metal foil to GND	Normal	--	0.011mA _{pk}	SS	60	ES1
		Abnormal Output overload	--	0.011mA _{pk}	SS	60	
		Single fault: EC101 SC	--	0.011mA _{pk}	SS	60	
Supplementary information:							
1) Type: Steady state (SS), Capacitance (CP), Single pulse (SP), Repetitive pulses (RP), etc,							
2) Additional Info: Frequency, Pulse duration, Pulse off time, Capacitance value, etc,							
3) USB 1 and USB 2 are in parallel.							

5.4.1.8	TABLE: Working voltage measurement				P
Location	RMS voltage (V)	Peak voltage (V)	Frequency (Hz)	Comments	

IEC 62368-1			
Clause	Requirement + Test	Result - Remark	Verdict

Primary pins to secondary pins				
CY101	112	200	60	--
CY102	107	184	60	--
CY103	109	200	60	--
CY104	109	200	60	--
CY106	181	372	60	--
U204	199	392	60	Max. RMS and Max. Peak
T101	287	500	147.9K	Max. RMS and Max. Peak
Supplementary information:				
Max. measured value shown on above table.				

5.4.1.10.2	TABLE: Vicat softening temperature of thermoplastics			N/A
Method.....:			ISO 306 / B50	—
Object/ Part No./Material	Manufacturer/trademark	Thickness (mm)	T softening (°C)	
--	--	--	--	
Supplementary information: --				

5.4.1.10.3	TABLE: Ball pressure test of thermoplastics			N/A
Allowed impression diameter (mm)		≤ 2 mm		—
Object/Part No./Material	Manufacturer/trademark	Thickness (mm)	Test temperature (°C)	Impression diameter (mm)
--	--	--	--	--
Supplementary information: --				

5.4.2, 5.4.3	TABLE: Minimum Clearances/Creepage distance							P
Clearance (cl) and creepage distance (cr) at/of/between:	U _p (V)	U _{rms} (V)	Freq ¹⁾ (kHz)	Required cl (mm)	cl (mm)	E, S, ²⁾ (V)	Required cr (mm)	cr (mm)
Primary pins to ferrite core of switching transformer T101 (B)	500	287	>30	3.0	25.0	--	5.8	25.0
Primary winding to ferrite core of switching transformer T101 (B)	500	287	>30	3.0	25.0	--	5.8	25.0

IEC 62368-1								
Clause	Requirement + Test				Result - Remark			Verdict
Secondary pin to primary pin of switching transformer T101 (R)	500	287	>30	3.0	33.7	--	5.8	33.7
Primary component EC103 to ferrite core of switching transformer T101 (B)	500	287	>30	3.0	9.2	--	5.8	9.2
Primary pins/ winding to ferrite core of PFC filter L101 (B)	≤420	≤240	>30	1.5	3.7	--	2.4	2.6
Secondary pins to ferrite core of PFC filter L101 (B)	≤420	≤240	>30	1.5	3.7	--	2.4	2.6
Different polarity trace of CY101/ CY102/ CY103/CY104/CY106 capacitor on PCB soldering side (R)	≤420	≤240	<30	3.0	7.4	--	4.8	7.4
Primary to secondary trace of optocoupler U204 on PCB soldering side (R)	≤420	≤240	<30	3.0	6.5	--	4.8	6.5
Primary to secondary trace on PCB soldering side (R) (Min.)	500	287	>30	3.0	7.1	--	5.8	7.1
Primary live parts to opening of rear enclosure (R)	500	287	>30	3.0	7.8	--	5.8	7.8
Primary live parts to secondary wire after 10N forced (R)	≤420	≤240	<30	3.0	>3.4	--	4.8	>6.3
Different poles of mains switch (B)	≤420	≤240	<30	1.5	3.5	--	2.4	3.5
Different polarity of line and neutral (B)	≤420	≤240	<30	1.5	2.6	--	2.4	2.6
Across mains fuse-link F101 (B)	≤420	≤240	<30	1.5	3.0	--	2.4	3.0
Supplementary information:								
1) Complete Electric Strength voltage (E, S, (V) when 5,4,2,4 applied)								
2) B=Basic insulation, S=Supplementary insulation, D=Double insulation, R=Reinforced insulation.								
3) Ferrite core was considered as secondary part.								

IEC 62368-1			
Clause	Requirement + Test	Result - Remark	Verdict

5.4.4.2	TABLE: Minimum distance through insulation				P
Distance through insulation (DTI) at/of		Peak voltage (V)	Insulation	Required DTI (mm)	Measured DTI (mm)
Plastic enclosure (R)		420	Reinforce	≥ 0.4	Min. 2.0
Bobbin of transformer T101		500	Reinforce	≥ 0.4	Min. 0.7
Supplementary information: --					

5.4.4.9	TABLE: Solid insulation at frequencies >30 kHz						P
Insulation material		E_P	Frequency (kHz)	K_R	Thickness d (mm)	Insulation	V_{PW} (Vpk)
Bobbin of transformer T101 (Phenolic)		17 kV/mm	147.9	0.71	0.7	Reinforce	500
Supplementary information: --							

5.4.9	TABLE: Electric strength tests			P
Test voltage applied between:		Voltage shape (Surge, Impulse, AC, DC, etc.)	Test voltage (V)	Breakdown Yes / No
L to N (fuse open)		DC	2500	No
L/N to output terminal		DC	4000	No
L/N to plastic enclosure with metal foil		DC	4000	No
L/N to metal enclosure		DC	4000	No
Primary to secondary of T101		DC	4000	No
Core to primary of T101		DC	4000	No
Supplementary information: --				

5.5.2.2	TABLE: Stored discharge on capacitors					P
Location	Supply voltage (V)	Operating and fault condition ¹⁾	Switch position	Measured voltage (Vpk)	ES Class	
Plug of L/N	264	Normal	/	16	ES1	
Plug of L/N	264	OC R127	/	32	ES1	
Supplementary information:						
X-capacitors installed for testing: see table 4.1.2						
[x] bleeding resistor rating: see table 4.1.2						
[] ICX:						
Normal operating condition (e.g., normal operation, or open fuse), SC= short circuit, OC= open circuit						

IEC 62368-1			
Clause	Requirement + Test	Result - Remark	Verdict

5.6.6	TABLE: Resistance of protective conductors and terminations				N/A
Location	Test current (A)	Duration (min)	Voltage drop (V)	Resistance (Ω)	
--	--	--	--	--	
Supplementary information: --					

5.7.4	TABLE: Unearthed accessible parts					P
Location	Operating and fault conditions	Supply Voltage (V)	Parameters			ES class
			Voltage (V _{rms} or V _{pk})	Current (A _{rms} or A _{peak})	Freq, (Hz)	
All accessible port and metal enclosure to earth	Normal	264Va.c	--	0.198mA _{pk}	60	ES1
	Abnormal Output overload	264Va.c	--	0.198mA _{pk}	60	ES1
	Single fault: EC101 SC	264Va.c	--	0.230mA _{pk}	60	ES1
Plastic enclosure to earth	Normal	264Va.c	--	0.011mA _{pk}	60	ES1
	Abnormal Output overload	264Va.c	--	0.011mA _{pk}	60	ES1
	Single fault: EC101 SC	264Va.c	--	0.011mA _{pk}	60	ES1
Supplementary information:						
Abbreviation: SC= short circuit; OC= open circuit						

5.7.5	TABLE: Earthed accessible conductive part			N/A
Supply voltage (V).....:	264VAC			—
Phase(s)	[x] Single Phase; [] Three Phase: [] Delta [] Wye			
Power Distribution System	☒ TN ☐ TT ☐ IT			
Location	Fault Condition No in IEC 60990 clause 6.2.2	Touch current (mA)	Comment	
--	--	--	--	
Supplementary Information:				

5.8	TABLE: Backfeed safeguard in battery backed up supplies					N/A
Location	Supply voltage (V)	Operating and fault condition	Time (s)	Open-circuit voltage (V)	Touch current (A)	ES Class
--	--	--	--	--	--	--

IEC 62368-1			
Clause	Requirement + Test	Result - Remark	Verdict

Supplementary information:

Abbreviation: SC= short circuit, OC= open circuit

6.2.2	TABLE: Power source circuit classifications					P
Location	Operating and fault condition	Voltage (V)	Current (A)	Max, Power ¹⁾ (W)	Time (S)	PS class
All circuits and components on power board	--	--	--	--	--	PS3
+12V output of transformer T101	Normal	10.80	5.9	64.1	5	PS2
	Single fault – EC212 SC	0	0	0	3	
LED circuit output of transformer T101	Normal	146.79	2.1	299.6	5	PS3
	Single fault – EC204 SC	0	0	0	3	
USB 1/2 output + to –	Normal	3.88	0.97	3.79	3	PS1
	Single fault: UU2 PIN 1-5 SC	4.84	0.5	2.41	3	
	Single fault: UD21 PIN 2-3 SC	0	0	0	3	
LNB output	Normal	17.42	0.68	11.85	3	PS1
	Single fault: UT101 1-2 SC	10.99	0.40	4.37	3	
Speaker output	Normal	2.63	0.44	1.15	3	PS1
	Abnormal-MAX	7.45	1.24	9.25	3	
	SC	0	0	0	3	
Supplementary information:						
Abbreviation: SC= short circuit; OC= open circuit; Max = Maximum attainable output power						
Measured after 3 s for PS1 and measured after 5 s for PS2 and PS3						
USB 1 and USB 2 are in parallel.						

6.2.3.1	TABLE: Determination of Arcing PIS	P
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IEC 62368-1			
Clause	Requirement + Test	Result - Remark	Verdict

Location	Open circuit voltage after 3 s (Vpk)	Measured r,m,s current (A)	Calculated value	Arcing PIS? Yes / No
All components/PCB trace located in PS3 circuits	--	--	--	Yes (declared)
Supplementary information: --				

6.2.3.2	TABLE: Determination of resistive PIS			P
Location		Operating and fault condition	Dissipate power (W)	Arcing PIS? Yes / No
All components/PCB trace located in PS3 circuits		--	--	Yes (declared)
Supplementary information:				
Abbreviation: SC= short circuit; OC= open circuit				

8.5.5	TABLE: High pressure lamp				N/A
Lamp manufacturer	Lamp type	Explosion method	Longest axis of glass particle (mm)	Particle found beyond 1 m Yes / No	
--	--	--	--	--	
Supplementary information: --					

9.6	TABLE: Temperature measurements for wireless power transmitters								N/A
Supply voltage (V)				--				—	
Max, transmit power of transmitter (W)				--				—	
Foreign objects	w/o receiver and direct contact		with receiver and direct contact		with receiver and at distance of 2 mm		with receiver and at distance of 5 mm		
	Object (°C)	Ambient (°C)	Object (°C)	Ambient (°C)	Object (°C)	Ambient (°C)	Object (°C)	Ambient (°C)	
-	--	--	--	--	--	--	--	--	
Supplementary information: --									

5.4.1.4, 9.3, B.1.5, B.2.6	TABLE: Temperature measurements					P
Supply voltage (V)		198(60Hz)	264(60Hz)	--	--	—
Ambient temperature during test T_{amb} (°C)		25.0	25.0	--	--	—
Maximum measured temperature T of part/at:		T (°C)				Allowed T_{max} (°C)
CON101		34.5	34.7	--	--	55=70- 40+25

Annexment 2: May 20, 2024

IEC 62368-1								
Clause		Requirement + Test			Result - Remark			Verdict
RV101		35.0		34.9	--	--	70=85-40+25	
LF103		40.3		37.0	--	--	115=130-40+25	
CX101		42.0		38.5	--	--	85=100-40+45	
LF101		65.3		42.1	--	--	115=130-4-+25	
PCB near BD101		74.4		55.9	--	--	115=130-40+25	
EC101 body		42.1		41.8	--	--	90=105-40+25	
L101 coil		71.7		56.2	--	--	115=130-40+25	
T101		64.9		64.7	--	--	95=110-40+45	
T101		60.8		60.8	--	--	95=110-40+45	
CY103 body		35.6		35.5	--	--	110=125-40+25	
CY106 body		52.8		50.9	--	--	110=125-40+25	
U204 body		38.5		38.6	--	--	85=100-40+25	
PCB near D204		69		69	--	--	115=130-40+25	
Mylar surface		38.5		38.5	--	--	ref	
Plastic enclosure inside near T101		36.5		36.6	--	--	ref	
Plastic enclosure outside near T101		32.5		32.7	--	--	77	
Metal enclosure outside		37.5		37.5	--	--	60	
Screen		34.0		34.1	--	--	77	
Temperature T of winding:		t ₁ (°C)	R ₁ (Ω)	t ₂ (°C)	R ₂ (Ω)	T (°C)	Allowed T _{max} (°C)	Insulation class
--		--	--	--	--	--	--	--
Supplementary information:								
Max, T _{ma} =40°C								

B.2.5		TABLE: Input test						P
U (V)	Hz	I (A)	I rated (A)	P (W)	P rated (W)	Fuse No	I fuse (A)	Condition/status
198	50	0.47	--	93.65	--	F101	0.47	LNB mode: USB1 and USB2
220	50	0.43	--	94.08	138	F101	0.43	

Amendment 2: May 20, 2023

IEC 62368-1								
Clause		Requirement + Test				Result - Remark		Verdict
240	50	0.40	--	92.43	138	F101	0.40	terminal loaded 5Vdc/500mA; LNB output loaded 13V/400mA
264	50	0.38	--	92.39	--	F101	0.38	
198	60	0.49	--	93.42	--	F101	0.49	
220	60	0.44	--	93.26	138	F101	0.44	
240	60	0.41	--	93.66	138	F101	0.41	
264	60	0.39	--	93.54	--	F101	0.39	HDMI Mode: 1/8 max. non-clipped output power USB loaded 5Vdc/500mA
198	50	0.43	--	84.47	--	F101	0.43	
220	50	0.40	--	85.74	138	F101	0.40	
240	50	0.36	--	83.35	138	F101	0.36	
264	50	0.35	--	83.29	--	F101	0.35	
198	60	0.44	--	84.24	--	F101	0.44	
220	60	0.40	--	84.06	138	F101	0.40	
240	60	0.37	--	84.58	138	F101	0.37	
264	60	0.36	--	84.45	--	F101	0.36	
Supplementary information: --								
The maximum output voltage of the LNB output can reach 18V, but when a signal is detected, it can only output 13V.								

B.3, B.4 TABLE: Abnormal operating and fault condition tests						P
Ambient temperature T_{amb} (°C)					25.0	—
Power source for EUT: Manufacturer, model/type, output rating....					--	—
Component No,	Condition	Supply voltage (V)	Test time	Fuse no,	Fuse current (A)	Observation
USB1/2 output “+” to “-”	OL	198	1hrs 57mins	F101	0.49→ 0.50→ 0.45	When USB2.0 output loaded with 0.94A, EUT attained steady conditions, and no hazard, then output current rise to 0.95A, the USB2.0 output can't be loaded, no higher temperature rise occurred. L101 coil: 60.3°C T101 coil: 56.8°C PCB near BD101: 54.4°C External plastic enclosure: 30.7 °C External metal enclosure: 41.3°C Screen: 31.1°C
LNB output “+” to “-”	OL	198	2hrs 8mins	F101	0.49 to 0.51 to 0.43	When LNB output loaded with 0.64A, EUT attained steady conditions, and no hazard, then output current rise to

IEC 62368-1						
Clause	Requirement + Test			Result - Remark		Verdict
						0.65A, the LNB output can't be loaded, no higher temperature rise occurred. L101 coil: 61.6°C T101 coil: 58.4°C PCB near BD101: 55.9°C External plastic enclosure: 32.3 °C External metal enclosure: 42.2°C Screen: 32.2°C
Speaker "R" output	SC	198	1hrs 35mins	FB1	0.658	The speaker couldn't work, recoverable, no component damaged, no hazards Winding surface of transformer T101:63.9 °C PCB near D204: 68.0°C External enclosure surface(plastic):32.2 °C External enclosure surface(metallic):37.0 °C Screen:33.7 °C
Speaker "R" output	Max	198	1hrs 36mins	FB1	0.841	EUT run for steady, No hazard. Winding surface of transformer T101:70.0 °C PCB near D204: 73.3°C External enclosure surface(plastic):33.1 °C External enclosure surface(metallic):37.3 °C Screen:33.9 °C
Transformer T101 output +12V circuit	OL	198	2hrs 53mins	F101	0.9→ 0.39	When Transformer T101 output +12V circuit current 3.5A for attaining steady conditions, and no hazard, then output current rise to 4.0A, the Transformer output +12V circuit can't be loaded, no higher temperature rise occurred. No component damaged, no hazards. The Max. temperature see below: PCB near Q102: 65.5°C T101 coil: 79°C T101 coil: 74.5°C External enclosure surface(plastic): 45.5°C External enclosure surface(metallic): 38.7°C

IEC 62368-1						
Clause	Requirement + Test			Result - Remark		Verdict
						Screen panel surface(plastic): 30.4°C Ambient: 25°C
Transformer T101 output LED circuit	OL	198	2hrs 7mins	F101	1.36→ 0.07	When Transformer T101 secondary (LED) output overload with 0.9A, and the highest temperature not exceeding its limit. And the output current exceeds 1A, the EUT can't work. Unrecoverable. PCB near Q102: 90.6°C T101 primary coil: 86.9°C T101 secondly coil: 75.1°C External enclosure surface(plastic): 31.5°C External enclosure surface(metallic): 47.9°C Screen panel surface(plastic): 31.2°C Ambient: 25°C
EUT	Openings of enclosure BL	198	1hrs 50mins	FB1	0.689	EUT run for steady, no component damaged, no hazards. Winding surface of transformer T101:68.7 °C PCB near D204: 74.5°C External enclosure surface(plastic):36.0 °C External enclosure surface(metallic):47.4 °C Screen:42.9 °C
U101 pin10-18	SC	264	10mins	F101	0.11	The unit shut down unrecoverable, no hazard, NB.
U101 pin12-18	SC	264	10mins	F101	0.11	The unit shut down unrecoverable, no hazard, NB.
U101 pin 8-18	SC	264	10mins	F101	0.11	The unit shut down unrecoverable, no hazard, NB.
Q101 Pin D-G	SC	198/264	1s	F101	0	Fuse opened immediately

IEC 62368-1						
Clause	Requirement + Test			Result - Remark		Verdict
						with fuse open current exceeding 2.75 times fuse link rated current, No hazard, NB.
Q101 Pin D-S	SC	198/264	1s	F101	0	Fuse opened immediately with fuse open current exceeding 2.75 times fuse link rated current, No hazard, NB.
Q101 Pin G-S	SC	264	10mins	F101	0.11	The unit shut down unrecoverable, no hazard, NB.
T101 pin 1-4	SC	264	1s	F101	0	Fuse opened immediately with fuse open current exceeding 2.75 times fuse link rated current, No hazard, NB.
T101 pin 2-3	SC	264	1s	F101	0	Fuse opened immediately with fuse open current exceeding 2.75 times fuse link rated current, No hazard, NB.
T101 pin 11-12	SC	264	10mins	F101	0.11	The unit shut down recoverable, no hazard, NB,
T101 pin 9-10	SC	264	10mins	F101	0.11	The unit shut down recoverable, no hazard, NB,
T101 pin 5-6	SC	264	10mins	F101	0.11	The unit shut down recoverable, no hazard, NB,
USB output + to -	SC	264	10mins	F101	0.71	The USB can't work, no component damaged, no hazard.
BD101	SC	264	1s	F101	0	Fuse opened immediately with fuse open current exceeding 2.75 times fuse link rated current, No hazard, NB.
EC102	SC	264	1s	F101	0	Fuse opened immediately with fuse open current exceeding 2.75 times fuse link rated current, No hazard, NB.
U204 pin3-4	SC	264	10mins	F101	0.10	The unit shut down unrecoverable, no hazard, NB.
U204 pin1-2	SC	264	10mins	F101	0.10	The unit shut down unrecoverable, no hazard, NB.
U204 pin1	OC	264	10mins	F101	0.10	The unit shut down unrecoverable, no hazard, NB.
Supplementary information:						

IEC 62368-1			
Clause	Requirement + Test	Result - Remark	Verdict

- Max: 100% of max, non-clipped output power; BL: Block; OL: Overload; SC=short circuit; OC= open circuit,
- Temperature limit for component: Transformer T101: 175+25-40-10=150°C, PCB: 300+25-40=285°C
- Comply with Hi-pot test including insulation component after the abnormal and single fault test
- USB 1 and USB 2 are in parallel.

M.3	TABLE: Protection circuits for batteries provided within the equipment						N/A
Is it possible to install the battery in a reverse polarity position?				--		—	
Equipment Specification	Charging						
	Voltage (V)			Current (A)			
	--			--			
Manufacturer/type	Battery specification						
	Non-rechargeable batteries		Rechargeable batteries				
	Discharging current (A)	Unintentional charging current (A)	Charging		Discharging current (A)	Reverse charging current (A)	
			Voltage (V)	Current (A)			
--	--	--	--	--	--	--	
Note: The tests of M.3.2 are applicable only when above appropriate data is not available,							
Specified battery temperature (°C).....				--			
Component No,	Fault condition	Charge/discharge mode	Test time	Temp, (°C)	Current (A)	Voltage (V)	Observation
--	--	--	--	--	--	--	--
Supplementary information:							
Abbreviation: SC= short circuit; OC= open circuit NL= no chemical leakage; NS= no spillage of liquid; NE= no explosion; NF= no emission of flame or expulsion of molten metal,							

M.4.2	TABLE: Charging safeguards for equipment containing a secondary lithium battery					N/A
Maximum specified charging voltage (V)					--	—
Maximum specified charging current (A)					--	—
Highest specified charging temperature (°C)					--	
Lowest specified charging temperature (°C)					--	
Battery manufacturer/type	Operating and fault condition	Measurement			Observation	
		Charging voltage (V)	Charging current (A)	Temp, (°C)		
--	--	--	--	--	--	
Supplementary information:						
Abbreviation: SC= short circuit; OC= open circuit; MSCV= maximum specified charging voltage; MSCC= maximum specified charging current; HSCT= highest specified charging temperature; LSCT= lowest						

IEC 62368-1			
Clause	Requirement + Test	Result - Remark	Verdict

specified charging temperature

Q.1	TABLE: Circuits intended for interconnection with building wiring (LPS)						P
Output Circuit	Condition	U _{oc} (V)	Time (s)	I _{sc} (A)		S (VA)	
				Meas,	Limit	Meas,	Limit
USB1/2 output "+" to "-"	Normal	5.05	5	0.97	8	3.79	100
	Single fault –SC UU2 PIN 1-5	4.98	5	2.09	8	8.95	100
	Single fault –SC UD21 PIN 2-3	0	5	0	8	0	100
LNB output "+" to "-"	Normal	18.39	5	0.68	8	11.85	100
	Single fault –SC UT101 1-2	11.86	5	5.33	8	30.52	100
Supplementary Information:							
SC=short circuit							
USB 1 and USB 2 are in parallel.							

T.2, T.3, T.4, T.5	TABLE: Steady force test						P
Part/Location	Material	Thickness (mm)	Probe	Force (N)	Test Duration (s)	Observation	
Internal wires and components	--	--	--	10	5	Cl/cr can still complying the requirement after force applied	
External rear enclosure	Plastic	See appended table 4.1.2	--	250	5	After test, no hazard.	
Supplementary information: --							

T.6, T.9	TABLE: Impact test				P
Location/part	Material	Thickness (mm)	Height (mm)	Observation	
Rear enclosure	Plastic	See appended table 4.1.2	1300	After test, no hazard.	
Supplementary information: --					

T.7	TABLE: Drop test	N/A
-----	------------------	-----

IEC 62368-1			
Clause	Requirement + Test	Result - Remark	Verdict

Location/part	Material	Thickness (mm)	Height (mm)	Observation
--	--	--	--	--
Supplementary information:--				

T.8	TABLE: Stress relief test					P
Location/Part	Material	Thickness (mm)	Oven Temperature (°C)	Duration (h)	Observation	
Enclosure	Plastic	See appended table 4.1.2	70	7	After test, no hazard.	
Supplementary information: --						

X	TABLE: Alternative method for determining minimum clearances distances				N/A
Clearance distanced between:	Peak of working voltage (V)	Required cl (mm)	Measured cl (mm)		
--	--	--	--		
Supplementary information: --					

IEC 62368-1			
Clause	Requirement + Test	Result - Remark	Verdict

4.1.2	TABLE: list of critical components and materials					P
Object / part No,	Manufacturer/ trademark	Type / model	Technical data	Standard	Mark(s) of conformity ¹⁾	
Power plug (EU type)	Shenzhen G-CINDA power Solution Co Ltd.	DRF-03	AC 250V, 2.5A	EN 50075	VDE DEKRA	
(alternative)	Shenzhen G-CINDA Power Solution Co Ltd	DRF-049	AC 250V, 2.5A	EN 50075	DEKRA	
(alternative)	Shenzhen Jili Dianye Co., Ltd.	JL-601	AC 250V, 2.5A	EN 50075	VDE	
(alternative)	Huizhou Yongda Telecommunication Industry Co., Ltd	YD-01	AC 250V, 2.5A	EN 50075	VDE	
(alternative)	Yutong Electronics (Huizhou) Co., Ltd	YT-05	AC 250V, 2.5A	EN 50075	VDE	
(alternative)	Dongguan Meimeng Electronic Co., Ltd	MM-031A	AC 250V, 2.5A	EN 50075	TUV	
(alternative)	Mainland Electric Wire & Cable Co., Ltd.	ML-207	AC 250V, 2.5A	EN 50075	VDE	
(alternative)	Shenzhen Shi Jiuzhou Rongsheng Technology Co., Ltd	JZ2P-A	AC 250V, 2.5A	EN 50075	VDE	
(alternative)	Hang Lee Industrial Co.	HL-77	AC 250V, 2.5A	EN 50075	VDE	
(alternative)	I-Sheng Electric Wire & Cable Co.,Ltd.	SP-021H, SP-021A	AC 250V, 2.5A	EN 50075	SGS	
(alternative)	Yip Chun Products Factory	YCV-010	AC 250V, 2.5A	EN 50075	VDE	
(alternative)	Interchangeable	Interchangeable	AC 250V, 2.5A	EN 50075	S & other EU Cert. marks	

IEC 62368-1			
Clause	Requirement + Test	Result - Remark	Verdict

4.1.2	TABLE: list of critical components and materials					P
Object / part No,	Manufacturer/ trademark	Type / model	Technical data	Standard	Mark(s) of conformity ¹⁾	
Power plug (BS type)	Shenzhen G-CINDA Power Solution Co., Ltd	GXD-018	AC 250 V 3/5/6/10/13A, (with fuse min 3A)	BS 1363-1	ASTA	
(alternative)	Dongguan Meimeng Electronic Co., Ltd.	MM-041A, MM-042A	AC 250 V 3/5/6/10/13A, (with fuse min 3A)	BS 1363-1	TUV Rheinland	
(alternative)	SHENZHEN JILI DIANYE CO., LTD	JL-313	AC 250 V 3/5/6/10A, (with fuse min 3A)	BS1363-1	ASTA	
(alternative)	Luen Tai (IP'S) Electrical Mfy	9518	AC 250 V 3/5/6/10/13A, (with fuse min 3A)	BS 1363-1	BSI	
(alternative)	I-Sheng Electric Wire & Cable Co., Ltd.	SP-62, SP-62P	AC 250 V 3/5/6/10/13A, (with fuse min 3A)	BS1363-1	ASTA	
(alternative)	Guangzhou Towell Electrical Appliance Co., Ltd	8802, 8801	AC 250 V 3/5/6/10/13A, (with fuse min 3A)	BS1363-1	ASTA	
(alternative)	Zhongshan City Xiangmeng Electric Co., Ltd	XM-013, XM-013A	AC 250 V 3/5/6/10/13A, (with fuse min 3A)	BS 1363-1	DEKRA	
(alternative)	KDJ Quality Electrical Company Ltd.	KDJ 828, KDJ 868	AC 250 V 3/5/6/10/13A, (with fuse min 3A)	BS 1363-1	BSI	
(alternative)	Mainland Electric Wire & Cable Co., Ltd.	ML-313, ML-213	AC 250 V 3/5/6/10/13A, (with fuse min 3A)	BS 1363-1	ASTA	
(alternative)	Yutong Electronics (Huizhou) Co., Ltd	YT-19	AC 250 V 3/5/6/10/13A, (with fuse min 3A)	BS1363-1	DEKRA	

IEC 62368-1			
Clause	Requirement + Test	Result - Remark	Verdict

4.1.2	TABLE: list of critical components and materials					P
Object / part No,	Manufacturer/ trademark	Type / model	Technical data	Standard	Mark(s) of conformity ¹⁾	
(alternative)	Dongguan Meimeng Electronic Co., Ltd.	MM-041A, MM-042A	AC 250 V 3/5/6/10/13A, (with fuse min 3A)	BS 1363-1	TUV Rheinland	
(alternative)	Yip Chun Products Factory	YCB-004	AC 250 V 3/5/6/10/13A, (with fuse min 3A)	BS 1363-1	ASTA	
(alternative)	Hang Lee Industrial Co.	HLB-13, HLB-23, HLB-33	AC 250 V 3/5/6/10/13A, (with fuse min 3A)	BS 1363-1	DEKRA	
(alternative)	Interchangeable	Interchangeable	AC 250 V 3/5/6/10/13A, (with fuse min 3A)	BS 1363-1	ASTA & other EU Cert. marks	
Power plug (Italy type)	Shenzhen G-CINDA Power Solution Co., Ltd	GXD-027	AC 250V, 10 A	CEI 23-50-II Ed.2007+V1:2008+V2:2011	IMQ	
(alternative)	Yip Chun Products Factory	YCI-004	AC 250V, 10 A	CEI 23-50-II Ed.2007+V1:2008+V2:2011	IMQ	
(alternative)	Hang Lee Industrial Co.	HL-19	AC 250V, 10 A	CEI 23-50-II Ed.2007+V1:2008+V2:2011	IMQ	
(alternative)	Yutong Electronics (Huizhou) Co., Ltd	YT-05	AC 250 V, 2.5 A	EN 50075	IMQ	
(alternative)	SHENZHEN JILI DIANYE CO., LTD	JL-316	AC 250 V, 10 A	EN 50075	IMQ	
(alternative)	Interchangeable	Interchangeable	AC 250V, min. 2.5 A	CEI 23-50-II Ed.2007+V1:2008+V2:2011, or EN 50075	IMQ certification marks	
Power cord (EU, BS, Italy type)	Shenzhen Tongyuan Industrial Co., Ltd	H05VVH2-F	2 X 0.75 mm ²	EN 50525-2-11	VDE	

IEC 62368-1			
Clause	Requirement + Test	Result - Remark	Verdict

4.1.2	TABLE: list of critical components and materials				P
Object / part No,	Manufacturer/ trademark	Type / model	Technical data	Standard	Mark(s) of conformity ¹⁾
(alternative)	Shenzhen G-CINDA Power Solution Co., Ltd	H05VVH2-F	2 X 0.75 mm ²	EN 50525-2-11	VDE
(alternative)	Shenzhen Baohing Electric Wire & Cable Manufacture Co., Ltd.	H05VVH2-F	2 X 0.75 mm ²	EN 50525-2-11	VDE
(alternative)	Yutong Electronics (Huizhou) Co., Ltd	H05VVH2-F	2 X 0.75 mm ²	EN 50525-2-11	VDE
(alternative)	SHENZHEN JILI DIANYE CO., LTD	H05VVH2-F	2 X 0.75 mm ²	EN 50525-2-11	VDE
(alternative)	Shenzhen Shi Jiuzhou Rongsheng Technology Co., Ltd.	H05VVH2-F	2 X 0.75 mm ²	EN 50525-2-11	VDE
(alternative)	MAINLAND ELECTRIC WIRE & CABLE CO., LTD.	H05VVH2-F	2 X 0.75 mm ²	EN 50525-2-11	VDE
(alternative)	Sheng Yi Electrical Factory	H05VVH2-F	2 X 0.75 mm ²	EN 50525-2-11	VDE
(alternative)	I-Sheng Electric Wire&Cable Co., Ltd.	H05VVH2-F	2 X 0.75 mm ²	EN 50525-2-11	VDE
(alternative)	Yip Chun Products Factory	H05VVH2-F	2 X 0.75 mm ²	EN 50525-2-11	VDE
(alternative)	Hang Lee Industrial Co.	H05VVH2-F	2 X 0.75 mm ²	EN 50525-2-11	VDE
(alternative)	Interchangeable	H05VVH2-F	2 X 0.75 mm ²	EN 50525-2-11	S & other EU Cert. marks
LED Panel	Innolux Corporation	V500XXX-XXX ("X" MAY BE 0-9, A-Z or blank)	50"	IEC/EN 62368-1	Tested with appliance

IEC 62368-1			
Clause	Requirement + Test	Result - Remark	Verdict

4.1.2	TABLE: list of critical components and materials					P
Object / part No,	Manufacturer/ trademark	Type / model	Technical data	Standard	Mark(s) of conformity ¹⁾	
(alternative)	AUO Corporation	T500XXXXX.X T500XXXX XX T500XXXX.XX ("X" MAY BE 0-9, A-Z or blank)	50"	IEC/EN 62368-1	Tested with appliance	
(alternative)	BOE Optoelectronics Technology	HX500XXX-XXX (X can from 0-9, A-Z or blank)	50"	IEC/EN 62368-1	Tested with appliance	
(alternative)	LG Display	LC500XXX-XXXXX (X can from 0-9, A-Z or blank)	50"	IEC/EN 62368-1	Tested with appliance	
(alternative)	Xianyang Caihong Optoelectronics Technology Co.,Ltd.	CV500XX-XXX (X can from 0-9, A-Z or blank)	50"	IEC/EN 62368-1	Tested with appliance	
AC Wire connector	Henjiu Industrial Co., Ltd	HA396XX-YYY ("X", "Y" can be from 0-9, A-Z or blank, X are different for marketing purpose and only model No. difference)	Min. 250V, 2.5A	UL 1977	UL	
(alternative)	ZHEJIANG HONGXING ELECTRICAL CO LTD	HX3960X-YYY ("X", "Y" can be from 0-9, A-Z or blank, X are different for marketing purpose and only model No. difference)	Min. 250V, 2.5A	UL 1977	UL	

IEC 62368-1			
Clause	Requirement + Test	Result - Remark	Verdict

4.1.2	TABLE: list of critical components and materials					P
Object / part No,	Manufacturer/ trademark	Type / model	Technical data	Standard	Mark(s) of conformity ¹⁾	
(alternative)	ZHEJIANG JIESHITAI ELECTRONICS CO LTD	A3965AWB, A3962 A-(02~14), A3962 AW-(02~14), A3962 AWG-(02~14), A3962 Y- (02~14)	Min. 250V, 2.5A	UL 1977	UL	
(alternative)	ZHEJIANG CHUANGJIA ELECTRONICS CO LTD	VH 3.96, VH 3.96 Belt Positioning 2P	Min. 250V, 2.5A	UL 1977	UL	
(alternative)	ZHEJIANG KUAILI ELECTRONICS CO., LTD	KL VH3.96	Min. 250V, 2.5A	UL 1977	UL	
Plastic material of AC Wire connector	TORAY INDUSTRIES INC	3004-V0(rr)	130 °C, V-0	UL 94	UL	
(alternative)	WENZHOU GUOCAI PLASTIC CEMENT CO LTD	GC 3004	66 °C, V-0	UL 94	UL	
(alternative)	ASAHI KASEI CORP	FR370(j)	66 °C, V-0	UL 94	UL	
(alternative)	GINAR TECHNOLOGY CO LTD	A0520FN(+)(f1)	105 °C, V-0	UL 94	UL	
(alternative)	Chi Mei Corporation	PA-766	60 °C, V-0	UL 94	UL	
(alternative)	ASAHI KASEI CORP	FG170	105 °C, V-0	UL 94	UL	
(alternative)	KUNSHAN LIZHAO PLASTICS CO LTD	FRPG0	75 °C, V-0	UL 94	UL	

IEC 62368-1			
Clause	Requirement + Test	Result - Remark	Verdict

4.1.2	TABLE: list of critical components and materials					P
Object / part No,	Manufacturer/ trademark	Type / model	Technical data	Standard	Mark(s) of conformity ¹⁾	
Main fuse (F101)	XC Electronics (Shen Zhen) Corp. Ltd.	5TR-Serie(s)	Min. AC 250 V, T5 A or Min. AC250 V, T5 AL	IEC/EN 60127-1 IEC/EN 60127-3	VDE	
(alternative)	Sunny East Enterprise Co., LTD	TMP, TSP	Min. AC 250 V, T5 A or Min. AC250 V, T5 AL	IEC/EN 60127-1 IEC/EN 60127-3	VDE	
(alternative)	Dongguan Better Electronic Technology Co., Ltd.	932	Min. AC 250 V, T5 A or Min. AC250 V, T5 AL	IEC/EN 60127-1 IEC/EN 60127-3	VDE	
(alternative)	Dongguan Reomax Electronics Co., Ltd.	MTS	Min. AC 250 V, T5 A or Min. AC250 V, T5 AL	IEC/EN 60127-1 IEC/EN 60127-3	VDE	
(alternative)	XC Electronics (Shen zhen) Corp. Ltd.	5TE	Min. AC 250 V, T5 A or Min. AC250 V, T5 AL	IEC/EN 60127-1 IEC/EN 60127-3	VDE	
(alternative)	Conquer Electronics Co., Ltd.	MET series, MST series	Min. AC 250 V, T5 A or Min. AC250 V, T5 AL	IEC/EN 60127-1 IEC/EN 60127-3	VDE	
(alternative)	Ever Island Electric Co., Ltd.	2000, 2010 Series	Min. AC 250 V, T5 A or Min. AC250 V, T5 AL	IEC/EN 60127-1 IEC/EN 60127-3	VDE	
(alternative)	SUZHOU WALTER ELECTRONIC CO LTD	2010, 2000	Min. AC 250 V, T5 A or Min. AC250 V, T5 AL	IEC/EN 60127-1 IEC/EN 60127-3	VDE	
(alternative)	ZHONG SHAN LANBAO ELECTRICAL APPLIANCES CO LTD	TB, TR	Min. AC 250 V, T5 A or Min. AC250 V, T5 AL	IEC/EN 60127-1 IEC/EN 60127-3	TUV	
(alternative)	Shenzhen Lanson Electronics Co. Ltd.	SMT	Min. AC 250 V, T5 A or Min. AC250 V, T5 AL	IEC/EN 60127-1 IEC/EN 60127-3	VDE	

IEC 62368-1			
Clause	Requirement + Test	Result - Remark	Verdict

4.1.2	TABLE: list of critical components and materials					P
Object / part No,	Manufacturer/ trademark	Type / model	Technical data	Standard	Mark(s) of conformity ¹⁾	
Y Cap, (CY101, CY102, CY104, CY106, CY103)	Shenzhen Teruixiang Electronic Co., Ltd.	TY-Series	Min AC 250V, 100pF for CY101 and CY102, 1000pF for CY106, 220pF for CY103 and CY104, Min 85°C, Y1	IEC 60384-14	VDE	
(alternative)	Xiamen Wanming Electronics Co., Ltd.	HJ	Min AC 250V, 100pF for CY101 and CY102, 1000pF for CY106, 220pF for CY103 and CY104, Min 85°C, Y1	IEC 60384-14	VDE	
(alternative)	Guangdong South Hongming Electronic Science and Technology Co., Ltd.	F	Min AC 250V, 100pF for CY101 and CY102, 1000pF for CY106, 220pF for CY103 and CY104, Min 85°C, Y1	IEC 60384-14	VDE	
(alternative)	Juhong Ele. Company	JB-Series	Min AC 250V, 100pF for CY101 and CY102, 1000pF for CY106, 220pF for CY103 and CY104, Min 85°C, Y1	IEC 60384-14	VDE	
(alternative)	Xiamen Sino Faith Electronic Technology Co., Ltd.	HCY Series	Min AC 250V, 100pF for CY101 and CY102, 1000pF for CY106, 220pF for CY103 and CY104, Min 85°C, Y1	IEC 60384-14	VDE	

IEC 62368-1			
Clause	Requirement + Test	Result - Remark	Verdict

4.1.2	TABLE: list of critical components and materials					P
Object / part No,	Manufacturer/ trademark	Type / model	Technical data	Standard	Mark(s) of conformity ¹⁾	
(alternative)	Anshan Kei Fat Electronic Ceramic Technical Co., Ltd.	CT7	Min AC 250V, 100pF for CY101 and CY102, 1000pF for CY106, 220pF for CY103 and CY104, Min 85°C, Y1	IEC 60384-14	VDE	
(alternative)	Zonkas Electronic Co., Ltd.	ZD	Min AC 250V, 100pF for CY101 and CY102, 1000pF for CY106, 220pF for CY103 and CY104, Min 85°C, Y1	IEC 60384-14	VDE	
(alternative)	Walsin Technology Corp.	AH	Min AC 250V, 100pF for CY101 and CY102, 1000pF for CY106, 220pF for CY103 and CY104, Min 85°C, Y1	IEC 60384-14	VDE	
(alternative)	Shenzhen Weidy Industrial Development Co., Ltd.	WYD	Min AC 250V, 100pF for CY101 and CY102, 1000pF for CY106, 220pF for CY103 and CY104, Min 85°C, Y1	IEC 60384-14	VDE	
(alternative)	Dongguan City Dafu Electronics Co., Ltd	CT7 Y1 Series	Min AC 250V, 100pF for CY101 and CY102, 1000pF for CY106, 220pF for CY103 and CY104, Min 85°C, Y1	IEC 60384-14	VDE	

IEC 62368-1			
Clause	Requirement + Test	Result - Remark	Verdict

4.1.2	TABLE: list of critical components and materials					P
Object / part No,	Manufacturer/ trademark	Type / model	Technical data	Standard	Mark(s) of conformity ¹⁾	
(alternative)	Shantou High-New Technology Dev. Zone Songtian Enterprise Co., Ltd	CD- Series	Min AC 250V, 100pF for CY101 and CY102, 1000pF for CY106, 220pF for CY103 and CY104, Min 85°C, Y1	IEC 60384-14	VDE	
(alternative)	Interchangeable	Interchangeable	Min AC 250V, 100pF for CY101 and CY102, 1000pF for CY106, 220pF for CY103 and CY104, Min 85°C, Y1	IEC 60384-14	S & other EU certification marks	
X capacitor (CX101, CX102)	Shenzhen Su Rong Capacitors Co., Ltd.	MPX, MKP	Min AC 275V, 0.47μF for each, Min 100°C, X2	IEC 60384-14	VDE	
(alternative)	Shenzhen Sincerity Technology Co., Ltd.	MPX, MKP	Min AC 275V, 0.47μF for each, Min 100°C, X2	IEC 60384-14	VDE	
(alternative)	Xiamen Wanming Electronics Co., Ltd.	MPX	Min AC 275V, 0.47μF for each, Min 100°C, X2	IEC 60384-14	VDE	
(alternative)	Shenzhen Shanrui Electronics Co., Ltd.	MPX	Min AC 275V, 0.47μF for each, Min 100°C, X2	IEC 60384-14	VDE	
(alternative)	Shenzhen Shenfeng Electronics Co., Ltd.	CBB62	Min AC 275V, 0.47μF for each, Min 100°C, X2	IEC 60384-14	VDE	
(alternative)	Shenzhen Jinghao Capacitor Co., Ltd.	CBB62B	Min AC 275V, 0.47μF for each, Min 100°C, X2	IEC 60384-14	VDE	

IEC 62368-1			
Clause	Requirement + Test	Result - Remark	Verdict

4.1.2	TABLE: list of critical components and materials					P
Object / part No,	Manufacturer/ trademark	Type / model	Technical data	Standard	Mark(s) of conformity ¹⁾	
(alternative)	Shenzhen Weidy Industrial Development Co., Ltd.	MKP	Min AC 275V, 0.47µF for each, Min 100°C, X2	IEC 60384-14	VDE	
(alternative)	Winday Electronic Industrial Co., Ltd	MPX	Min AC 275V, 0.47µF for each, Min 100°C, X2	IEC 60384-14	VDE	
(alternative)	Dongguan Champion Electric Technology Co., Ltd	MPX Series	Min AC 275V, 0.47µF for each, Min 100°C, X2	IEC 60384-14	VDE	
(alternative)	Fuxin Pan Ocean Electronic Ltd.	MPX-X2	Min AC 275V, 0.47µF for each, Min 100°C, X2	IEC 60384-14	VDE	
(alternative)	Hongzhi Enterprises Ltd.	MPX	Min AC 275V, 0.47µF for each, Min 100°C, X2	IEC 60384-14	VDE	
(alternative)	WINDAY ELECTRONIC (DONG GUAN) CO., LTD	MPX Serie(s)	Min AC 275V, 0.47µF for each, Min 100°C, X2	IEC 60384-14	VDE	
(alternative)	Dongguan City Dafu Electronics Co. Ltd.	MPX Series	Min AC 275V, 0.47µF for each, Min 100°C, X2	IEC 60384-14	VDE	
(alternative)	Interchangeable	Interchangeable	Min AC 275V, 0.47µF for each, Min 100°C, X2	IEC 60384-14	S & other EU cert. marks	
Varistor (RV101)	Brightking (Shenzhen) Co., Ltd.	14D561K (561KD14 marked on the body)	Min.300V, 6kV/3kA, coating: V-0, 40/85/56	IEC 61051-1, IEC 61051-2, IEC 61051-2-2, IEC 60950-1 Annex Q	VDE	
(alternative)	BestBright Electronics Co. Ltd	561KD14, 14D561K	Min.300V, 6kV/3kA, coating: V-0, 40/85/56	IEC 61051-1, IEC 61051-2, IEC 61051-2-2, IEC 60950-1 Annex Q	VDE	

IEC 62368-1			
Clause	Requirement + Test	Result - Remark	Verdict

4.1.2	TABLE: list of critical components and materials					P
Object / part No,	Manufacturer/ trademark	Type / model	Technical data	Standard	Mark(s) of conformity ¹⁾	
(alternative)	Dongguan City Dafu Electronics Co. Ltd.	14D561K	Min.300V, 6kV/3kA, coating: V-0, 40/85/56	IEC 61051-1, IEC 61051-2, IEC 61051-2-2, IEC 60950-1 Annex Q	VDE	
(alternative)	Guangdong South Hongming Electronic Science and Technology Co., Ltd.	ZVR14D561 (14D561K marked on the body)	Min.300V, 6kV/3kA, coating: V-0, 40/85/56	IEC 61051-1, IEC 61051-2, IEC 61051-2-2, IEC 60950-1 Annex Q	VDE	
(alternative)	Shantou High-New Technology Dev. Zone Songtian Enterprise Co., Ltd	14D561K	Min.300V, 6kV/3kA, coating: V-0, 40/85/56	IEC 61051-1, IEC 61051-2, IEC 61051-2-2, IEC 60950-1 Annex Q	VDE	
(alternative)	Shenzhen Weidy Industrial Development Co., Ltd.	WZV14D561KX @ (WZV14D561KT marked on the body) WZV14D561K(X) (WZV14D561KT marked on the body) V-561K-14DEH (WZV14D561KT marked on the body)	Min.300V, 6kV/3kA, coating: V-0, 40/85/56	IEC 61051-1, IEC 61051-2, IEC 61051-2-2, IEC 60950-1 Annex Q	VDE	
Heat shrinkable tube	Shenzhen Woer Heat-shrinkable material Co., Ltd.	RSFR-H	600V, 125°C, VW-1	UL 224	UL	
(alternative)	Changyuan Electronics Group Co., Ltd	CB-HFT	600V, 125°C, VW-1	UL 224	UL	
(alternative)	SHENZHEN WOLIDA TRADING CO LTD	RSFR-H	600V, 125°C, VW-1	UL 224	UL	

IEC 62368-1			
Clause	Requirement + Test	Result - Remark	Verdict

4.1.2	TABLE: list of critical components and materials					P
Object / part No,	Manufacturer/ trademark	Type / model	Technical data	Standard	Mark(s) of conformity ¹⁾	
(alternative)	DONGGUAN QUANTAI INDUSTRIAL CO LTD	T-2, QT-1	600V, 125°C, VW-1	UL 224	UL	
(alternative)	DONGGUAN LING FREE HARDWARE PLASTICS PRODUCT CO LTD	LFSET-003, FST-002	600V, 125°C, VW-1	UL 224	UL	
(alternative)	DONGGUAN LING FREE HARDWARE PLASTICS PRODUCT CO LTD	LFST-001	300V, 125°C, VW-1	UL 224	UL	
(alternative)	Interchangeable	Interchangeable	Min. 300V, VW-1, 105 °C or better	UL 224	UL	
Switching transformer (T101)	Huizhou MVT New Power Technology Co., Ltd.	ZY-T0239	Class B, See appendix 3 for details	IEC/EN 62368-1	Tested with appliance	
Insulation system for T101	HUIZHOU MVT NEW POWER TECHNOLOGY CO LTD	ZY-T	Class 130 (B) electrical insulation system	UL 1446	UL	
- Bobbin material	CHANGCHUN PLASTICS CO., LTD	T375HF, T200NA	Phenolic, Min. thickness 0.70mm, 150 °C, V-0	UL 94	UL	
(alternative)	SUMIDUREZ SINGAPORE PTE LTD	PM9830, PM9850	Phenolic, Min. thickness 0.70mm, 150 °C, V-0	UL 94	UL	
- Insulation tape	JINGJIANG YAHUA PRESSURE SENSITIVE GLUE CO LTD	CT* (b)(g)	130 °C, flame retardant	UL 510	UL	

IEC 62368-1			
Clause	Requirement + Test	Result - Remark	Verdict

4.1.2	TABLE: list of critical components and materials					P
Object / part No,	Manufacturer/ trademark	Type / model	Technical data	Standard	Mark(s) of conformity ¹⁾	
(alternative)	Shenzhen Shengdafeng Electronic Material Co Ltd	SDF-312	130 °C, flame retardant	UL 510	UL	
(alternative)	DONGGUAN SHIN YAHUA ELECTRONIC MATERIAL CO LTD	CT*(b)(g), CT*(c)(g)	130 °C, flame retardant	UL 510	UL	
- Magnet wire	Interchangeable	Interchangeable	Polyurethane Copper Wires used as windings, at least 130°C.	UL 1446	UL	
- Insulation tube	CHANGYUAN ELECTRONICS GROUP CO LTD	CB-TT-L	200 °C, VW-1	UL 224	UL	
(alternative)	HUIZHOU DONGJU FLUO TECH PLASTIC CO LTD	TFL	200 °C, VW-1	UL 224	UL	
(alternative)	GREAT HOLDING INDUSTRIAL CO LTD	TFL	200 °C, VW-1	UL 224	UL	
- Triple insulated wire	HUIZHOU DONGJU FLUO TECH PLASTIC CO LTD	TIW-BXXyy	130°C	IEC/EN 62368-1	VDE	

IEC 62368-1			
Clause	Requirement + Test	Result - Remark	Verdict

4.1.2	TABLE: list of critical components and materials				P
Object / part No,	Manufacturer/ trademark	Type / model	Technical data	Standard	Mark(s) of conformity ¹⁾
PFC filter (L101)	Huizhou MVT New Power Technology CO., LTD	ZY-T0164	130°C Primary winding: N1 Pin 1-3: 2UEW Ø0.10mm*20 x 1P x 50Ts. Secondary: winding: N2 Pin 4-5: 2UEW Ø0.30mm*1 x 3Ts;	IEC/EN 62368-1	Tested with appliance
- Bobbin material	Changchun Plastics Co., Ltd.	T375J, T375HF, T200HF	Phenolic, Min. thickness 0.40mm, 150 °C, V-0	UL 94	UL
(alternative)	SUMITOMA BAKELITE CO.,LTD	PM-9820, PM-9825	Phenolic, Min. thickness 0.40mm, 150 °C, V-0	UL 94	UL
- Insulation tape	Jingjiang Yahua Pressure Sensitive Glue Co., Ltd	CT* (c)(g)	130°C, flame retardant	UL 510	UL
(alternative)	Chang Shu Liang Yi Tape Industry Co., Ltd	LY-XX(a)(b)	130°C, flame retardant	UL 510	UL
(alternative)	Xinyu Shengdafeng Electric Material Co., Ltd.	SDF-312	130°C, flame retardant	UL 510	UL
Magnet Wire	HOI LUEN ELECTRICAL MFR CO LTD	2UEW	130°C	UL 1446	UL
(Alternative)	ZHEJIANG HONGBO TECHNOLOGY CO LTD	xUEW/130	130°C	UL 1446	UL

IEC 62368-1			
Clause	Requirement + Test	Result - Remark	Verdict

4.1.2	TABLE: list of critical components and materials					P
Object / part No,	Manufacturer/ trademark	Type / model	Technical data	Standard	Mark(s) of conformity ¹⁾	
- Insulation tube	Changyuan Electronics Group Co., Ltd.	CB-TT-L	200°C, VW-1	UL 224	UL	
(alternative)	Fluo Tech Industries Co., Ltd	TFL	200°C, VW-1	UL 224	UL	
(alternative)	Great Holding Industrial Co., Ltd	TFL	200°C, VW-1	UL 224	UL	
Line filter (LF103)	Huizhou MVT New Power Technology Co., Ltd	TC-140404-102	100µH, (130°C)	IEC/EN 62368-1	Tested with appliance	
- Bobbin material	CHANGCHUN PLASTICS CO., LTD	T375J, T375HF	Phenolic, Min. thickness 0.40mm, 150 °C, V-0	UL 94	UL	
- Triple insulated wire	Great Leoflon Industrial Co., Ltd	TRW(B)*	130°C	IEC/EN 62368-1	VDE	
(alternative)	Suzhou Yusheng Electronic CO.,LTD	TIW-B+@	130°C	IEC/EN 62368-1	VDE	
- Insulation tube	CHANGYUAN ELECTRONIC	CB-TT-L	200 °C, VW-1	UL 224	UL	
(alternative)	FLUO TECH INDUSTRIES CO., LTD	TFL	200 °C, VW-1	UL 224	UL	
(alternative)	GREAT HOLDING INDUSRLAL CO., LTD	TFL	200 °C, VW-1	UL 224	UL	
- Magnet wire	Interchangeable	Interchangeable	130°C	UL 1446	UL	
Line filter (LF101, LF102)	Huizhou MVT New Power Technology Co., Ltd.	SYUC19-L017mA	17mH - 30%/+50% (130°C)	IEC/EN 62368-1	Tested with appliance	

IEC 62368-1			
Clause	Requirement + Test	Result - Remark	Verdict

4.1.2	TABLE: list of critical components and materials					P
Object / part No,	Manufacturer/ trademark	Type / model	Technical data	Standard	Mark(s) of conformity ¹⁾	
- Bobbin material	CHANGCHUN PLASTICS CO., LTD	T375J, T375HF	Phenolic, Min. thickness 0.40mm, 150 °C, V-0	UL 94	UL	
- Magnet wire	Interchangeable	Interchangeable	130°C	UL 1446	UL	
Optocoupler (U204)	Everlight Electronics Co., Ltd.	EL817V, EL817	Dti ≥0.4mm, Ex. Cl. & Cr. ≥7.6mm, 110°C	IEC/EN 60747-5-5	VDE	
(alternative)	Bright Led Electronics Corp	BPC-817(A, B, C, D, L), BPC-817M, BPC-817 S	Dti ≥0.4mm, Ex. Cl. & Cr. ≥7.6mm, 110°C	IEC/EN 60747-5-5	VDE	
(alternative)	China Resources Semiconductor (ShenZhen) Ltd.	PC817 (A-Z) (blank;M;S), 817 (A-Z) (blank;M;S)	Dti ≥0.4mm, Ex. Cl. & Cr. ≥7.62mm, 110°C	IEC/EN 60747-5-5	VDE	
(alternative)	Renesas Electronics Corp.	PS2561-1, PS2561AL1-1	Dti ≥0.4mm, Ex. Cl. & Cr. ≥7.0mm, 100°C	IEC/EN 60747-5-5	VDE	
(alternative)	Lite-On Technology Corporation	LTV-817	Dti ≥0.4mm, Ex. Cl. & Cr. ≥7.0mm, 115°C	IEC/EN 60747-5-5	VDE	
(alternative)	Changzhou Galaxy Century Micro-electronics Co., Ltd	BL817 (A, B, C, D, L), BL817M (A, B, C, D, L) BL817S, BL817S (A, B, C, D, L)	Dti ≥0.4mm, Ex. Cl. & Cr. ≥7.5mm, 100°C	IEC/EN 60747-5-5	VDE	
(alternative)	Shenzhen Orient Components Co. Ltd.	ORPC-817 x, ORPC-817M x (x can from 0 to 9, A to Z or blank)	Dti ≥0.4mm, Ex. Cl. & Cr. ≥7.6mm, 110°C	IEC/EN 60747-5-5	VDE	
(alternative)	COSMO Electronics Corporation	K1010	Dti ≥0.4mm, Ex. Cl. & Cr. ≥6.5mm, 115°C	IEC/EN 60747-5-5	VDE	
(alternative)	CRM ICBG (Wuxi) Co., Ltd.	PC817x x-Where suffix x indicates one capital letter A to Z.	Dti ≥0.4mm, Ex. Cl. & Cr. ≥7.62mm, 110°C	IEC/EN 60747-5-5	VDE	

IEC 62368-1			
Clause	Requirement + Test	Result - Remark	Verdict

4.1.2	TABLE: list of critical components and materials					P
Object / part No,	Manufacturer/ trademark	Type / model	Technical data	Standard	Mark(s) of conformity ¹⁾	
PCB (Power bord)	Shandong Jinbao Electronics Co Ltd	ZD-95(G)F	V-0, 130°C	UL 94, UL 796	UL	
(alternative)	SHANTOU YUEDONG LIANFENG ELECTRONIC INDUSTRY CO LTD	LF-D	V-0, 130°C	UL 94, UL 796	UL	
(alternative)	Shenzhen Zhong Luo Electronics Co Ltd	ZL-03, ZL-03A, ZL-01	V-0, 130°C	UL 94, UL 796	UL	
(alternative)	Guangdong Hongtai Electronic Inc Co., Ltd	HH-1, HH-2, HH-3	V-0, 130°C	UL 94, UL 796	UL	
(alternative)	CANYON CIRCUIT TECHNOLOGY (HUIZHOU) CO LTD	SH-04, SH-05	V-0, 130°C	UL 94, UL 796	UL	
(alternative)	CHENG HO POWER CORP	24B, 24B-6, 4B-1	V-0, 130°C	UL 94, UL 796	UL	
(alternative)	ZHUHAI JINGLIHUA PCB CO LTD	94-V0D, JLH94-V0, JLH94-V0C	V-0, 130°C	UL 94, UL 796	UL	
(alternative)	EASTOP INTERNATIONAL LTD	ET-005	V-0, 130°C	UL 94, UL 796	UL	
(alternative)	DONGGUAN Shi PINSHENG ELECTRONIC CO.,LTD	PS-2	V-0, 130°C	UL 94, UL 796	UL	
(alternative)	FUJIAN MILKY-WAY PRINTED CIRCUIT BOARD INDUSTRIAL CO LTD	MW-2, MW-CEM1	V-0, 130°C	UL 94, UL 796	UL	

IEC 62368-1			
Clause	Requirement + Test	Result - Remark	Verdict

4.1.2	TABLE: list of critical components and materials					P
Object / part No,	Manufacturer/ trademark	Type / model	Technical data	Standard	Mark(s) of conformity ¹⁾	
(alternative)	WODE ELECTRONICS TECHNOLOGY(ZHUHAI) CO LTD	WD-12	V-0, 130°C	UL 94, UL 796	UL	
(alternative)	DONGGUAN WANNIENFU ELECTRONIC CO LTD	101V0, 05V0	V-0, 130°C	UL 94, UL 796	UL	
(alternative)	Nippon(Bo Luo) Electronics Co Ltd	S4	V-0, 130°C	UL 94, UL 796	UL	
(alternative)	XINFENG WENFENG ELECTRONIC TECHNOLOGY CO LTD	XFWF-01	V-0, 130°C	UL 94, UL 796	UL	
(alternative)	ZHUHAI KINGSUN ELECTRONICS AND TECHNOLOGY CO LTD	KS-D, KS-D1	V-0, 130°C	UL 94, UL 796	UL	
(alternative)	SHENZHEN BRIGHTPCB CO LTD	BR-DL-1, BR-DL-2,	V-0, 130°C	UL 94, UL 796	UL	
(alternative)	EASTECH GROUP CO LTD	XLX-D	V-0, 130°C	UL 94, UL 796	UL	
(alternative)	JIANGXI XUSHENG ELECTRONICS CO	XS-D	V-0, 130°C	UL 94, UL 796	UL	
(alternative)	MEIZHOU DINGTAI P C B CO LTD	DT-4, DT-2, AL01	V-0, 130°C	UL 94, UL 796	UL	
(alternative)	ZHEJIANG JINSHUN INDUSTRY CORP	LT-FR1 \$\$	V-0, 130°C	UL 94, UL 796	UL	

IEC 62368-1			
Clause	Requirement + Test	Result - Remark	Verdict

4.1.2	TABLE: list of critical components and materials					P
Object / part No,	Manufacturer/ trademark	Type / model	Technical data	Standard	Mark(s) of conformity ¹⁾	
(alternative)	HENAN HAI LE ELECTRONICS CO., LTD.	HL-0, HL-01, HL-02	V-0, 130°C	UL 94, UL 796	UL	
(alternative)	Interchangeable	Interchangeable	V-0, 130°C or better	UL 94, UL 796	UL	
PCB (Main board)	SHENZHEN RUOMEI ELECTRONICS CO LTD	RM-01	V-0, 130°C	UL 94, UL 796	UL	
(alternative)	Cheung Hung Technology International Ltd	CH-M, CH-D	V-0, 130°C	UL 94, UL 796	UL	
(alternative)	GUANGDONG CHAMPION ASIA ELECTRONICS CO LTD	F-M, F-D	V-0, 130°C	UL 94, UL 796	UL	
(alternative)	ZHUHAI JINGLIHUA PCB CO LTD	94-V0D, JLH94-V0, JLH94-V0C	V-0, 130°C	UL 94, UL 796	UL	
(alternative)	FUJIAN MILKY-WAY TECHNOLOGY CO LTD	MW-2, MW-R4, MW-CEM1	V-0, 130°C	UL 94, UL 796	UL	
(alternative)	AOSHIKANG PRECISION CIRCUIT (HUIZHOU) CO LTD	A-2; K-2; S-2; K-3	V-0, 130°C	UL 94, UL 796	UL	
(alternative)	BAOYUEJIA Electronics Co Ltd	BYJ-3	V-0, 130°C	UL 94, UL 796	UL	
(alternative)	ZHUHAI KINGSUN ELECTRONICS AND TECHNOLOGY CO LTD	KS-M1, KS-AL, KS-AL1, KS-D, KS-D1	V-0, 130°C	UL 94, UL 796	UL	

IEC 62368-1			
Clause	Requirement + Test	Result - Remark	Verdict

4.1.2	TABLE: list of critical components and materials					P
Object / part No,	Manufacturer/ trademark	Type / model	Technical data	Standard	Mark(s) of conformity ¹⁾	
(alternative)	JIANGXI XUSHENG ELECTRONICS CO	XS-D, XS-M	V-0, 130°C	UL 94, UL 796	UL	
(alternative)	LONG YAN JINSHIYU ELECTRONIC LTD	JSY-1, JSY-3,	V-0, 130°C	UL 94, UL 796	UL	
(alternative)	MEIZHOU DINGTAI P C B CO LTD	DT-4, DT-2, AL01	V-0, 130°C	UL 94, UL 796	UL	
(alternative)	ZHEJIANG JINSHUN INDUSTRY CORP	LT-FR1 \$\$	V-0, 130°C	UL 94, UL 796	UL	
(alternative)	HENAN HAI LE ELECTRONICS CO., LTD.	HL-0, HL-01, HL-02	V-0, 130°C	UL 94, UL 796	UL	
(alternative)	Hunan Lier Printed Circuit Board Co Ltd	F-M, L-M, F-D, L-D	V-0, 130°C	UL 94, UL 796	UL	
(alternative)	XING FENG FU CHANG FA ELECTRONIC CO LTD	FCF-3, FCF-2, FCF-4	V-0, 130°C	UL 94, UL 796	UL	
(alternative)	Ganzhou ZhongShenglong Electronic Co Ltd	ZSL-1, ZSL-1A, ZSL-2, ZSL-2A	V-0, 130°C	UL 94, UL 796	UL	
(alternative)	Jiangxi Xinyi Technology Co., Ltd.	XY-D, XY-M	V-0, 130°C	UL 94, UL 796	UL	
(alternative)	Interchangeable	Interchangeable	V-0, 130°C or better	UL 94, UL 796	UL	
(alternative)	Longnan Champion Asia Electronic Technology Co Ltd.	F-D	V-0, 130°C	UL 94, UL 796	UL	

IEC 62368-1			
Clause	Requirement + Test	Result - Remark	Verdict

4.1.2	TABLE: list of critical components and materials					P
Object / part No,	Manufacturer/ trademark	Type / model	Technical data	Standard	Mark(s) of conformity ¹⁾	
(alternative)	JIANGMEN BENLIDA PRINTED CIRCUIT CO.,LTD	BLD-B, BLD-D	V-0, 130°C	UL 94, UL 796	UL	
(alternative)	TOPSEARCH PRINTED CIRCUITS (HK) LTD	TS-D-8V03C SG TS-D-7V04 SG	V-0, 130°C	UL 94, UL 796	UL	
(alternative)	SHENZHEN WUZHU TECH CO LTD	WZ-6(@)	V-0, 130°C	UL 94, UL 796	UL	
(alternative)	SHEN ZHEN MANKUN TECHNOLOGY CO., LTD	MK-D, MK-D600, MK-DC	V-0, 130°C	UL 94, UL 796	UL	
(alternative)	VICTORY GIANT TECHNOLOGY (HUIZHOU) CO LTD	SH1, SH9	V-0, 130°C	UL 94, UL 796	UL	
(alternative)	RONG HUI ELECTRONICS (HUIZHOU) CO LTD	RH-3	V-0, 130°C	UL 94, UL 796	UL	
(alternative)	XIN FENG FU CHANG FA ELECTRONIC CO LTD	FCF-3	V-0, 130°C	UL 94, UL 796	UL	
(alternative)	MEIZHOU DINGTAI P C B CO LTD	DT-2	V-0, 130°C	UL 94, UL 796	UL	
(alternative)	ZHUHAI KINGSUN ELECTRONICS AND TECHNOLOGY CO LTD	KS-D, KS-D1, KS-D2	V-0, 130°C	UL 94, UL 796	UL	

IEC 62368-1			
Clause	Requirement + Test	Result - Remark	Verdict

4.1.2	TABLE: list of critical components and materials					P
Object / part No,	Manufacturer/ trademark	Type / model	Technical data	Standard	Mark(s) of conformity ¹⁾	
(alternative)	Huizhou China Eagle Electronic Technology Co Ltd	CA-F120	V-0, 130°C	UL 94, UL 796	UL	
(alternative)	ELEC & ELTEK MULTILAYER PCB LTD	E3330E	V-0, 130°C	UL 94, UL 796	UL	
(alternative)	SHENZHEN KINWONG ELECTRONIC CO LTD	5、5C(%)、10、10B	V-0, 130°C	UL 94, UL 796	UL	
(alternative)	SHEN ZHEN SUN & LYNN CIRCUITS CO LTD	SL-2, SL-D, SL-HD	V-0, 130°C	UL 94, UL 796	UL	
(alternative)	HUIZHOU GLORYSKY ELECTRONICS CO LTD	GS-D, GS-D1, GS-D3	V-0, 130°C	UL 94, UL 796	UL	
(alternative)	YIYANG MINGZHENGHONG ELECTRONICS CO LTD	BR-CL-1, BR-DL-1, BR-DL-2	V-0, 130°C	UL 94, UL 796	UL	
(alternative)	SHENZHEN BOMIN ELECTRONIC CO LTD	BM2, BM-2, BM6, BM6-1	V-0, 130°C	UL 94, UL 796	UL	
(alternative)	AOSHIKANG TECHNOLOGY CO LTD	K-2	V-0, 130°C	UL 94, UL 796	UL	
(alternative)	HUIZHOU WELGAO ELECTRONICS CO LTD	W-1, W-5	V-0, 130°C	UL 94, UL 796	UL	
(alternative)	GUANG DONG XING DA HONG YE ELECTRONIC CO LTD	XD-102	V-0, 130°C	UL 94, UL 796	UL	

IEC 62368-1			
Clause	Requirement + Test	Result - Remark	Verdict

4.1.2	TABLE: list of critical components and materials					P
Object / part No,	Manufacturer/ trademark	Type / model	Technical data	Standard	Mark(s) of conformity ¹⁾	
(alternative)	Shenzhen Xunjiexing Technology Co Ltd	JX01	V-0, 130°C	UL 94, UL 796	UL	
(alternative)	JIANGXI ZHIBOXIN TECHNOLOGY CO LTD	DS01, DS02	V-0, 130°C	UL 94, UL 796	UL	
(alternative)	SHENZHEN STARIVER CIRCUITS CO LTD	SR-01A	V-0, 130°C	UL 94, UL 796	UL	
(alternative)	GuangDong Kingshine Electronic Technology Co Ltd	D0, D1	V-0, 130°C	UL 94, UL 796	UL	
(alternative)	GanZhou YiHao New Materials Co Ltd	YH-D	V-0, 130°C	UL 94, UL 796	UL	
(alternative)	CHANGZHOU AOHONG ELECTRONICS CO LTD	AOH-2, AOH-4	V-0, 130°C	UL 94, UL 796	UL	
(alternative)	Interchangeable	Interchangeable	V-0, 130°C or better	UL 94, UL 796	UL	
Plastic material of rear enclosure (for top or bottom enclosure of rear enclosure, refer to configuration s in general product information field)	Dongguan BaoHua Plastic Raw Material Co., Ltd	BH-8114	PS, V-0, Min. thickness 2.0mm	UL 94	UL	

IEC 62368-1			
Clause	Requirement + Test	Result - Remark	Verdict

4.1.2	TABLE: list of critical components and materials					P
Object / part No,	Manufacturer/ trademark	Type / model	Technical data	Standard	Mark(s) of conformity ¹⁾	
(alternative)	Dong Guan Zhi Shang Plastics Technology Co Ltd	ZS960	PC/ABS, V-0, Min. thickness 2.0mm	UL 94	UL	
(alternative)	QINGDAO GUOEN TECHNOLOGY CO LTD	PC/ABS40(xx)G-(ys)(xx) -Denotes a two digit number, (ys) - Denotes two digit letter by A-Z and two or three digit number	PC/ABS, V-0, Min. thickness 2.0mm	UL 94	UL	
(alternative)	QINGDAO HAIER NEW MATERIAL R & D CO LTD	NH-9012	PC/ABS, V-0, Min. thickness 2.0mm	UL 94	UL	
(alternative)	SHENZHEN FUHENG NEW MATERIAL CO LTD	HIPS-R503	PS, V-0, Min. thickness 2.0mm	UL 94	UL	
(alternative)	GUANGDONG GUOLI SCIENCE AND TECHNOLOGY CO LTD	FRHIPS-620BK	PS, V-0, Min. thickness 2.0mm	UL 94	UL	
(alternative)	QINGDAO HAIER NEW MATERIAL R & D CO LTD	FR-3007	PS, V-0, Min. thickness 2.0mm	UL 94	UL	
(alternative)	QINGDAO HAIER NEW MATERIAL R & D CO LTD	FR-6003GX	PA6, V-0, Min. thickness 2.0mm	UL 94	UL	
(alternative)	QINGDAO GUOEN TECHNOLOGY CO LTD	PS3001Z-600A(Y) Y - will be replaced by a-z, to denote different color	PS, V-0, Min. thickness 2.0mm	UL 94	UL	

IEC 62368-1			
Clause	Requirement + Test	Result - Remark	Verdict

4.1.2	TABLE: list of critical components and materials				P
Object / part No,	Manufacturer/ trademark	Type / model	Technical data	Standard	Mark(s) of conformity ¹⁾
(alternative)	SHENZHEN FUHENG NEW MATERIAL CO LTD	PC-RG(XX)1 (XX) -Replaced by two digits from 00 to 50 to represent glass fiber content.	PC, V-0, Min. thickness 2.0mm	UL 94	UL
(alternative)	SHENZHEN FUHENG NEW MATERIAL CO LTD	PC/ABS-205FR, PCA-(XX)2RG (XX) -Replaced by two digits from 00 to 50 to represent glass fiber content.	PC/ABS, V-0, Min. thickness 2.0mm	UL 94	UL
(alternative)	LG Chemical Ltd. LG	LUPOY GN-5001RF(T), LUPOY GN5001RF(T)	PC, V-0, Min. thickness 2.0mm	UL 94	UL
(alternative)	QINGDAO GUOEN TECHNOLOGY CO LTD	PC40(xx)Z(xx) - Denotes a two digit number, 01-99 inclusive, representing a customer code.	PC, V-0, Min. thickness 2.0mm	UL 94	UL
(alternative)	QINGDAO GUOEN TECHNOLOGY CO LTD	PC/ABS40(xx)G-(ys)(xx) -Denotes a two digit number, (ys) - Denotes two digit letter by A-Z and two or three digit number	PC/ABS, V-0, 50 °C, min. thickness 1.6mm	UL 94	UL
(alternative)	QINGDAO HAIER NEW MATERIAL R & D CO LTD	NH-9012	PC/ABS, V-0, 60 °C, min. thickness 1.5mm	UL 94	UL
(alternative)	CHI MEI CORPORATION	PA-765A(+)	PC/ABS, V-0, 85 °C, min. thickness 1.5mm	UL 94	UL

IEC 62368-1			
Clause	Requirement + Test	Result - Remark	Verdict

4.1.2	TABLE: list of critical components and materials					P
Object / part No,	Manufacturer/ trademark	Type / model	Technical data	Standard	Mark(s) of conformity ¹⁾	
(alternative)	QING DAO GON TECHNOLOGY CO.,LTD.	PC/ABS4034ZH	PC/ABS, V-0, 60 °C, min. thickness 1.5mm	UL 94	UL	
(alternative)	DongGuan BaoHua Plastic Raw Material Co Ltd	BH-9379	PC/ABS, V-0, 60 °C, min. thickness 1.5mm	UL 94	UL	
(alternative)	KINGFA SCI & TECH CO LTD	JH960-6300 (##) (##) - Replaced by ten digits maximum by a combination of letters and/or numbers as color code.	PC/ABS, V-0, 60 °C, min. thickness 1.5mm	UL 94	UL	
(alternative)	China Science Baoyi Vision Technology (Jiangsu) Co Ltd	6240T	PC/ABS, V-0, 60°C, Min. thickness 2.0mm	UL 94	UL	
(alternative)	JINMA VIET NAM PLASTIC CO., LTD	PC/ABS4034ZH BK, PC/ABS4034ZH WT	PC/ABS, V-0, 60°C, Min. thickness 2.0mm	UL 94	UL	
Plastic material of front enclosure (or bottom enclosure of rear enclosure, refer to configuration s in general product information field)	QINGDAO GUOEN TECHNOLOGY CO LTD	HIPS 33(xx) (xx) can from 01 to 99)	HB, 50 °C Min. thickness 2.0mm	UL 94	UL	

IEC 62368-1			
Clause	Requirement + Test	Result - Remark	Verdict

4.1.2	TABLE: list of critical components and materials					P
Object / part No,	Manufacturer/ trademark	Type / model	Technical data	Standard	Mark(s) of conformity ¹⁾	
(alternative)	FORMOSA CHEMICALS & FIBRE CORP PLASTICS DIV	AG15AB	HB, 60 °C, Min. thickness 2.0mm	UL 94	UL	
(alternative)	SHENZHEN FUHENG NEW MATERIAL CO LTD	HIPS-568	HB, 50 °C, Min. thickness 2.0mm	UL 94	UL	
(alternative)	SHENZHEN FUHENG NEW MATERIAL CO LTD	ABS-600	HB, 60°C Min. thickness 2.0mm	UL 94	UL	
(alternative)	QINGDAO HAIER NEW MATERIAL R & D CO LTD	HRABS-HG, RF-1020G10	HB, 60 °C Min. thickness 2.0mm	UL 94	UL	
(alternative)	QINGDAO HAIER NEW MATERIAL R & D CO LTD	CR-3003	HB, 50 °C Min. thickness 2.0mm	UL 94	UL	
(alternative)	LG Chemical Ltd.	HI121H	HB, 95 °C Min. thickness 2.0mm	UL 94	UL	
(alternative)	Chi Mei Corporation	PA-757(+)	HB, 85 °C Min. thickness 2.0mm	UL 94	UL	
(alternative)	Chi Mei Corporation	PA-707(+)	HB, 60 °C Min. thickness 2.0mm	UL 94	UL	
(alternative)	KINGFA SCI & TECH CO LTD	HIPS-5197	HB, 50 °C Min. thickness 2.0mm	UL 94	UL	
(alternative)	Dongguan BaoHua Plastic Raw Material Co., Ltd	BH-8114	PS, V-0, Min. thickness 2.0mm	UL 94	UL	

IEC 62368-1			
Clause	Requirement + Test	Result - Remark	Verdict

4.1.2	TABLE: list of critical components and materials					P
Object / part No,	Manufacturer/ trademark	Type / model	Technical data	Standard	Mark(s) of conformity ¹⁾	
(alternative)	Dong Guan Zhi Shang Plastics Technology Co Ltd	ZS960	PC/ABS, V-0, Min. thickness 2.0mm	UL 94	UL	
(alternative)	QINGDAO GUOEN TECHNOLOGY CO LTD	PC/ABS40(xx)G-(ys)(xx) -Denotes a two digit number, (ys) - Denotes two digit letter by A-Z and two or three digit number	PC/ABS, V-0, Min. thickness 2.0mm	UL 94	UL	
(alternative)	QINGDAO HAIER NEW MATERIAL R & D CO LTD	NH-9012	PC/ABS, V-0, Min. thickness 2.0mm	UL 94	UL	
(alternative)	SHENZHEN FUHENG NEW MATERIAL CO LTD	HIPS-R503	PS, V-0, Min. thickness 2.0mm	UL 94	UL	
(alternative)	GUANGDONG GUOLI SCIENCE AND TECHNOLOGY CO LTD	FRHIPS-620BK	PS, V-0, Min. thickness 2.0mm	UL 94	UL	
(alternative)	QINGDAO HAIER NEW MATERIAL R & D CO LTD	FR-3007	PS, V-0, Min. thickness 2.0mm	UL 94	UL	
(alternative)	QINGDAO HAIER NEW MATERIAL R & D CO LTD	FR-6003GX	PA6, V-0, Min. thickness 2.0mm	UL 94	UL	
(alternative)	QINGDAO GUOEN TECHNOLOGY CO LTD	PS3001Z600A(Y) Y - will be replaced by a-z, to denote different color	PS, V-0, Min. thickness 2.0mm	UL 94	UL	

IEC 62368-1			
Clause	Requirement + Test	Result - Remark	Verdict

4.1.2	TABLE: list of critical components and materials				P
Object / part No,	Manufacturer/ trademark	Type / model	Technical data	Standard	Mark(s) of conformity ¹⁾
(alternative)	SHENZHEN FUHENG NEW MATERIAL CO LTD	PC-RG(XX)1(XX) -Replaced by two digits from 00 to 50 to represent glass fiber content.	PC, V-0, Min. thickness 2.0mm	UL 94	UL
(alternative)	SHENZHEN FUHENG NEW MATERIAL CO LTD	PC/ABS-205FR, PCA-(XX)2RG (XX) -Replaced by two digits from 00 to 50 to represent glass fiber content.	PCABS, V-0, Min. thickness 2.0mm	UL 94	UL
(alternative)	LG Chemical Ltd. LG	LUPOY GN-5001RF(T); LUPOY GN5001RF(T)	PC, V-0, Min. thickness 2.0mm	UL 94	UL
(alternative)	QINGDAO GUOEN TECHNOLOGY CO LTD	PC40(xx)Z (xx) - Denotes a two digit number, 01-99 inclusive, representing a customer code.	PC, V-0, Min. thickness 2.0mm	UL 94	UL
(alternative)	QINGDAO GUOEN TECHNOLOGY CO LTD	PC/ABS40(xx)G-(ys) (xx) - Denotes a two digit number, (ys) - Denotes two digit letter by A-Z and two or three digit number	PC/ABS, V-0, 50 °C, min. thickness 1.6mm	UL 94	UL
(alternative)	QINGDAO HAIER NEW MATERIAL R & D CO LTD	NH-9012	PC/ABS, V-0, 60 °C, min. thickness 1.5mm	UL 94	UL
(alternative)	DongGuan BaoHua Plastic Raw Material Co Ltd	BH-15B	HB, 60 °C Min. thickness 1.6mm	UL 94	UL

IEC 62368-1			
Clause	Requirement + Test	Result - Remark	Verdict

4.1.2	TABLE: list of critical components and materials					P
Object / part No,	Manufacturer/ trademark	Type / model	Technical data	Standard	Mark(s) of conformity ¹⁾	
Speaker plastic enclosure	FORMOSA CHEMICALS & FIBRE CORP PLASTICS DIV	AC310(+)	PC/ABS, Rated V-0, 60°C, Min. thickness 1.0mm	UL 94	UL	
Internal speaker	Interchangeable	Interchangeable	Two provided, 6Ω 12W for each one.	IEC/EN 62368-1	Tested with appliance	
Base	Interchangeable	Interchangeable	Metal material	IEC/EN 62368-1	Tested with appliance	
Insulation sheet (only use for configuration 1)	Mianyang Longhua Film Co., Ltd.	PP-(i)(j)	V-0, 100°C, min thickness 0.4mm	UL 94	UL	
(alternative)	SICHUAN LONGHUA FILM CO LTD	PC-870	V-0, 100°C, min thickness 0.4mm	UL 94	UL	
(alternative)	Chengdu Kanglongxin Plastics Co., Ltd	KLX PP BK-10; KLX PP WT-10	V-0, 80°C, min. 0,4mm	UL 94	UL	
(alternative)	Sichuan Dongfang Insulating Material Co., Ltd	DFR3738A(d), DFR3732A (d)	V-0, 110°C, min thickness 0.4mm	UL 94	UL	
(alternative)	Hunan Dobesty Optical Material Co Ltd	DB9842B, DB328AP	V-0, 110°C, min thickness 0.4mm	UL 94	UL	
(alternative)	Ningbo Exciport New Material Co., Ltd	GZEFR99A	V-0, 80°C, min thickness 0.4mm	UL 94	UL	
Supplementary information:						
1) Provided evidence ensures the agreed level of compliance. See OD-CB2039.						
2) License available upon request for all the certified components.						

IEC62368_1E - ATTACHMENT			
Clause	Requirement + Test	Result - Remark	Verdict
ATTACHMENT TO TEST REPORT IEC 62368-1 EUROPEAN GROUP DIFFERENCES AND NATIONAL DIFFERENCES (Audio/video, information and communication technology equipment - Part 1: Safety requirements)			
Differences according to: EN IEC 62368-1:2020+A11:2020			
Attachment Form No ,: EU_GD_IEC62368_1E			
Attachment Originator: UL(Demko)			
Master Attachment: Dated 2021-02-04			
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	CENELEC COMMON MODIFICATIONS (EN)		--
	Clause numbers in the cells that are shaded light grey are clause references in EN IEC 62368-1:2020+A11:2020, All other clause numbers in that column, except for those in the paragraph below, refers to IEC 62368-1:2018, Clauses, subclauses, notes, tables, figures and annexes which are additional to those in IEC 62368-1:2018 are prefixed "Z",		--
	Add the following annexes: Annex ZA (normative) Normative references to international publications with their corresponding European publications Annex ZB (normative) Special national conditions Annex ZC (informative) A-deviations Annex ZD (informative) IEC and CENELEC code designations for flexible cords		--
1	Modification to Clause 3,		N/A
3.3.19	Sound exposure <i>Replace 3.3.19 of IEC 62368-1 with the following definitions:</i>		N/A
3.3.19.1	momentary exposure level, MEL metric for estimating 1 s sound exposure level from the HD 483-1 S2 test signal applied to both channels, based on EN 50332-1:2013, 4.2, Note 1 to entry: MEL is measured as A-weighted levels in dB, Note 2 to entry: See B,3 of EN 50332-3:2017 for additional information,		N/A

IEC62368_1E - ATTACHMENT			
Clause	Requirement + Test	Result - Remark	Verdict
3.3.19.3	sound exposure, E A-weighted sound pressure (p) squared and integrated over a stated period of time, T Note 1 to entry: The SI unit is $\text{Pa}^2 \text{s}$, $E = \int_0^T p(t)^2 dt$		N/A
3.3.19.4	sound exposure level, SEL logarithmic measure of sound exposure relative to a reference value, E_0, typically the 1 kHz threshold of hearing in humans, Note 1 to entry: SEL is measured as A-weighted levels in dB, $SEL = 10 \lg \left(\frac{E}{E_0} \right) \text{ dB}$ Note 2 to entry: See B,4 of EN 50332-3:2017 for additional information,		N/A
3.3.19.5	digital signal level relative to full scale, dBFS levels reported in dBFS are always r,m,s, Full scale level, 0 dBFS, is the level of a dc-free 997-Hz sine wave whose undithered positive peak value is positive digital full scale, leaving the code corresponding to negative digital full scale unused Note 1 to entry: It is invalid to use dBFS for non-r,m,s, levels, Because the definition of full scale is based on a sine wave, the level of signals with a crest factor lower than that of a sine wave may exceed 0 dBFS, In particular, square wave signals may reach +3.01 dBFS,		N/A

IEC62368_1E - ATTACHMENT			
Clause	Requirement + Test	Result - Remark	Verdict
2	Modification to Clause 10		N/A
10.6	Safeguards against acoustic energy sources Replace 10.6 of IEC 62368-1 with the following:		N/A
10.6.1.1	Introduction Safeguard requirements for protection against long-term exposure to excessive sound pressure levels from personal music players closely coupled to the ear are specified below, Requirements for earphones and headphones intended for use with personal music players are also covered, A personal music player is a portable equipment intended for use by an ordinary person, that: – is designed to allow the user to listen to audio or audiovisual content / material; and – uses a listening device, such as headphones or earphones that can be worn in or on or around the ears; and – has a player that can be body worn (of a size suitable to be carried in a clothing pocket) and is intended for the user to walk around with while in continuous use (for example, on a street, in a subway, at an airport, etc.), EXAMPLES Portable CD players, MP3 audio players, mobile phones with MP3 type features, PDAs or similar equipment, Personal music players shall comply with the requirements of either 10.6.2 or 10.6.3, NOTE 1 Protection against acoustic energy sources from telecom applications is referenced to ITU-T P,360, NOTE 2 It is the intention of the Committee to allow the alternative methods for now, but to only use the dose measurement method as given in 10.6.5 in future, Therefore, manufacturers are encouraged to implement 10.6.5 as soon as possible, Listening devices sold separately shall comply with the requirements of 10.6.6, These requirements are valid for music or video mode only, The requirements do not apply to: – professional equipment; NOTE 3 Professional equipment is equipment sold through special sales channels, All products sold through normal electronics stores are considered not to be professional equipment, – hearing aid equipment and other devices for assistive listening; – the following type of analogue personal music players: • long distance radio receiver (for example, a multiband radio receiver or world band radio receiver, an AM radio receiver), and		N/A

IEC62368_1E - ATTACHMENT			
Clause	Requirement + Test	Result - Remark	Verdict
	• cassette player/recorder;		
Cont'd	NOTE 4 This exemption has been allowed because this technology is falling out of use and it is expected that within a few years it will no longer exist, This exemption will not be extended to other technologies, – a player while connected to an external amplifier that does not allow the user to walk around while in use, For equipment that is clearly designed or intended primarily for use by children, the limits of the relevant toy standards may apply, The relevant requirements are given in EN 71-1: 2011, 4.20 and the related tests methods and measurement distances apply,		N/A
10.6.1.2	Non-ionizing radiation from radio frequencies in the range 0 to 300 GHz The amount of non-ionizing radiation is regulated by European Council Recommendation 1999/519/EC of 12 July 1999 on the limitation of exposure of the general public to electromagnetic fields (0 Hz to 300 GHz), For intentional radiators, ICNIRP guidelines should be taken into account for Limiting Exposure to Time-Varying Electric, Magnetic, and Electromagnetic Fields (up to 300 GHz), For hand-held and body mounted devices, attention is drawn to EN 50360 and EN 50566,		N/A
10.6.2	Classification of devices without the capacity to estimate sound dose		N/A
10.6.2.1	General This standard is transitioning from short-term based (30 s) requirements to long-term based (40 hour) requirements, These clauses remain in effect only for devices that do not comply with sound dose estimation as stipulated in EN 50332-3, For classifying the acoustic output $L_{Aeq,T}$, measurements are based on the A-weighted equivalent sound pressure level over a 30 s period, For music where the average sound pressure (long term $L_{Aeq,T}$) measured over the duration of the song is lower than the average produced by the programme simulation noise, measurements may be done over the duration of the complete song, In this case, T becomes the duration of the song, NOTE Classical music, acoustic music and broadcast typically has an average sound pressure (long term $L_{Aeq,T}$) which is much lower than the average programme simulation noise, Therefore, if the player is capable to analyse the content and compare it with the programme simulation noise, the warning does not need to be given as long as the average sound pressure of the song does not exceed the required limit, For example, if the player is set with the programme simulation noise to 85 dB, but the average music level of the song is only 65 dB,		N/A

IEC62368_1E - ATTACHMENT			
Clause	Requirement + Test	Result - Remark	Verdict
	there is no need to give a warning or ask an acknowledgement as long as the average sound level of the song is not above the basic limit of 85 dB,		
10.6.2.2	RS1 limits (to be superseded, see 10.6.3.2) RS1 is a class 1 acoustic energy source that does not exceed the following: – for equipment provided as a package (player with its listening device), and with a proprietary connector between the player and its listening device, or where the combination of player and listening device is known by other means such as setting or automatic detection, the $L_{Aeq,T}$ acoustic output shall be ≤ 85 dB when playing the fixed “programme simulation noise” described in EN 50332-1, – for equipment provided with a standardized connector (for example, a 3.5 phone jack) that allows connection to a listening device for general use, the unweighted r.m.s. output voltage shall be ≤ 27 mV (analogue interface) or -25 dBFS (digital interface) when playing the fixed “programme simulation noise” described in EN 50332-1, – The RS1 limits will be updated for all devices as per 10.6.3.2,		N/A
10.6.2.3	RS2 limits (to be superseded, see 10.6.3.3) RS2 is a class 2 acoustic energy source that does not exceed the following: – for equipment provided as a package (player with its listening device), and with a proprietary connector between the player and its listening device, or when the combination of player and listening device is known by other means such as setting or automatic 130 detection, the $L_{Aeq,T}$ acoustic output shall be ≤ 100 dB(A) when playing the fixed “programme simulation noise” as described in EN 50332-1, – for equipment provided with a standardized connector (for example, a 3.5 phone jack) that allows connection to a listening device for general use, the unweighted r.m.s. output voltage shall be ≤ 150 mV (analogue interface) or -10 dBFS (digital interface) when playing the fixed “programme simulation noise” as described in EN 50332-1,		N/A
10.6.2.4	RS3 limits RS3 is a class 3 acoustic energy source that exceeds RS2 limits,		N/A
10.6.3	Classification of devices (new)		N/A
10.6.3.1	General		N/A

IEC62368_1E - ATTACHMENT			
Clause	Requirement + Test	Result - Remark	Verdict
	Previous limits (10.6.2) created abundant false negative and false positive PMP sound level warnings, New limits, compliant with The Commission Decision of 23 June 2009, are given below,		
10.6.3.2	RS1 limits (new) RS1 is a class 1 acoustic energy source that does not exceed the following: – for equipment provided as a package (player with its listening device), and with a proprietary connector between the player and its listening device, or where the combination of player and listening device is known by other means such as setting or automatic detection, the $L_{Aeq,T}$ acoustic output shall be ≤ 80 dB when playing the fixed “programme simulation noise” described in EN 50332-1, – for equipment provided with a standardized connector (for example, a 3,5 phone jack) that allows connection to a listening device for general use, the unweighted r,m,s, output voltage shall be ≤ 15 mV (analogue interface) or -30 dBFS (digital interface) when playing the fixed “programme simulation noise” described in EN 50332-1,		N/A
10.6.3.3	RS2 limits (new) RS2 is a class 2 acoustic energy source that does not exceed the following: – for equipment provided as a package (player with its listening device), and with a proprietary connector between the player and its listening device, or where the combination of player and listening device is known by other means such as setting or automatic detection, the weekly sound exposure level, as described in EN 50332-3, shall be ≤ 80 dB when playing the fixed “programme simulation noise” described in EN 50332-1, – for equipment provided with a standardized connector (for example, a 3.5 phone jack) that allows connection to a listening device for general use, the unweighted r,m,s, output level, integrated over one week, as described in EN50332-3, shall be ≤ 15 mV (analogue interface) or -30 dBFS (digital interface) when playing the fixed “programme simulation noise” described in EN 50332-1,		N/A
10.6.4	Requirements for maximum sound exposure		N/A
10.6.4.1	Measurement methods All volume controls shall be turned to maximum during tests, Measurements shall be made in accordance with EN 50332-1 or EN 50332-2 as applicable,		N/A

IEC62368_1E - ATTACHMENT			
Clause	Requirement + Test	Result - Remark	Verdict
10.6.4.2	Protection of persons Except as given below, protection requirements for parts accessible to ordinary persons, instructed persons and skilled persons are given in 4.3, NOTE 1 Volume control is not considered a safeguard, Between RS2 and an ordinary person, the basic safeguard may be replaced by an instructional safeguard in accordance with Clause F.5, except that the instructional safeguard shall be placed on the equipment, or on the packaging, or in the instruction manual, Alternatively, the instructional safeguard may be given through the equipment display during use, The elements of the instructional safeguard shall be as follows: – element 1a: the symbol , IEC 60417-6044 (2011-01) – element 2: "High sound pressure" or equivalent wording – element 3: "Hearing damage risk" or equivalent wording – element 4: "Do not listen at high volume levels for long periods," or equivalent wording An equipment safeguard shall prevent exposure of an ordinary person to an RS2 source without intentional physical action from the ordinary person and shall automatically return to an output level not exceeding what is specified for an RS1 source when the power is switched off, The equipment shall provide a means to actively inform the user of the increased sound level when the equipment is operated with an output exceeding RS1, Any means used shall be acknowledged by the user before activating a mode of operation which allows for an output exceeding RS1, The acknowledgement does not need to be repeated more than once every 20 h of cumulative listening time, NOTE 2 Examples of means include visual or audible signals, Action from the user is always needed, NOTE 3 The 20 h listening time is the accumulative listening time, independent of how often and how long the personal music player has been switched off, A skilled person shall not be unintentionally exposed to RS3,		N/A

IEC62368_1E - ATTACHMENT			
Clause	Requirement + Test	Result - Remark	Verdict
10.6.5	Requirements for dose-based systems		N/A
10.6.5.1	General requirements Personal music players shall give the warnings as provided below when tested according to EN 50332-3, using the limits from this clause, The manufacturer may offer optional settings to allow the users to modify when and how they wish to receive the notifications and warnings to promote a better user experience without defeating the safeguards, This allows the users to be informed in a method that best meets their physical capabilities and device usage needs, If such optional settings are offered, an administrator (for example, parental restrictions, business/educational administrators, etc,) shall be able to lock any optional settings into a specific configuration, The personal music player shall be supplied with easy to understand explanation to the user of the dose management system, the risks involved, and how to use the system safely, The user shall be made aware that other sources may significantly contribute to their sound exposure, for example work, transportation, concerts, clubs, cinema, car races, etc,		N/A
10.6.5.2	Dose-based warning and requirements When a dose of 100 % CSD is reached, and at least at every 100 % further increase of CSD, the device shall warn the user and require an acknowledgement, In case the user does not acknowledge, the output level shall automatically decrease to compliance with class RS1, The warning shall at least clearly indicate that listening above 100 % CSD leads to the risk of hearing damage or loss,		N/A

IEC62368_1E - ATTACHMENT			
Clause	Requirement + Test	Result - Remark	Verdict
10.6.5.3	Exposure-based requirements With only dose-based requirements, cause and effect could be far separated in time, defying the purpose of educating users about safe listening practice, In addition to dose-based requirements, a PMP shall therefore also put a limit to the short-term sound level a user can listen at, The exposure-based limiter (EL) shall automatically reduce the sound level not to exceed 100 dB(A) or 150 mV integrated over the past 180 s, based on methodology defined in EN 50332-3, The EL settling time (time from starting level reduction to reaching target output) shall be 10 s or faster, Test of EL functionality is conducted according to EN 50332-3, using the limits from this clause, For equipment provided as a package (player with its listening device), the level integrated over 180 s shall be 100 dB or lower, For equipment provided with a standardized connector, the unweighted level integrated over 180 s shall be no more than 150 mV for an analogue interface and no more than -10 dBFS for a digital interface, NOTE In case the source is known not to be music (or test signal), the EL may be disabled,		N/A
10.6.6	Requirements for listening devices (headphones, earphones, etc.)		N/A
10.6.6.1	Corded listening devices with analogue input With 94 dB L_{Aeq} acoustic pressure output of the listening device, and with the volume and sound settings in the listening device (for example, built-in volume level control, additional sound features like equalization, etc.) set to the combination of positions that maximize the measured acoustic output, the input voltage of the listening device when playing the fixed “programme simulation noise” as described in EN 50332-1 shall be ≥ 75 mV, NOTE The values of 94 dB and 75 mV correspond with 85 dB and 27 mV or 100 dB and 150 mV,		N/A
10.6.6.2	Corded listening devices with digital input With any playing device playing the fixed “programme simulation noise” described in EN 50332-1, and with the volume and sound settings in the listening device (for example, built-in volume level control, additional sound features like equalization, etc.) set to the combination of positions that maximize the measured acoustic output, the $L_{Aeq, \tau}$ acoustic output of the listening device shall be ≤ 100 dB with an input signal of -10 dBFS,		N/A

Annexment 2: May 20, 2020

IEC62368_1E - ATTACHMENT							
Clause	Requirement + Test				Result - Remark		Verdict
10.6.6.3	Cordless listening devices In cordless mode, – with any playing and transmitting device playing the fixed programme simulation noise described in EN 50332-1; and – respecting the cordless transmission standards, where an air interface standard exists that specifies the equivalent acoustic level; and – with volume and sound settings in the receiving device (for example, built-in volume level control, additional sound features like equalization, etc.) set to the combination of positions that maximize the measured acoustic output for the above mentioned programme simulation noise, the $L_{Aeq,T}$ acoustic output of the listening device shall be ≤ 100 dB with an input signal of -10 dBFS,						N/A
10.6.6.4	Measurement method <i>Measurements shall be made in accordance with EN 50332-2 as applicable,</i>						N/A
3	Modification to the whole document						N/A
	Delete all the “country” notes in the reference document according to the following						N/A
	0.2.1	Note 1 and 2	1	Note 4 and 5	3.3.8.1	Note 2	
	3.3.8.3	Note 1	4.1.15	Note	4.7.3	Note 1 and 2	
	5.2.2.2	Note	5.4.2.3.2.2 Table 12	Note c	5.4.2.3.2.4	Note 1 and 3	
	5.4.2.3.2.4 Table 13	Note 2	5.4.2.5	Note 2	5.4.5.1	Note	
	5.4.10.2.1	Note	5.4.10.2.2	Note	5.4.10.2.3	Note	
	5.5.2.1	Note	5.5.6	Note	5.6.4.2.1	Note 2 and 3 and 4	
	5.6.8	Note 2	5.7.6	Note	5.7.7.1	Note 1 and Note 2	
	8.5.4.2.3	Note	10.2.1 Table 39	Note 3 and 4 and 5	10.5.3	Note 2	
	10.6.4	Note 3	F.3.3.6	Note 3	Y.4.1	Note	
	Y.4.5	Note					
	list:						

IEC62368_1E - ATTACHMENT			
Clause	Requirement + Test	Result - Remark	Verdict
4	Modification to Clause 1		N/A
1	Add the following note: <i>NOTE Z1 The use of certain substances in electrical and electronic equipment is restricted within the EU: see Directive 2011/65/EU,</i>		N/A
5	Modification to 4,Z1		N/A
4.Z1	Add the following new subclause after 4.9: To protect against excessive current, short-circuits and earth faults in circuits connected to an a,c, mains, protective devices shall be included either as integral parts of the equipment or as parts of the building installation, subject to the following, a), b) and c): a) except as detailed in b) and c), protective devices necessary to comply with the requirements of B.3.1 and B.4 shall be included as parts of the equipment; b) for components in series with the mains input to the equipment such as the supply cord, appliance coupler, r,f,i, filter and switch, short-circuit and earth fault protection may be provided by protective devices in the building installation; c) it is permitted for pluggable equipment type B or permanently connected equipment, to rely on dedicated overcurrent and short-circuit protection in the building installation, provided that the means of protection, e.g, fuses or circuit breakers, is fully specified in the installation instructions, If reliance is placed on protection in the building installation, the installation instructions shall so state, except that for pluggable equipment type A the building installation shall be regarded as providing protection in accordance with the rating of the wall socket outlet,		N/A
6	Modification to 5.4.2.3.2.4		N/A
5.4.2.3.2.4	Add the following to the end of this subclause: The requirement for interconnection with external circuit is in addition given in EN 50491-3:2009,		N/A
7	Modification to 10.2.1		N/A
10.2.1	Add the following to ^{c)} and ^{d)} in table 39: For additional requirements, see 10.5.1,		N/A

IEC62368_1E - ATTACHMENT			
Clause	Requirement + Test	Result - Remark	Verdict
8	Modification to 10.5.1		N/A
10.5.1	Add the following after the first paragraph: For RS 1 compliance is checked by measurement under the following conditions: In addition to the normal operating conditions, all controls adjustable from the outside by hand, by any object such as a tool or a coin, and those internal adjustments or pre-sets which are not locked in a reliable manner, are adjusted so as to give maximum radiation whilst maintaining an intelligible picture for 1 h, at the end of which the measurement is made, NOTE Z1 Soldered joints and paint lockings are examples of adequate locking, The dose-rate is determined by means of a radiation monitor with an effective area of 10 cm², at any point 10 cm from the outer surface of the apparatus, Moreover, the measurement shall be made under fault conditions causing an increase of the high voltage, provided an intelligible picture is maintained for 1 h, at the end of which the measurement is made, For RS1, the dose-rate shall not exceed 1 µSv/h taking account of the background level, NOTE Z2 These values appear in Directive 96/29/Euratom of 13 May 1996,		N/A
9	Modification to G.7.1		N/A
G.7.1	Add the following note: NOTE Z1 The harmonized code designations corresponding to the IEC cord types are given in Annex ZD,		N/A

IEC62368_1E - ATTACHMENT			
Clause	Requirement + Test	Result - Remark	Verdict
10	Modification to Bibliography		N/A
	Add the following notes for the standards indicated: IEC 60130-9 NOTE Harmonized as EN 60130-9. IEC 60269-2 NOTE Harmonized as HD 60269-2. IEC 60309-1 NOTE Harmonized as EN 60309-1. IEC 60364 NOTE some parts harmonized in HD 384/HD 60364 series. IEC 60601-2-4 NOTE Harmonized as EN 60601-2-4. IEC 60664-5 NOTE Harmonized as EN 60664-5. IEC 61032:1997 NOTE Harmonized as EN 61032:1998 (not modified). IEC 61508-1 NOTE Harmonized as EN 61508-1. IEC 61558-2-1 NOTE Harmonized as EN 61558-2-1. IEC 61558-2-4 NOTE Harmonized as EN 61558-2-4. IEC 61558-2-6 NOTE Harmonized as EN 61558-2-6. IEC 61643-1 NOTE Harmonized as EN 61643-1. IEC 61643-21 NOTE Harmonized as EN 61643-21. IEC 61643-311 NOTE Harmonized as EN 61643-311. IEC 61643-321 NOTE Harmonized as EN 61643-321. IEC 61643-331 NOTE Harmonized as EN 61643-331.		N/A
11	ADDITION OF ANNEXES		P
ZB	ANNEX ZB, SPECIAL NATIONAL CONDITIONS (EN)		P
4.1.15	Denmark, Finland, Norway and Sweden To the end of the subclause the following is added: Class I pluggable equipment type A intended for connection to other equipment or a network shall, if safety relies on connection to reliable earthing or if surge suppressors are connected between the network terminals and accessible parts, have a marking stating that the equipment shall be connected to an earthed mains socket-outlet, The marking text in the applicable countries shall be as follows: In Denmark: "Apparatets stikprop skal tilsluttes en stikkontakt med jord som giver forbindelse til stikproppens jord," In Finland: "Laite on liitettävä suojakoskettimilla varustettuun pistorasiaan" In Norway: "Apparatet må tilkoples jordet stikkontakt" In Sweden: "Apparaten skall anslutas till jordat uttag"		N/A
4.7.3	United Kingdom To the end of the subclause the following is added: The torque test is performed using a socket-outlet complying with BS 1363, and the plug part shall be assessed to the relevant clauses of BS 1363, Also see Annex G.4.2 of this annex		N/A

IEC62368_1E - ATTACHMENT			
Clause	Requirement + Test	Result - Remark	Verdict
5.2.2.2	Denmark After the 2nd paragraph add the following: A warning (marking safeguard) for high touch current is required if the touch current exceeds the limits of 3.5 mA a.c, or 10 mA d.c,		N/A
5.4.11.1 and Annex G	Finland and Sweden To the end of the subclause the following is added: For separation of the telecommunication network from earth the following is applicable: If this insulation is solid, including insulation forming part of a component, it shall at least consist of either • two layers of thin sheet material, each of which shall pass the electric strength test below, or • one layer having a distance through insulation of at least 0.4 mm, which shall pass the electric strength test below, If this insulation forms part of a semiconductor component (e.g, an optocoupler), there is no distance through insulation requirement for the insulation consisting of an insulating compound completely filling the casing, so that clearances and creepage distances do not exist, if the component passes the electric strength test in accordance with the compliance clause below and in addition • passes the tests and inspection criteria of 5.4.8 with an electric strength test of 1.5 kV multiplied by 1.6 (the electric strength test of 5.4.9 shall be performed using 1.5 kV), and • is subject to routine testing for electric strength during manufacturing, using a test voltage of 1.5 kV, It is permitted to bridge this insulation with a capacitor complying with EN 60384-14:2005, subclass Y2, A capacitor classified Y3 according to EN 60384-14: 2005, may bridge this insulation under the following conditions: • the insulation requirements are satisfied by having a capacitor classified Y3 as defined by EN 60384-14, which in addition to the Y3 testing, is tested with an impulse test of 2.5 kV defined in 5.4.11; • the additional testing shall be performed on all the test specimens as described in EN 60384-14; the impulse test of 2.5 kV is to be performed before the endurance test in EN 60384-14, in the sequence of tests as described in EN 60384-14,		N/A

IEC62368_1E - ATTACHMENT			
Clause	Requirement + Test	Result - Remark	Verdict
5.5.2.1	Norway After the 3rd paragraph the following is added: Due to the IT power system used, capacitors are required to be rated for the applicable line-to-line voltage (230 V),		P
5.5.6	Finland, Norway and Sweden To the end of the subclause the following is added: Resistors used as basic safeguard or bridging basic insulation in class I pluggable equipment type A shall comply with G.10.1 and the test of G.10.2,	No such resistor used	N/A
5.6.1	Denmark Add to the end of the subclause Due to many existing installations where the socket-outlets can be protected with fuses with higher rating than the rating of the socket-outlets the protection for pluggable equipment type A shall be an integral part of the equipment, <i>Justification:</i> In Denmark an existing 13 A socket outlet can be protected by a 20 A fuse,		P
5.6.4.2.1	Ireland and United Kingdom After the indent for pluggable equipment type A , the following is added: – the protective current rating is taken to be 13 A, this being the largest rating of fuse used in the mains plug,		N/A
5.6.4.2.1	France After the indent for pluggable equipment type A , the following is added: – in certain cases, the protective current rating of the circuit supplied from the mains is taken as 20 A instead of 16 A,		N/A
5.6.5.1	To the second paragraph the following is added: The range of conductor sizes of flexible cords to be accepted by terminals for equipment with a rated current over 10 A and up to and including 13 A is: 1.25 mm ² to 1.5 mm ² in cross-sectional area,		N/A
5.6.8	Norway To the end of the subclause the following is added: Equipment connected with an earthed mains plug is classified as class I equipment , See the Norway marking requirement in 4.1.15, The symbol IEC 60417-6092, as specified in F.3.6.2, is accepted,		N/A
5.7.6	Denmark To the end of the subclause the following is added: The installation instruction shall be affixed to the		N/A

IEC62368_1E - ATTACHMENT			
Clause	Requirement + Test	Result - Remark	Verdict
	equipment if the protective conductor current exceeds the limits of 3.5 mA a.c, or 10 mA d.c,		
5.7.6.2	Denmark To the end of the subclause the following is added: The warning (marking safeguard) for high touch current is required if the touch current or the protective current exceed the limits of 3.5 mA,		N/A
5.7.7.1	Norway and Sweden To the end of the subclause the following is added: The screen of the television distribution system is normally not earthed at the entrance of the building and there is normally no equipotential bonding system within the building, Therefore the protective earthing of the building installation needs to be isolated from the screen of a cable distribution system, It is however accepted to provide the insulation external to the equipment by an adapter or an interconnection cable with galvanic isolator, which may be provided by a retailer, for example, The user manual shall then have the following or similar information in Norwegian and Swedish language respectively, depending on in what country the equipment is intended to be used in: “Apparatus connected to the protective earthing of the building installation through the mains connection or through other apparatus with a connection to protective earthing – and to a television distribution system using coaxial cable, may in some circumstances create a fire hazard, Connection to a television distribution system therefore has to be provided through a device providing electrical isolation below a certain frequency range (galvanic isolator, see EN 60728-11)” NOTE In Norway, due to regulation for CATV-installations, and in Sweden, a galvanic isolator shall provide electrical insulation below 5 MHz, The insulation shall withstand a dielectric strength of 1,5 kV r,m,s., 50 Hz or 60 Hz, for 1 min, Translation to Norwegian (the Swedish text will also be accepted in Norway): “Apparater som er koplet til beskyttelsesjord via nettplugg og/eller via annet jordtilkoplet utstyr – og er tilkoplet et koaksialbasert kabel-TV nett, kan forårsake brannfare, For å unngå dette skal det ved tilkopling av apparater til kabel-TV nett installeres en galvanisk isolator mellom apparatet og kabel-TV nettet,” Translation to Swedish: ”Apparater som är kopplad till skyddsjord via jordat vägguttag och/eller via annan utrustning och samtidigt är kopplad till kabel-TV nät kan		N/A

IEC62368_1E - ATTACHMENT			
Clause	Requirement + Test	Result - Remark	Verdict
	i vissa fall medföra risk för brand, För att undvika detta skall vid anslutning av apparaten till kabel-TV nät galvanisk isolator finnas mellan apparaten och kabel-TV nätet, ”,		
8.5.4.2.3	United Kingdom Add the following after the 2 nd dash bullet in 3 ^d paragraph: An emergency stop system complying with the requirements of IEC 60204-1 and ISO 13850 is required where there is a risk of personal injury,		N/A
B.3.1 and B.4	Ireland and United Kingdom The following is applicable: To protect against excessive currents and short-circuits in the primary circuit of direct plug-in equipment , tests according to Annexes B,3,1 and B,4 shall be conducted using an external miniature circuit breaker complying with EN 60898-1, Type B, rated 32A, If the equipment does not pass these tests, suitable protective devices shall be included as an integral part of the direct plug-in equipment , until the requirements of Annexes B,3,1 and B,4 are met		N/A

IEC62368_1E - ATTACHMENT			
Clause	Requirement + Test	Result - Remark	Verdict
G.4.2	Denmark To the end of the subclause the following is added: Supply cords of single phase appliances having a rated current not exceeding 13 A shall be provided with a plug according to DS 60884-2-D1:2011, CLASS I EQUIPMENT provided with socket-outlets with earth contacts or which are intended to be used in locations where protection against indirect contact is required according to the wiring rules shall be provided with a plug in accordance with standard sheet DK 2-1a or DK 2-5a, If a single-phase equipment having a RATED CURRENT exceeding 13 A or if a polyphase equipment is provided with a supply cord with a plug, this plug shall be in accordance with the standard sheets DK 6-1a in DS 60884-2-D1 or EN 60309-2, Mains socket outlets intended for providing power to Class II apparatus with a rated current of 2,5 A shall be in accordance DS 60884-2-D1:2011 standard sheet DKA 1-4a, Other current rating socket outlets shall be in compliance with Standard Sheet DKA 1-3a or DKA 1-1c, Mains socket-outlets with earth shall be in compliance with DS 60884-2-D1:2011 Standard Sheet DK 1-3a, DK 1-1c, DK1-1d, DK 1-5a or DK 1-7a <i>Justification:</i> Heavy Current Regulations, Section 6c		N/A
G.4.2	United Kingdom To the end of the subclause the following is added: The plug part of direct plug-in equipment shall be assessed to BS 1363: Part 1, 12.1, 12.2, 12.3, 12.9, 12.11, 12.12, 12.13, 12.16, and 12.17, except that the test of 12.17 is performed at not less than 125 °C, Where the metal earth pin is replaced by an Insulated Shutter Opening Device (ISOD), the requirements of clauses 22.2 and 23 also apply,		P

IEC62368_1E - ATTACHMENT			
Clause	Requirement + Test	Result - Remark	Verdict
G.7.1	United Kingdom To the first paragraph the following is added: Equipment which is fitted with a flexible cable or cord and is designed to be connected to a mains socket conforming to BS 1363 by means of that flexible cable or cord shall be fitted with a 'standard plug' in accordance with the Plugs and Sockets etc, (Safety) Regulations 1994, Statutory Instrument 1994 No, 1768, unless exempted by those regulations, NOTE "Standard plug" is defined in SI 1768:1994 and essentially means an approved plug conforming to BS 1363 or an approved conversion plug,		P
G.7.1	Ireland To the first paragraph the following is added: Apparatus which is fitted with a flexible cable or cord shall be provided with a plug in accordance with Statutory Instrument 525: 1997, "13 A Plugs and Conversion Adapters for Domestic Use Regulations: 1997, S.I, 525 provides for the recognition of a standard of another Member State which is equivalent to the relevant Irish Standard		N/A
G.7.2	Ireland and United Kingdom To the first paragraph the following is added: A power supply cord with a conductor of 1,25 mm² is allowed for equipment which is rated over 10 A and up to and including 13 A,		P

IEC62368_1E - ATTACHMENT																																																								
Clause	Requirement + Test	Result - Remark	Verdict																																																					
ZC	ANNEX ZC, NATIONAL DEVIATIONS (EN)		N/A																																																					
10.5.2	Germany The following requirement applies: For the operation of any cathode ray tube intended for the display of visual images operating at an acceleration voltage exceeding 40 kV, authorization is required, or application of type approval (Bauartzulassung) and marking, <i>Justification:</i> German ministerial decree against ionizing radiation (Röntgenverordnung), in force since 2002-07-01, implementing the European Directive 96/29/EURATOM, NOTE Contact address: Physikalisch-Technische Bundesanstalt, Bundesallee 100, D-38116 Braunschweig, Tel.: Int+49-531-592-6320, Internet: http://www.ptb.de		N/A																																																					
ZD	IEC and CENELEC CODE DESIGNATIONS FOR FLEXIBLE CORDS (EN)		P																																																					
	<table><tr><th rowspan="2">Type of flexible cord</th><th colspan="2">Code designations</th></tr><tr><th>IEC</th><th>CENELEC</th></tr><tr><td colspan="3">PVC insulated cords</td></tr><tr><td>Flat twin tinsel cord</td><td>60227 IEC 41</td><td>H03VH-Y</td></tr><tr><td>Light polyvinyl chloride sheathed flexible cord</td><td>60227 IEC 52</td><td>H03VV-F H03VVH2-F</td></tr><tr><td>Ordinary polyvinyl chloride sheathed flexible cord</td><td>60227 IEC 53</td><td>H05VV-F H05VVH2-F</td></tr><tr><td colspan="3">Rubber insulated cords</td></tr><tr><td>Braided cord</td><td>60245 IEC 51</td><td>H03RT-F</td></tr><tr><td>Ordinary tough rubber sheathed flexible cord</td><td>60245 IEC 53</td><td>H05RR-F</td></tr><tr><td>Ordinary polychloroprene sheathed flexible cord</td><td>60245 IEC 57</td><td>H05RN-F</td></tr><tr><td>Heavy polychloroprene sheathed flexible cord</td><td>60245 IEC 66</td><td>H07RN-F</td></tr><tr><td colspan="3">Cords having high flexibility</td></tr><tr><td>Rubber insulated and sheathed cord</td><td>60245 IEC 86</td><td>H03RR-H</td></tr><tr><td>Rubber insulated, crosslinked PVC sheathed cord</td><td>60245 IEC 87</td><td>H03RV4-H</td></tr><tr><td>Crosslinked PVC insulated and sheathed cord</td><td>60245 IEC 88</td><td>H03V4V4-H</td></tr><tr><td colspan="3">Cords insulated and sheathed with halogen-free thermoplastic compounds</td></tr><tr><td>Light halogen-free thermoplastic insulated and sheathed flexible cords</td><td></td><td>H03Z1Z1-F H03Z1Z1H2-F</td></tr><tr><td>Ordinary halogen-free thermoplastic insulated and sheathed flexible cords</td><td></td><td>H05Z1Z1-F H05Z1Z1H2-F</td></tr></table>		Type of flexible cord	Code designations		IEC	CENELEC	PVC insulated cords			Flat twin tinsel cord	60227 IEC 41	H03VH-Y	Light polyvinyl chloride sheathed flexible cord	60227 IEC 52	H03VV-F H03VVH2-F	Ordinary polyvinyl chloride sheathed flexible cord	60227 IEC 53	H05VV-F H05VVH2-F	Rubber insulated cords			Braided cord	60245 IEC 51	H03RT-F	Ordinary tough rubber sheathed flexible cord	60245 IEC 53	H05RR-F	Ordinary polychloroprene sheathed flexible cord	60245 IEC 57	H05RN-F	Heavy polychloroprene sheathed flexible cord	60245 IEC 66	H07RN-F	Cords having high flexibility			Rubber insulated and sheathed cord	60245 IEC 86	H03RR-H	Rubber insulated, crosslinked PVC sheathed cord	60245 IEC 87	H03RV4-H	Crosslinked PVC insulated and sheathed cord	60245 IEC 88	H03V4V4-H	Cords insulated and sheathed with halogen-free thermoplastic compounds			Light halogen-free thermoplastic insulated and sheathed flexible cords		H03Z1Z1-F H03Z1Z1H2-F	Ordinary halogen-free thermoplastic insulated and sheathed flexible cords		H05Z1Z1-F H05Z1Z1H2-F	P
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Fig. 1: Front view of the equipment



Fig. 2: Rear view of the equipment

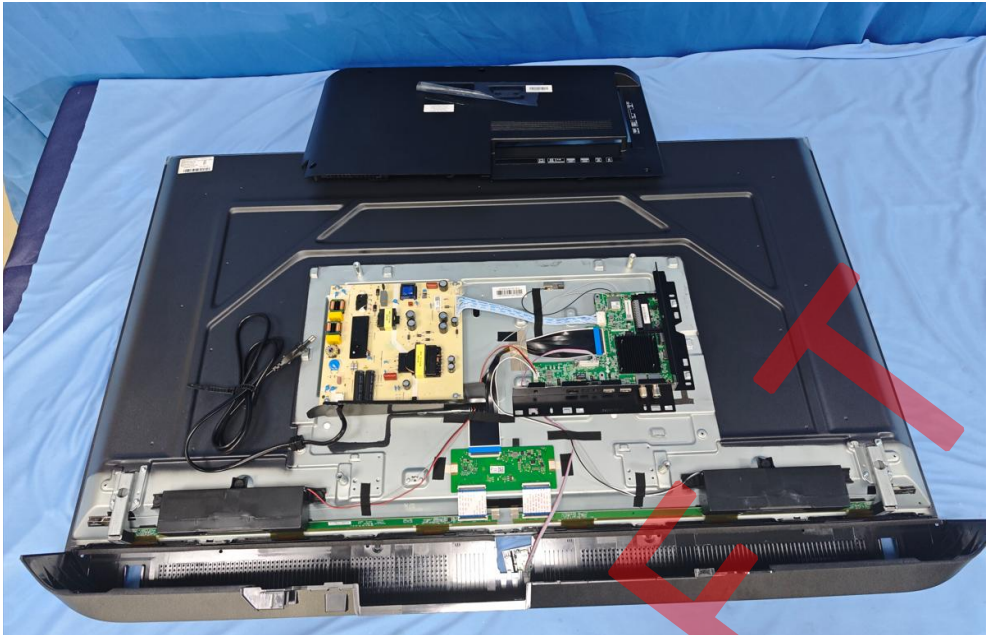


Fig. 3: Internal view-1 of the equipment for configuration A (configuration 1)

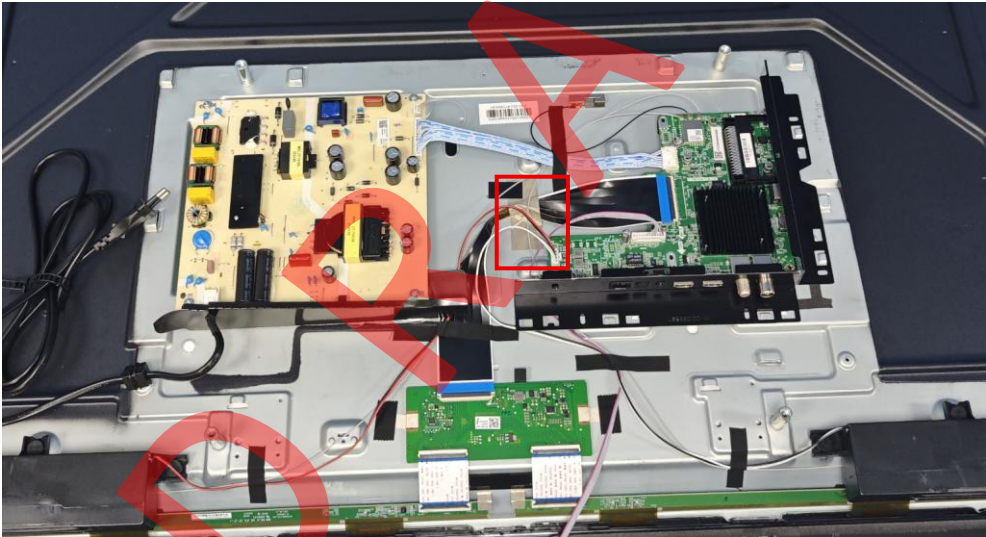


Fig. 4: Internal view-2 of the equipment for configuration A (configuration 1)

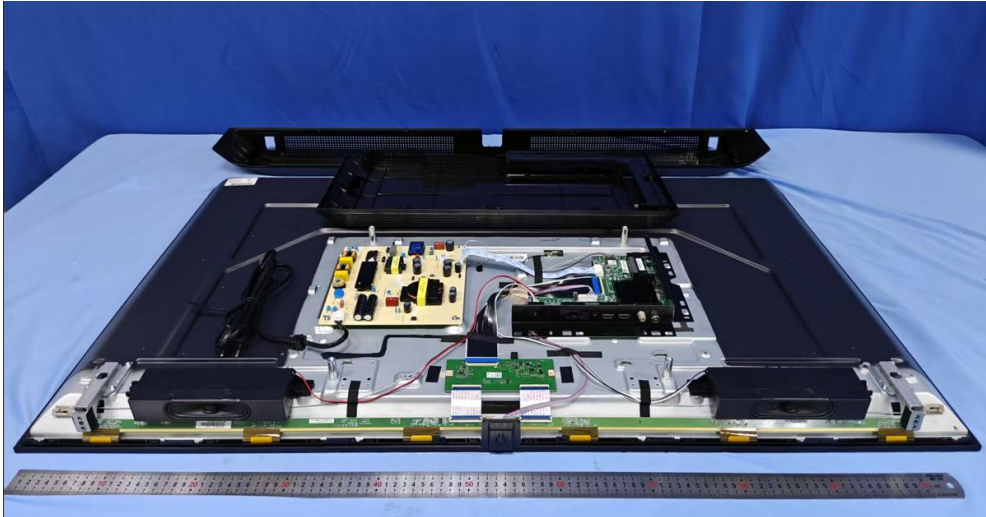


Fig. 5: Internal view-3 of the equipment for configuration A (configuration 2)

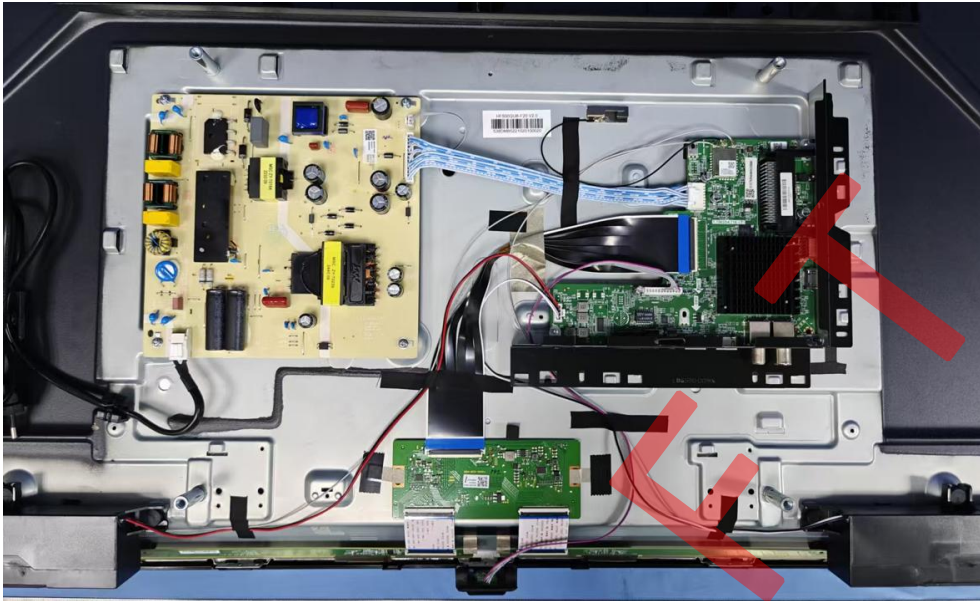


Fig. 6: Internal view-4 of the equipment for configuration A (configuration 2)

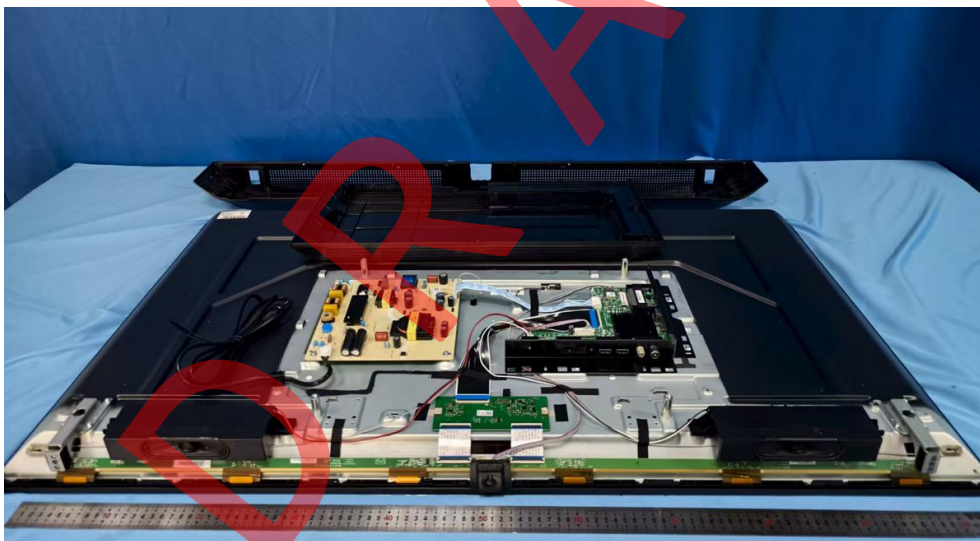


Fig. 7: Internal view-5 of the equipment for configuration B (configuration 2)

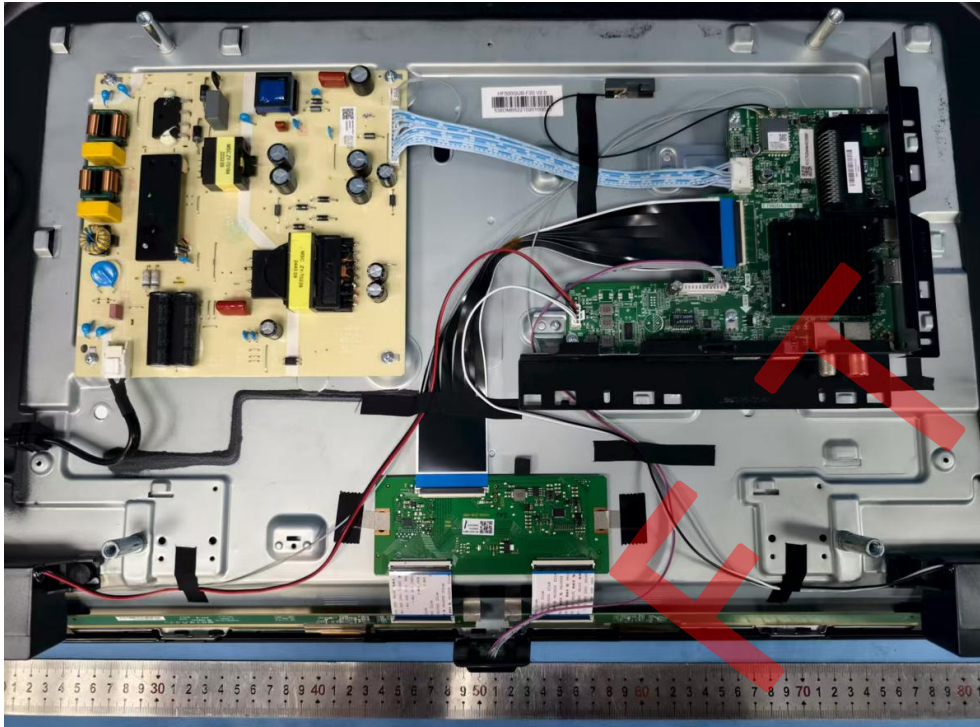


Fig. 8: Internal view-6 of the equipment for configuration B (configuration 2)

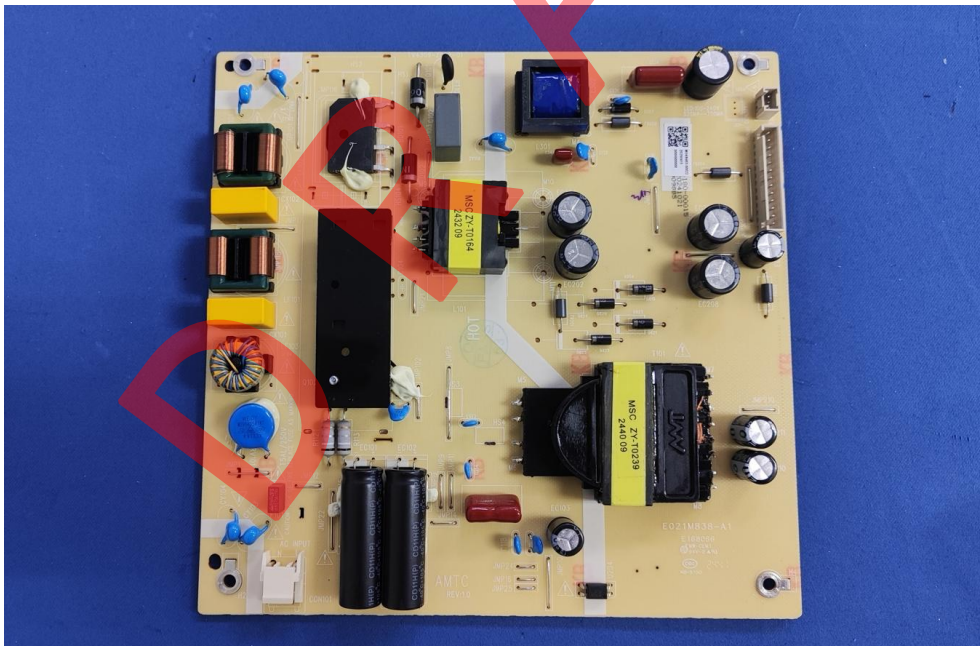


Fig. 9: Component side view of power board

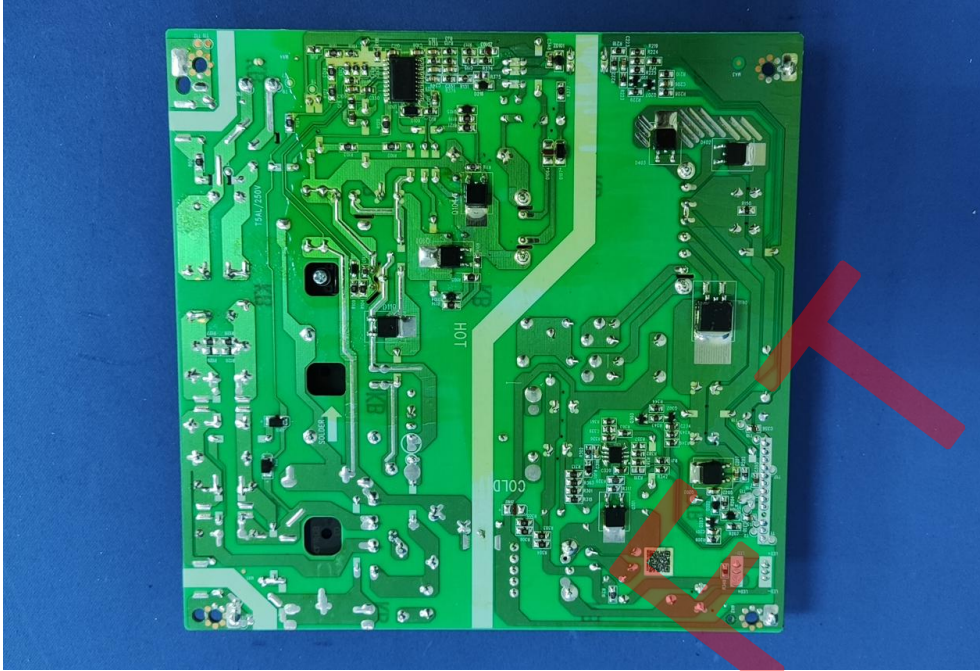


Fig. 10: Soldering side view of power board

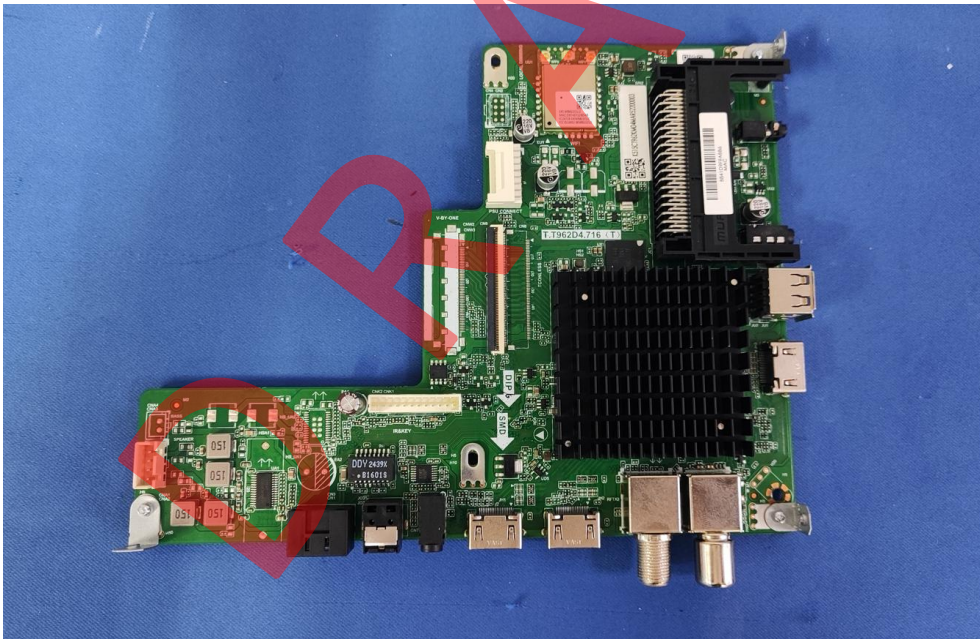


Fig. 11: Component side view of main board

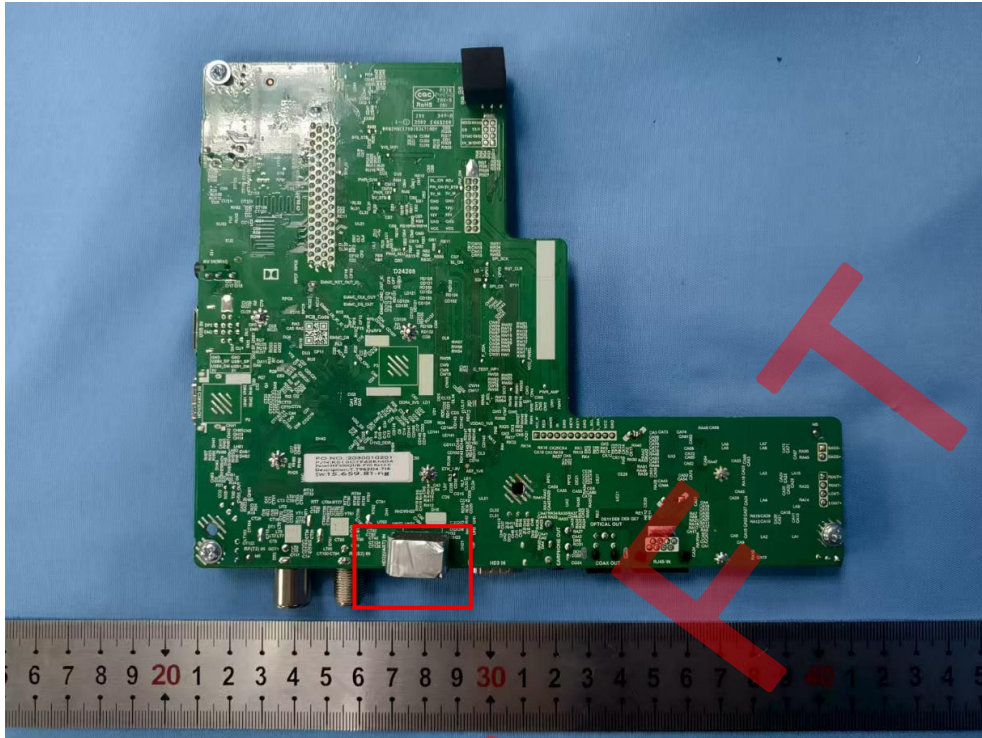


Fig. 12: Soldering side view_1 of main board for configuration A

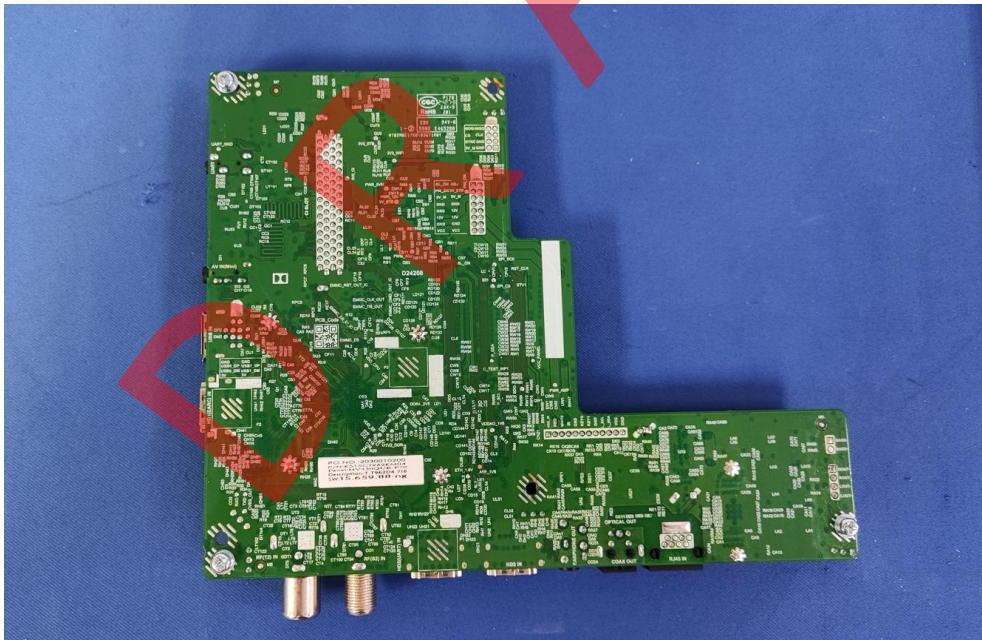


Fig. 13: Soldering side view_2 of main board for configuration B

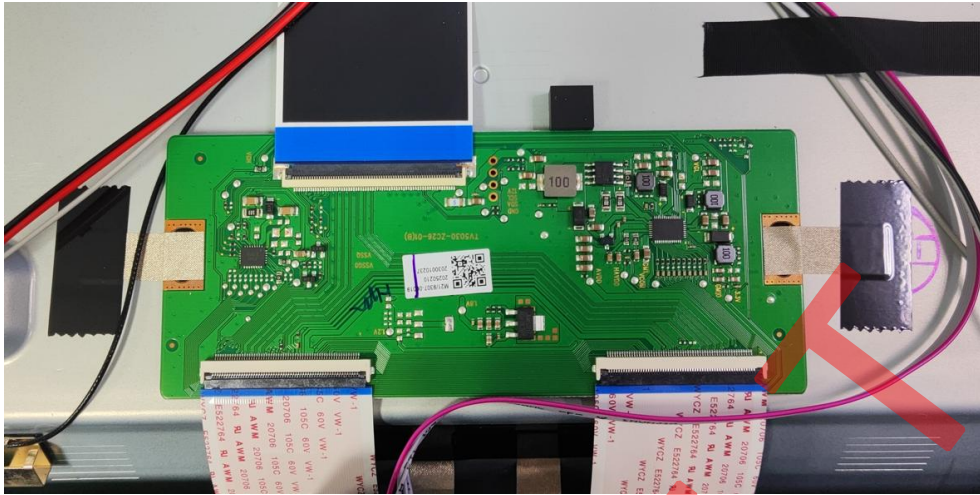


Fig. 14: Component side view of control board



Fig. 15: External view-1 of transformer T101

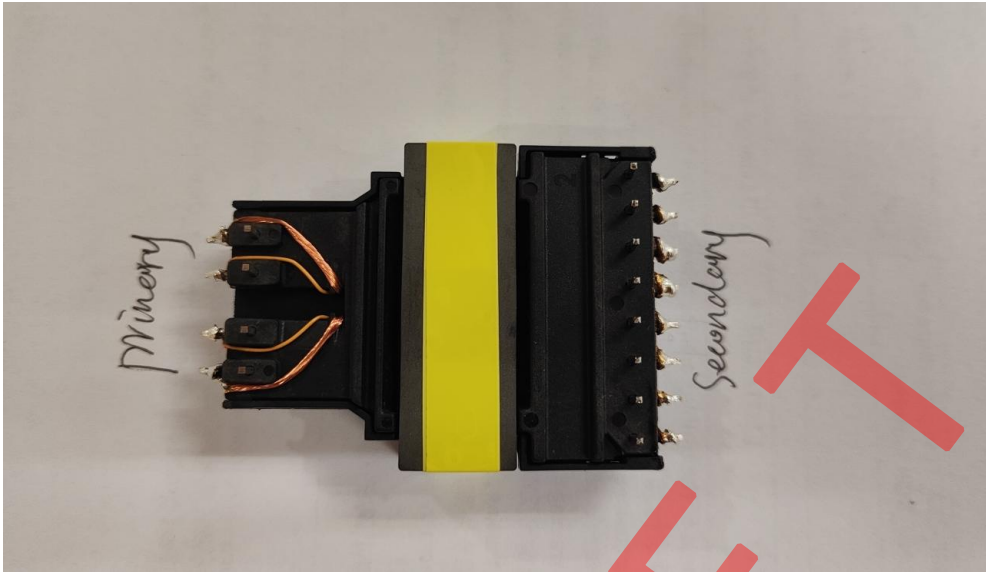


Fig. 16: External view-2 of transformer T101

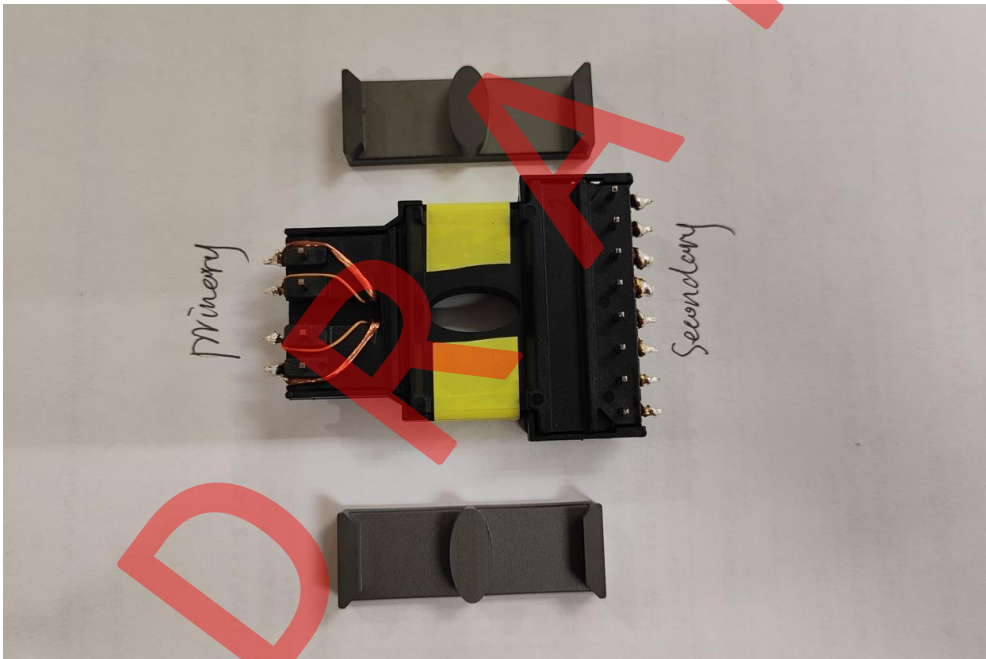


Fig. 17: Internal view-1 of transformer T101

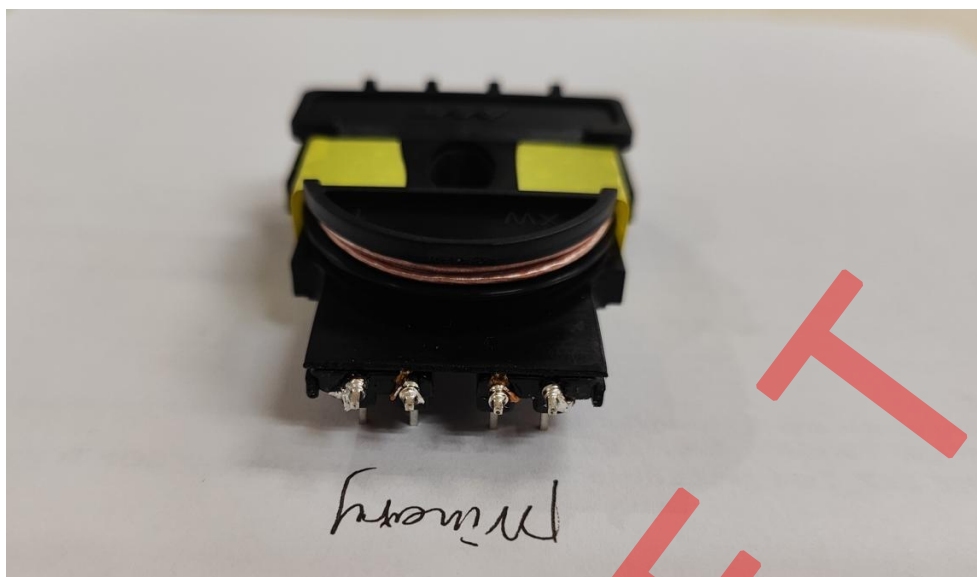


Fig. 18: Internal view-2 of transformer T101

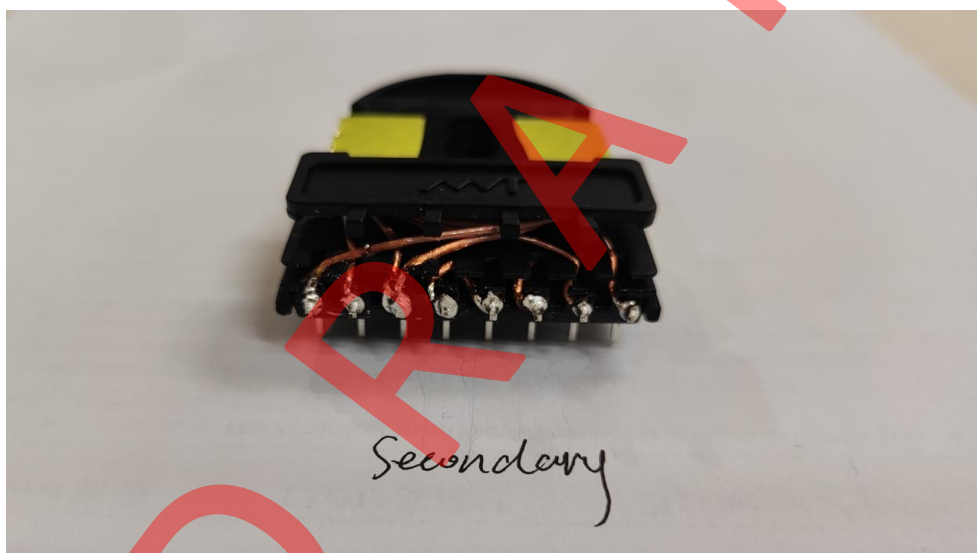


Fig. 19: Internal view-3 of transformer T101



Fig. 20: Front view with stand base B



Fig. 21: Rear view with stand base B

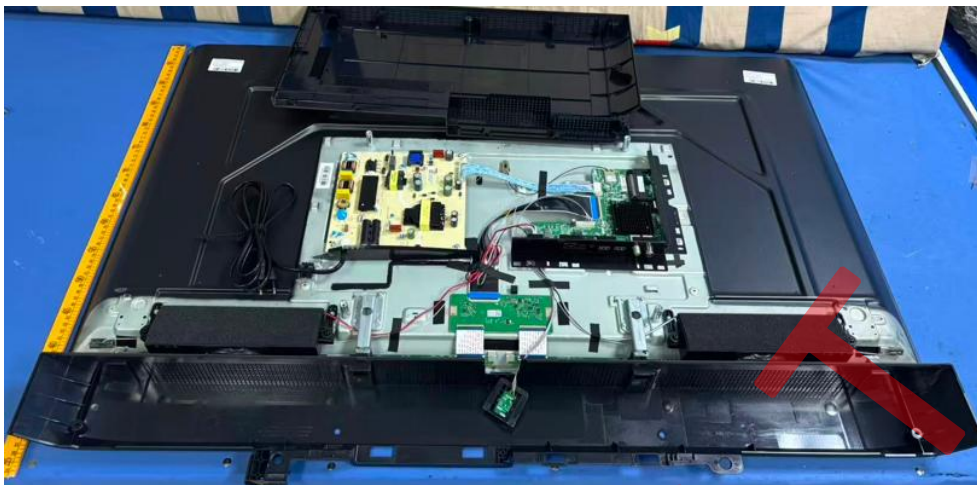


Fig. 22: Internal view_1 of equipment



Fig. 23: Internal view_1 of equipment

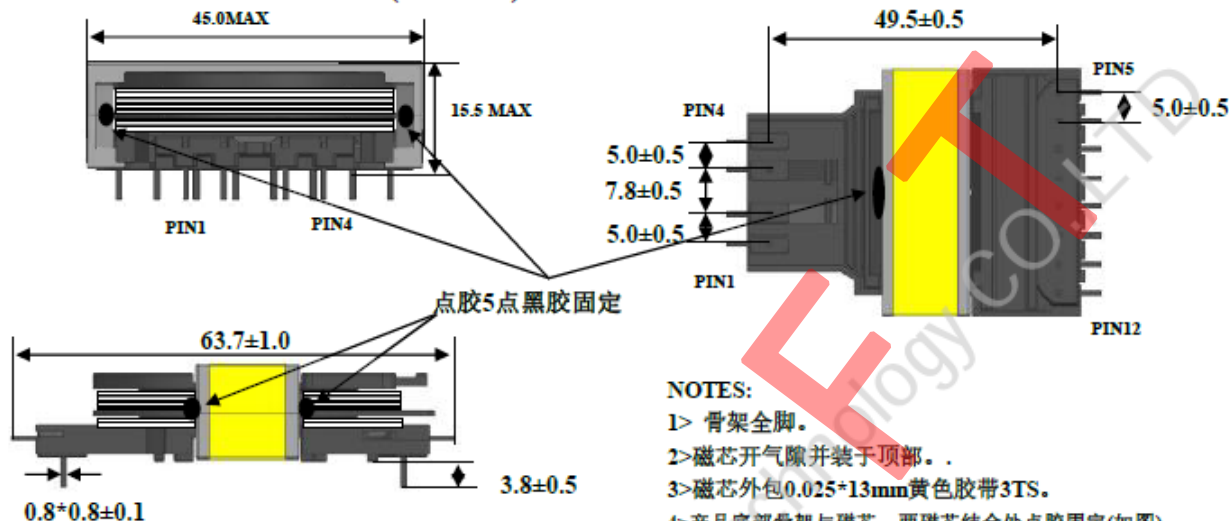
1. IDENTIFICATION

PART TYPE:ED43变压器

PART NAME:ZY-T0239 A9

MANUFACTURER : Huizhou MVT New Power Technology CO.,LTD

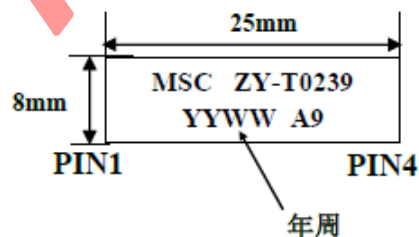
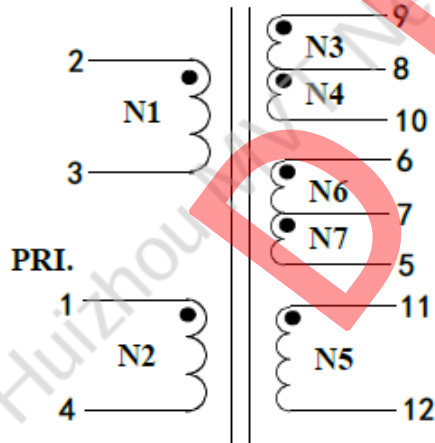
2. MECHANICAL DIMENSION(UNIT:mm)



NOTES:

- 1> 骨架全脚。
- 2> 磁芯开气隙并装于顶部。
- 3> 磁芯外包0.025*13mm黄色胶带3TS。
- 4> 产品底部骨架与磁芯、两磁芯结合处点胶固定(如图)。
- 5> 脚长 3.8 ± 0.5 不包括锡尖。
- 6> 磁芯中柱填气隙胶作业
- 7> 标签:贴标签 (透明底黑字)或者喷码于产品顶部:

3. SCHEMATICS :



SEC. 7/8为可连焊脚位

说明: 连焊脚位实际制作中, 可连锡;
也可不连锡, 机板连接在一起不影响使用!

SAME PHASE ●

TEFLON TUBE: ☐

4. WINDING CONSTRUCTION

BOBBIN装夹方向

从PIN脚侧边看，产品逆时针绕线

5. WINDING	TERMINAL	WIRE SIZE	TURNS	TAPE LAYERS
N1	2—3	TRW(B)Φ0.30mm±1	4TS	
N2	1—4	2UEWΦ0.10mm±35*1 (绞合线)	33TS	
N3	9—8	2UEWΦ0.10mm±50*1 (丝包线)	3TS	同层并绕
N4	8—10	2UEWΦ0.10mm±50*1 (丝包线)	3TS	
N5	11—12	2UEWΦ0.10mm±20*1 (丝包线)	24TS	
N6	6—7	2UEWΦ0.10mm±50*1 (丝包线)	2TS	同层并绕
N7	7—5	2UEWΦ0.10mm±50*1 (丝包线)	2TS	

注意：1、圈数参考测试条件：20KHz, 1V。

2、N1N2绕在本体骨架,N3、N4、N5、N6、N7本体装上外壳后再绕制。

3、N1绕在本体第一槽，N2绕在第二槽，N3N4绕在第三槽,N5N6N7绕在第四槽

4、7/8为可连焊脚位。

6. ELECTRICAL CHARACTERISTICS (AT 25 ±3℃)

1) INDUCTANCE:(100KHz,1.0V,内阻 100Ω, 串联测试)

L(1~4)=550uH±10%

2)DC SUPERPOSITION:(100KHz,1.0V,内阻 100Ω, 串联测试) (抽检)

L(1.2A)≥L*80% (25℃); L(1.2A)≥L*80% (110℃)

3) LEAKAGE INDUCTANCE:(100KHz,1.0V,内阻 100Ω, 串联测试)

Lk(1~4)=48±10%uH (短路N3、N4、N5、N6、N7绕组)

7. HI-POT TEST

APPLY AC/3600V,3mA BETWEEN PRI. TO SEC. FOR 60 SECOND.

APPLY AC/500V,3mA BETWEEN SEC AND PRI TO CORE. FOR 60 SECOND.

APPLY AC/3000V,3mA BETWEEN PRI. TO CORE. FOR 60 SECOND.

8. INSULATION RESISTANCE

APPLY DC/500V, BETWEEN PRI TO SEC FOR 100MOHM MIN

9. DC.RESISTANCE (AT25 ±3℃)

R(1~4)=160mΩMAX R(2~3)=110mΩ MAX R(6~5)=35mΩ MAX

R(9~10)=45mΩ MAX R(11~12) =320mΩ MAX

Specification_2 of Transformer